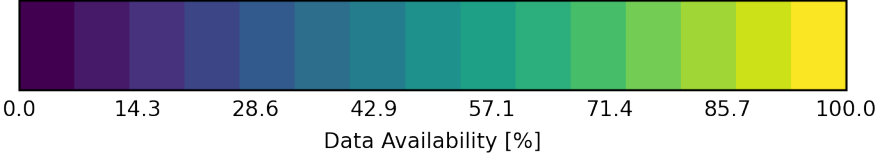


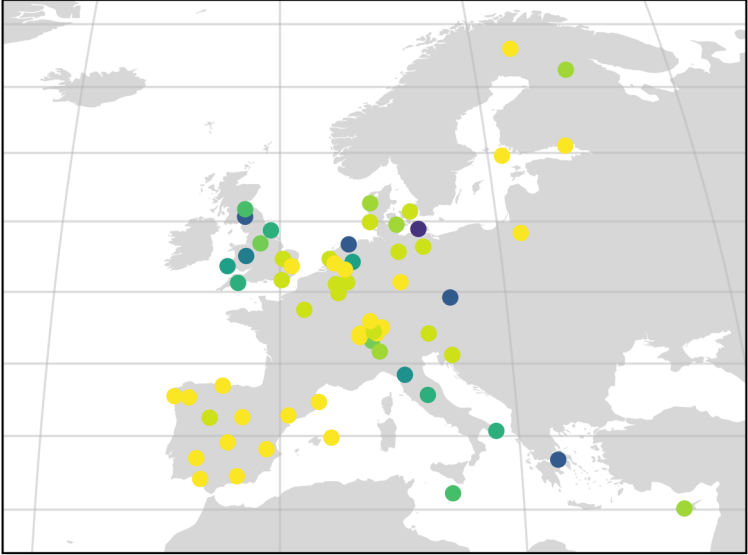
EVALUATION REPORT

Network: EBAS
Species: O3, NO2
Temporal resolution: Hourly
Date range: 01/01/2018 - 01/04/2018
Models: MONARCH
Subsections: LOCAL·All, LOCAL·France, LOCAL·Germany, LOCAL
·Spain, LOCAL·UK

Map Data Availability (Summary)
EBAS|sconco3



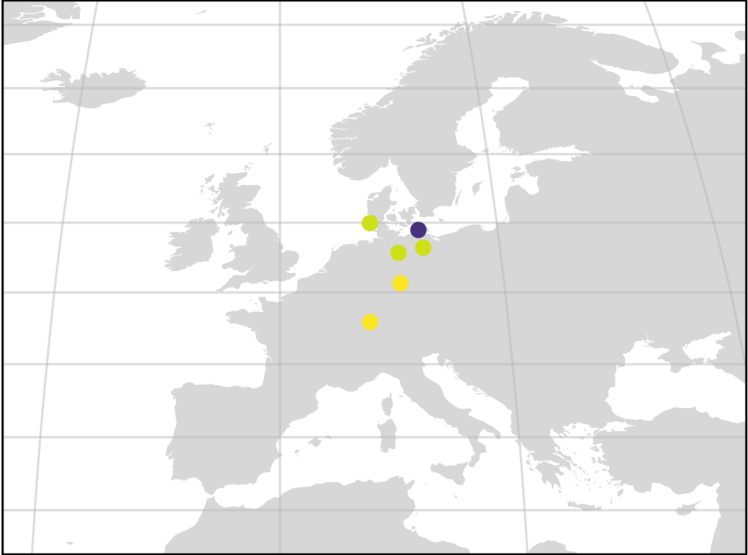
observations
LOCAL·All (62 stations)



observations
LOCAL·France (1 station)



observations
LOCAL·Germany (6 stations)

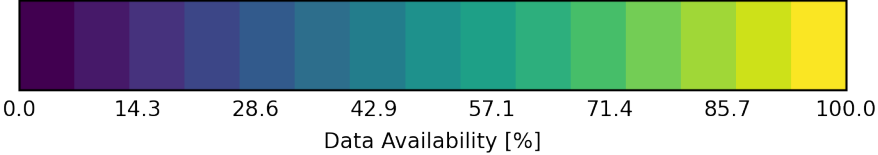


observations
LOCAL·Spain (13 stations)

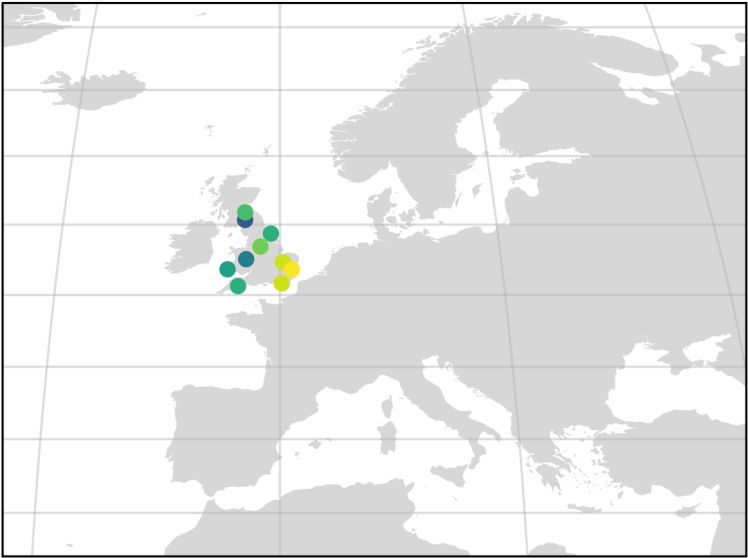


Map Data Availability (Summary)

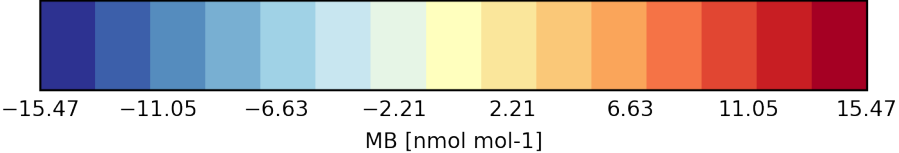
EBAS|sconco3



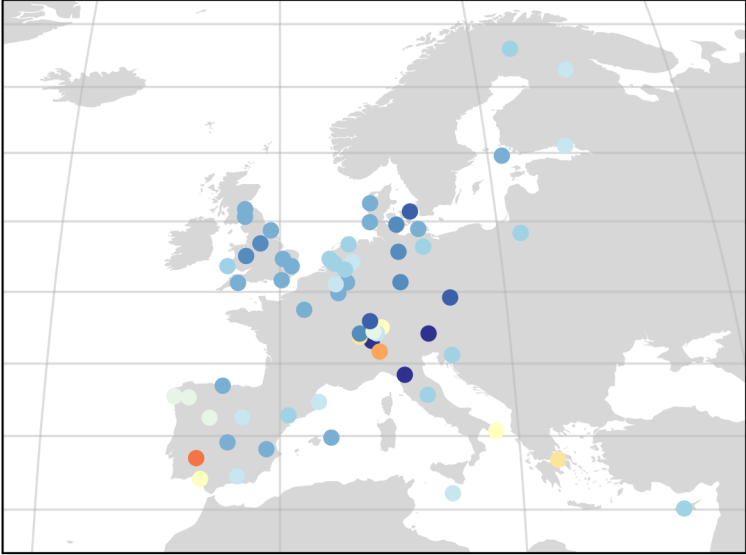
observations
LOCAL·UK (10 stations)



Map MB (Summary)
EBAS|sconco3



MONARCH
LOCAL·All (62 stations)



MONARCH
LOCAL·France (1 station)



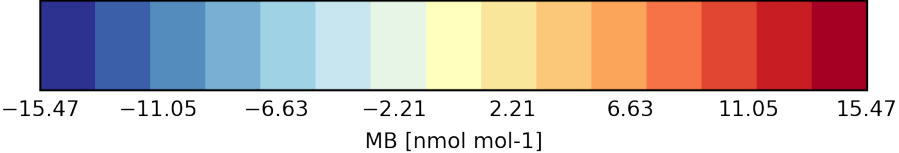
MONARCH
LOCAL·Germany (6 stations)



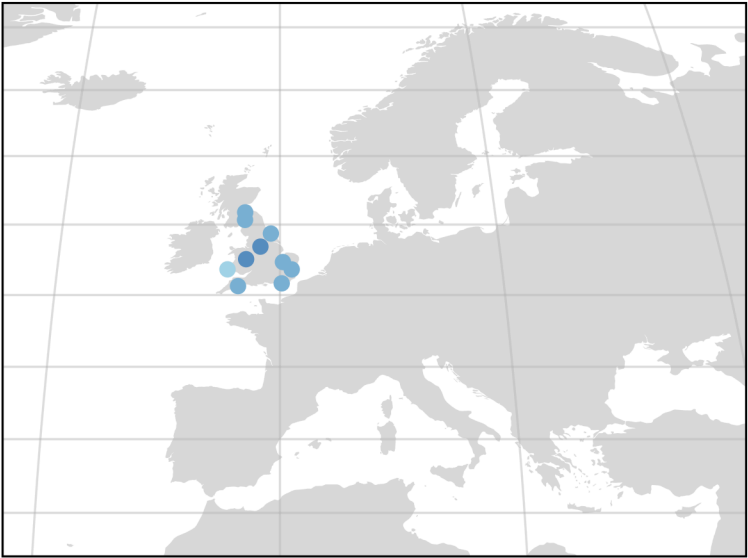
MONARCH
LOCAL·Spain (13 stations)



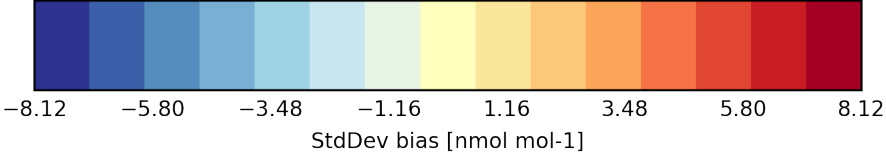
Map MB (Summary)
EBAS|sconco3



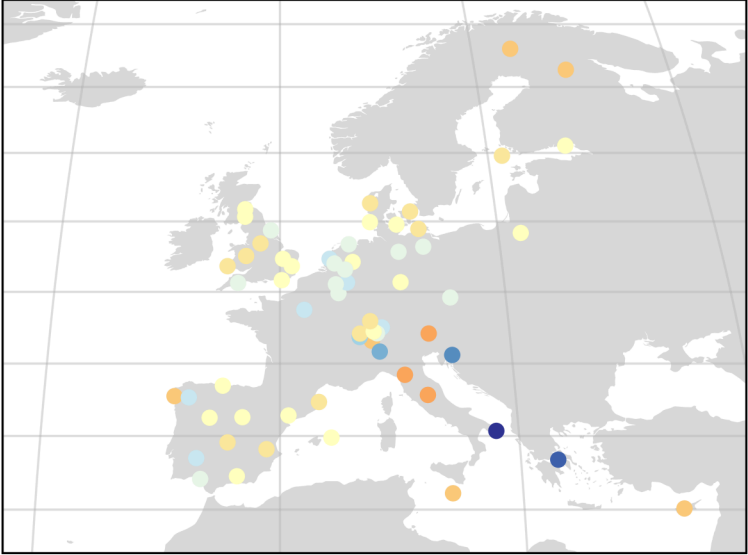
MONARCH
LOCAL·UK (10 stations)



Map StdDev bias (Summary)
EBAS|sconco3



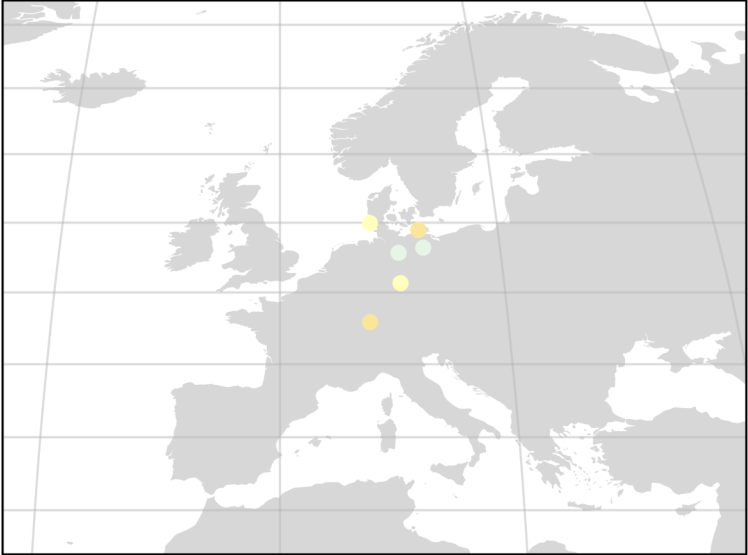
MONARCH
LOCAL·All (62 stations)



MONARCH
LOCAL·France (1 station)



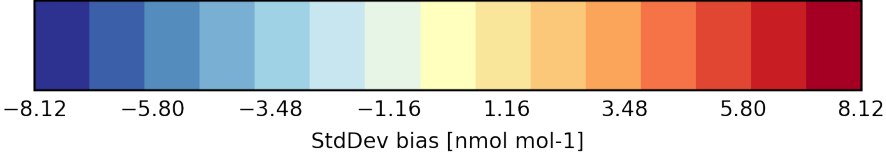
MONARCH
LOCAL·Germany (6 stations)



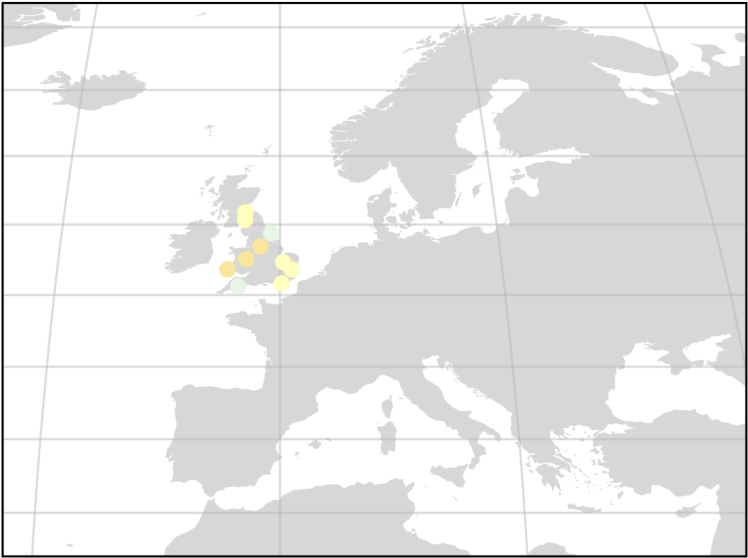
MONARCH
LOCAL·Spain (13 stations)



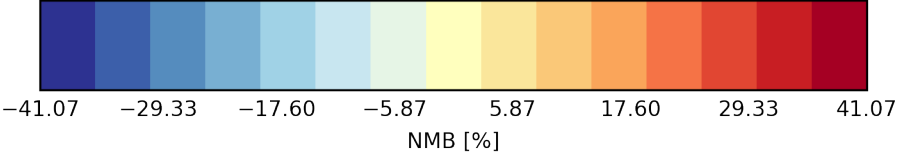
Map StdDev bias (Summary)
EBAS|sconco3



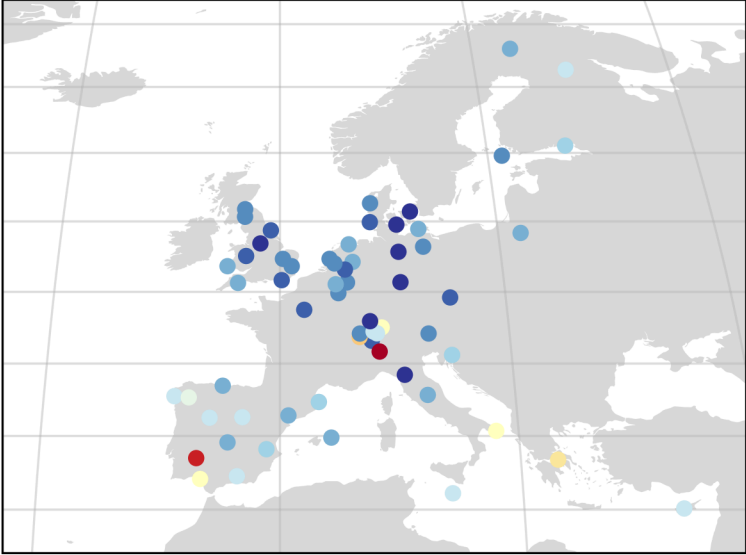
MONARCH
LOCAL·UK (10 stations)



Map NMB (Summary)
EBAS|sconco3



MONARCH
LOCAL·All (62 stations)



MONARCH
LOCAL·France (1 station)



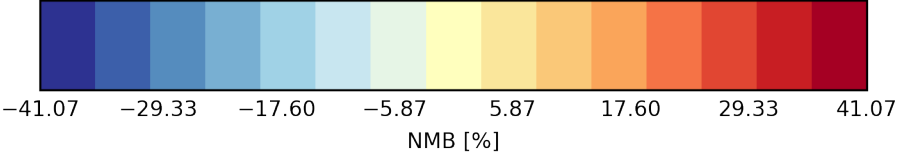
MONARCH
LOCAL·Germany (6 stations)



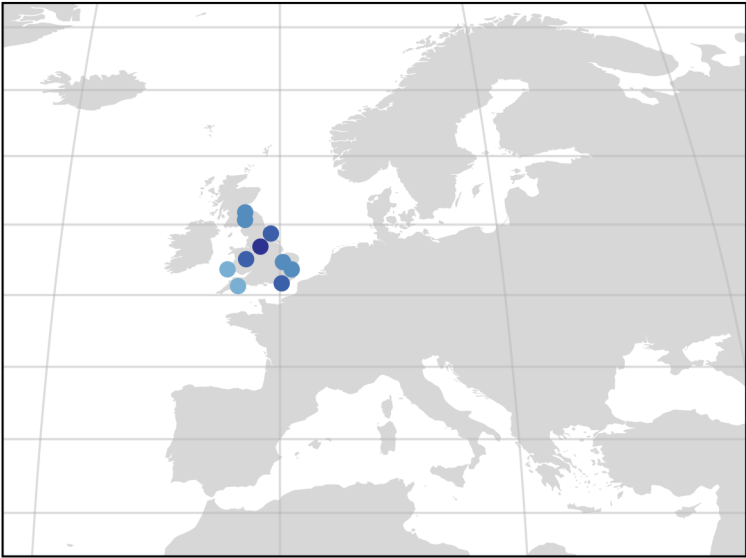
MONARCH
LOCAL·Spain (13 stations)



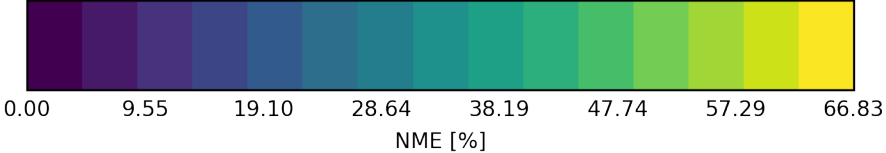
Map NMB (Summary)
EBAS|sconco3



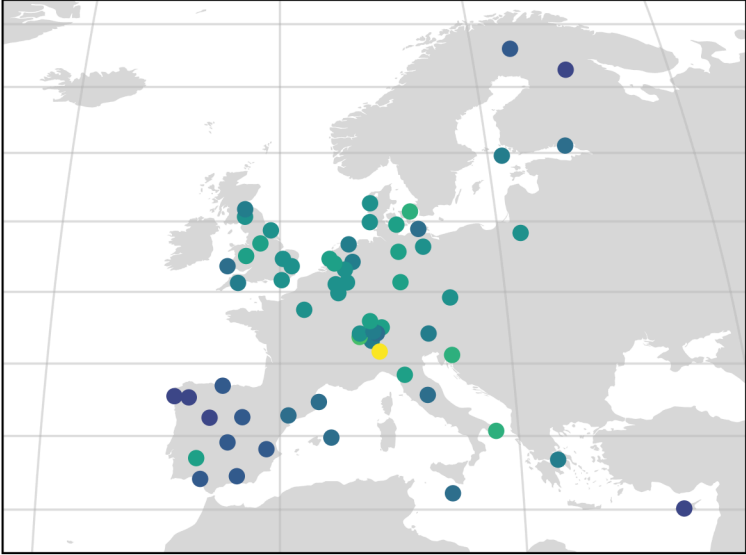
MONARCH
LOCAL-UK (10 stations)



Map NME (Summary)
EBAS|sconco3



MONARCH
LOCAL·All (62 stations)



MONARCH
LOCAL·France (1 station)



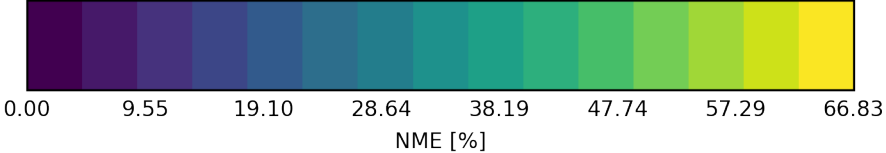
MONARCH
LOCAL·Germany (6 stations)



MONARCH
LOCAL·Spain (13 stations)



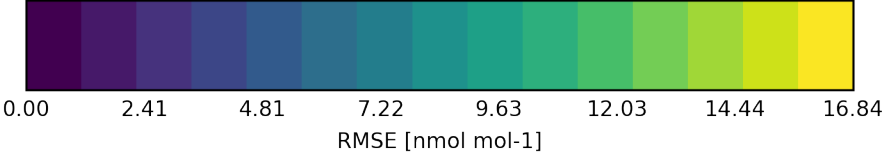
Map NME (Summary)
EBAS|sconco3



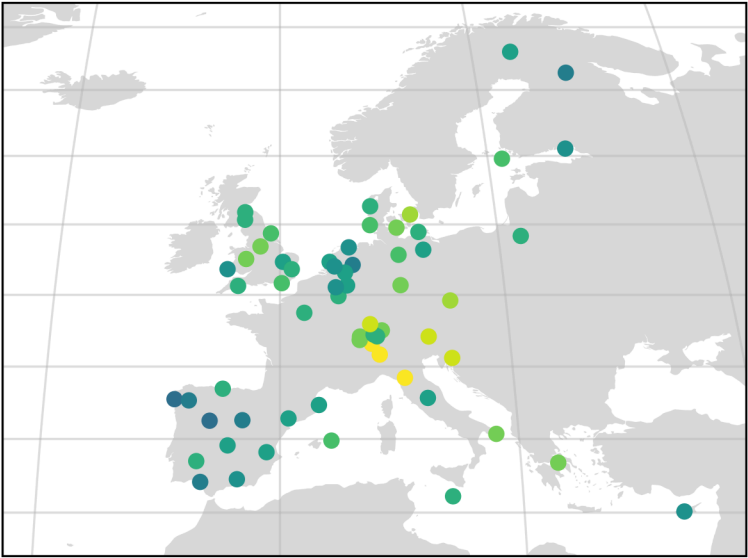
MONARCH
LOCAL·UK (10 stations)



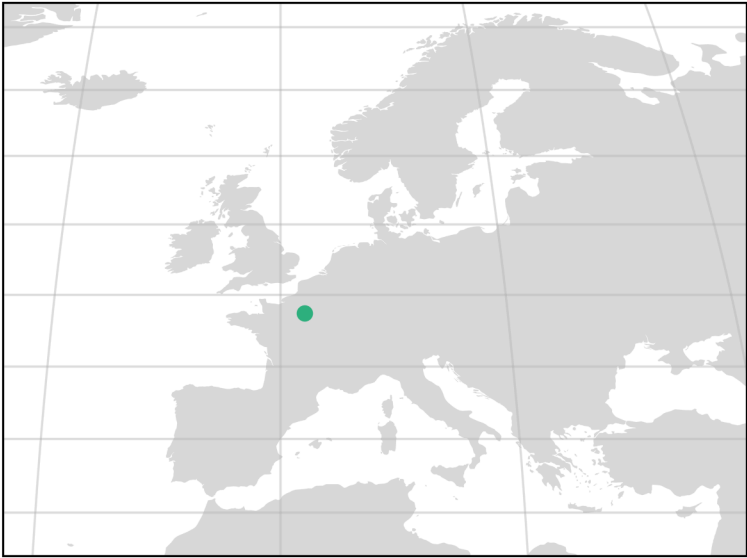
Map RMSE (Summary)
EBAS|sconco3



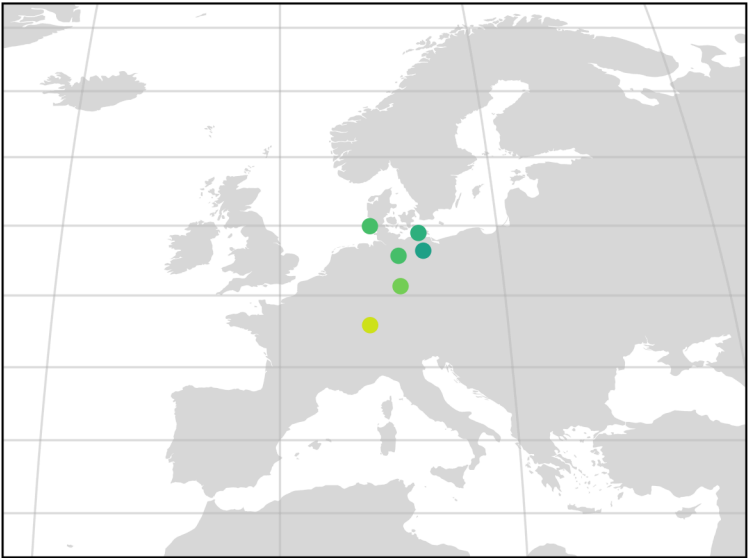
MONARCH
LOCAL·All (62 stations)



MONARCH
LOCAL·France (1 station)



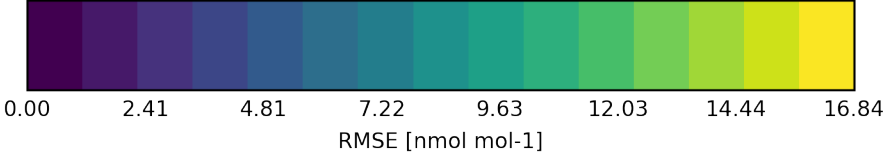
MONARCH
LOCAL·Germany (6 stations)



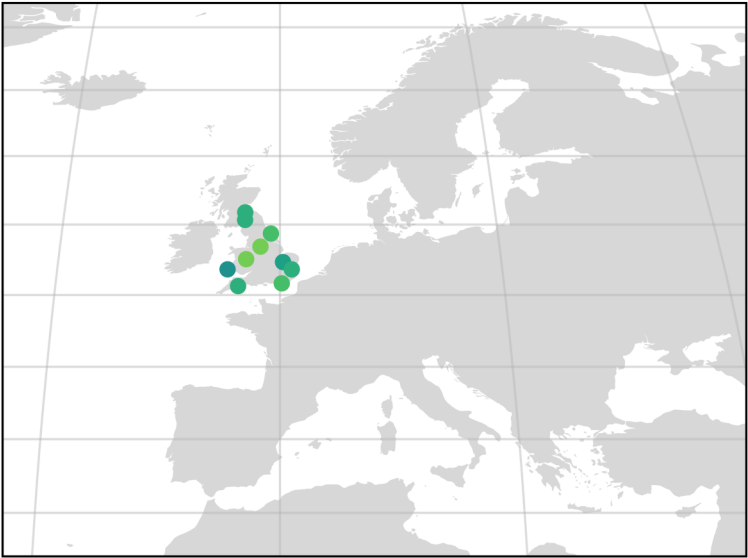
MONARCH
LOCAL·Spain (13 stations)



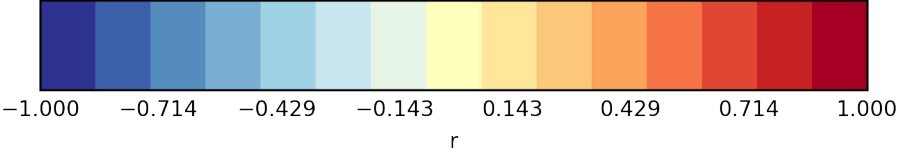
Map RMSE (Summary)
EBAS|sconco3



MONARCH
LOCAL·UK (10 stations)



Map r (Summary)
EBAS|sconco3



MONARCH
LOCAL·All (62 stations)



MONARCH
LOCAL·France (1 station)



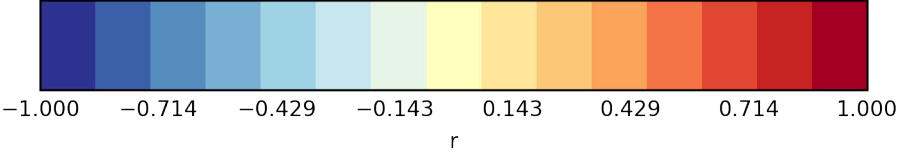
MONARCH
LOCAL·Germany (6 stations)



MONARCH
LOCAL·Spain (13 stations)

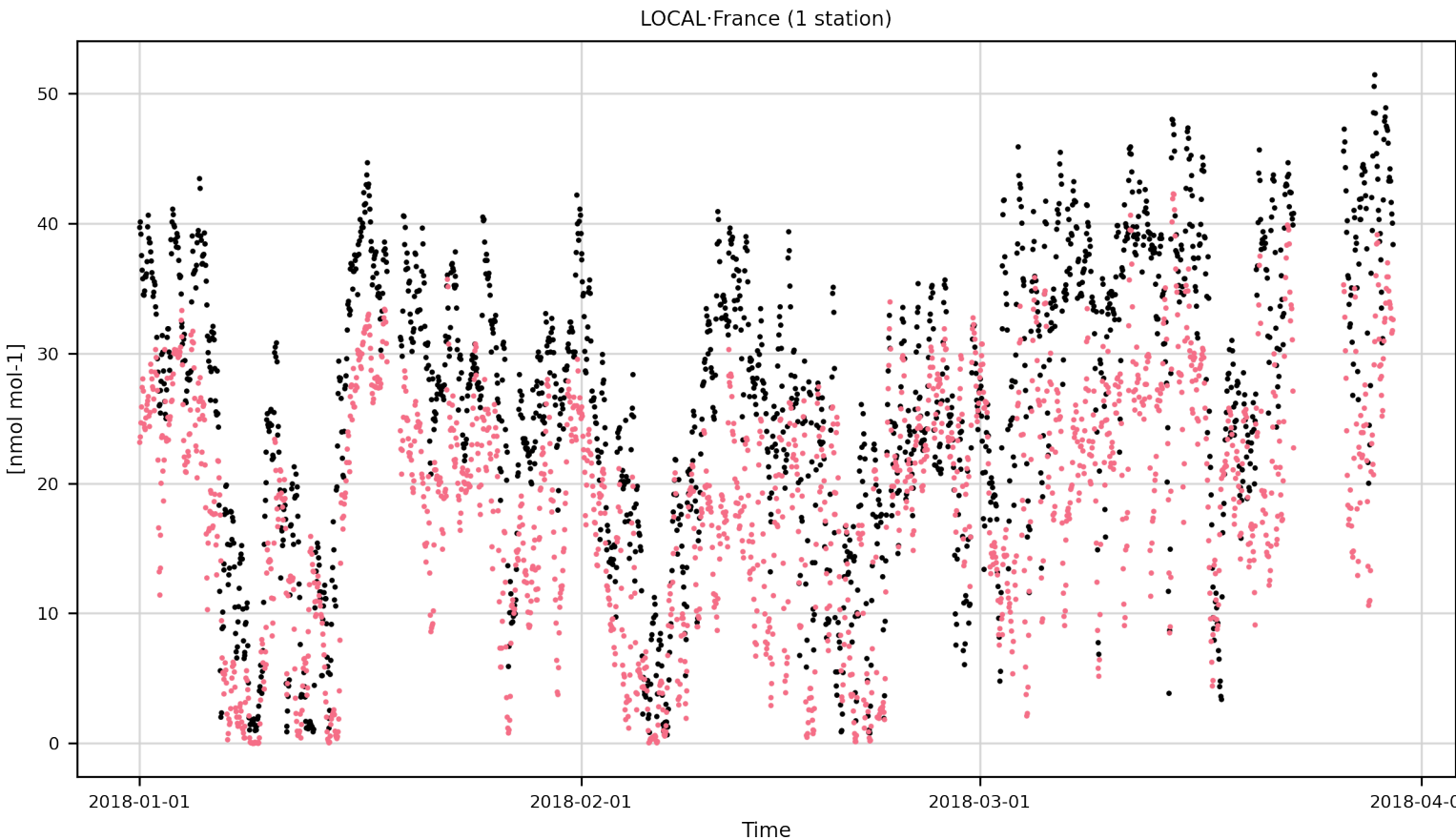
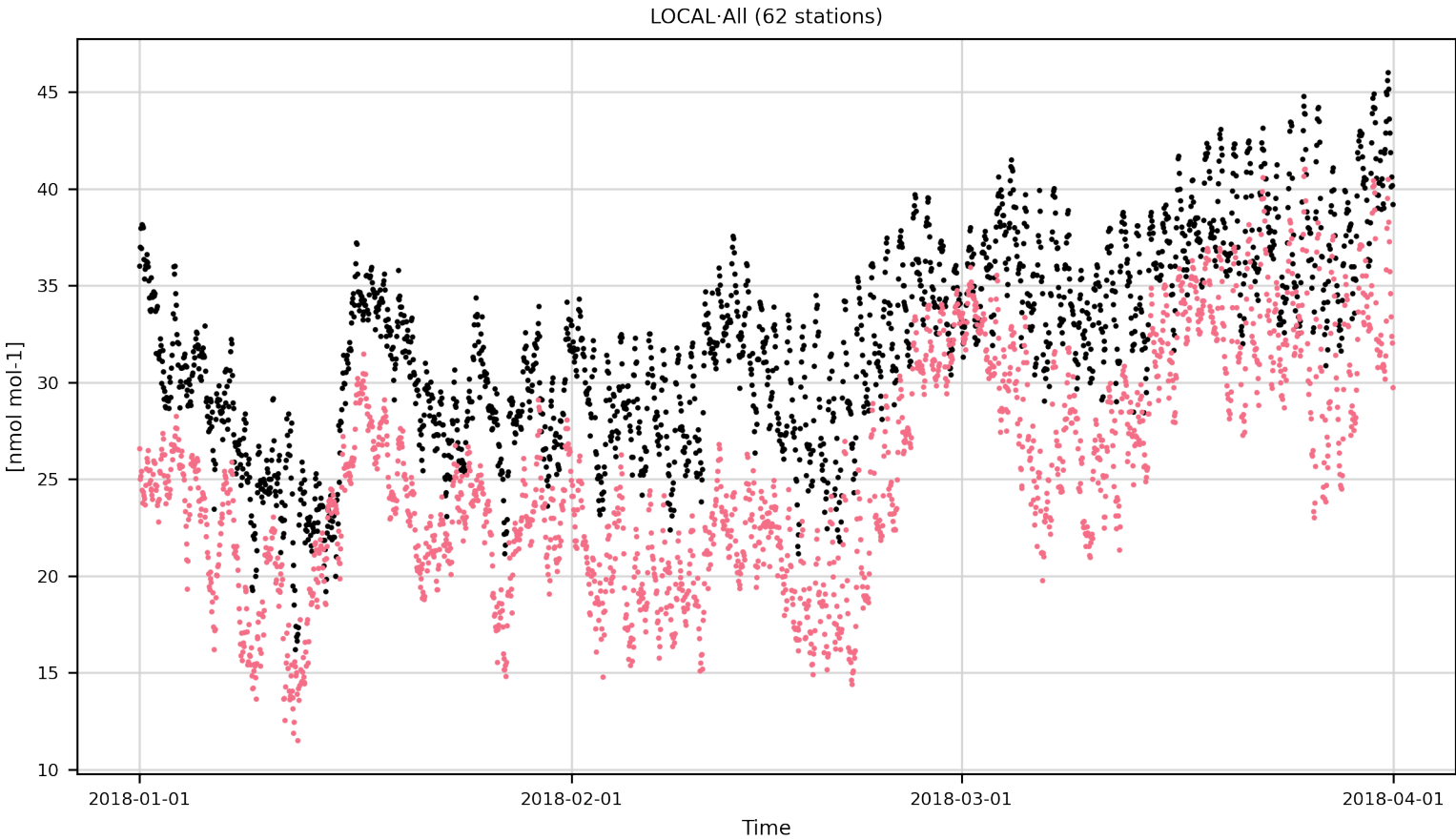


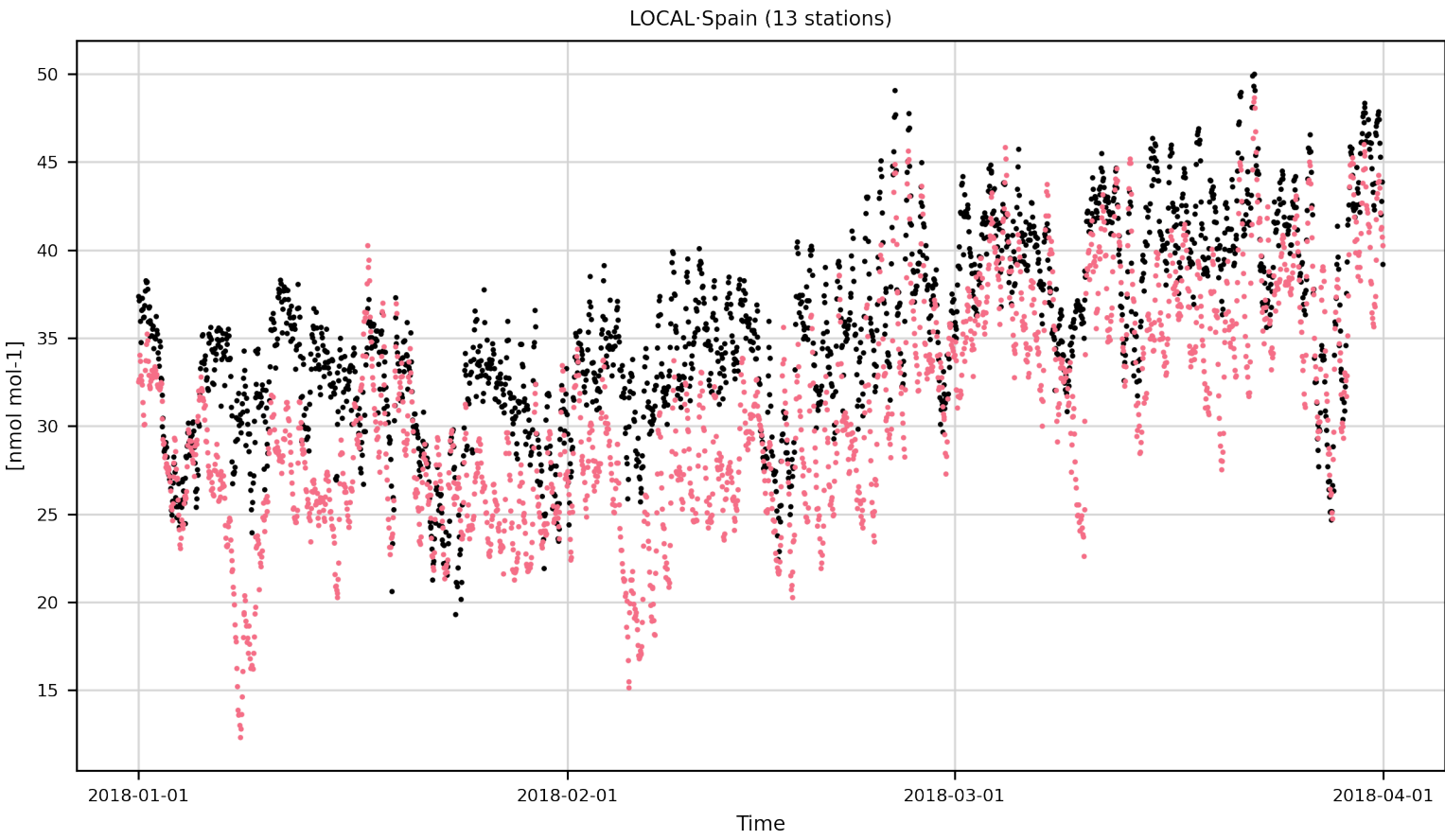
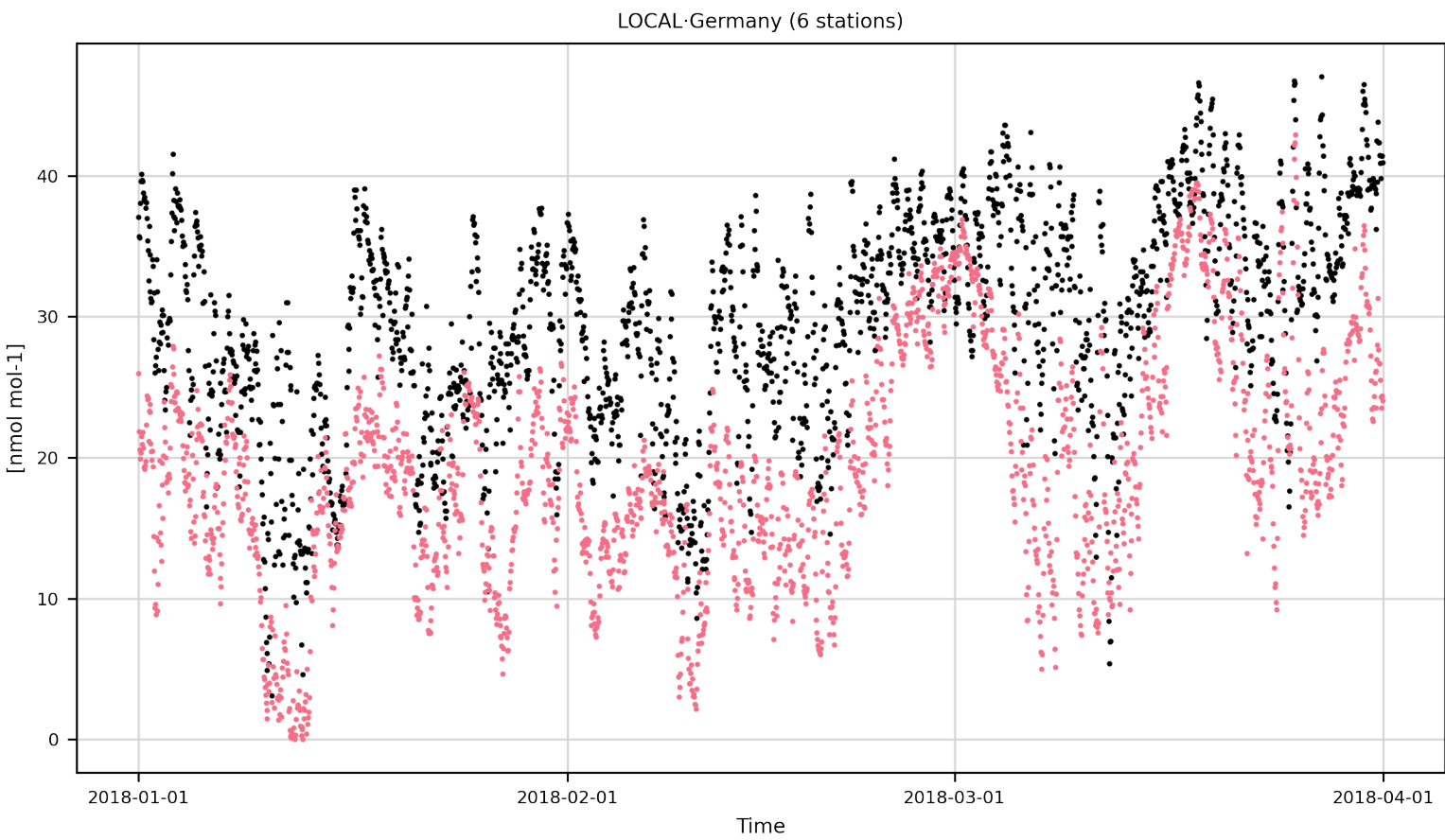
Map r (Summary)
EBAS|sconco3

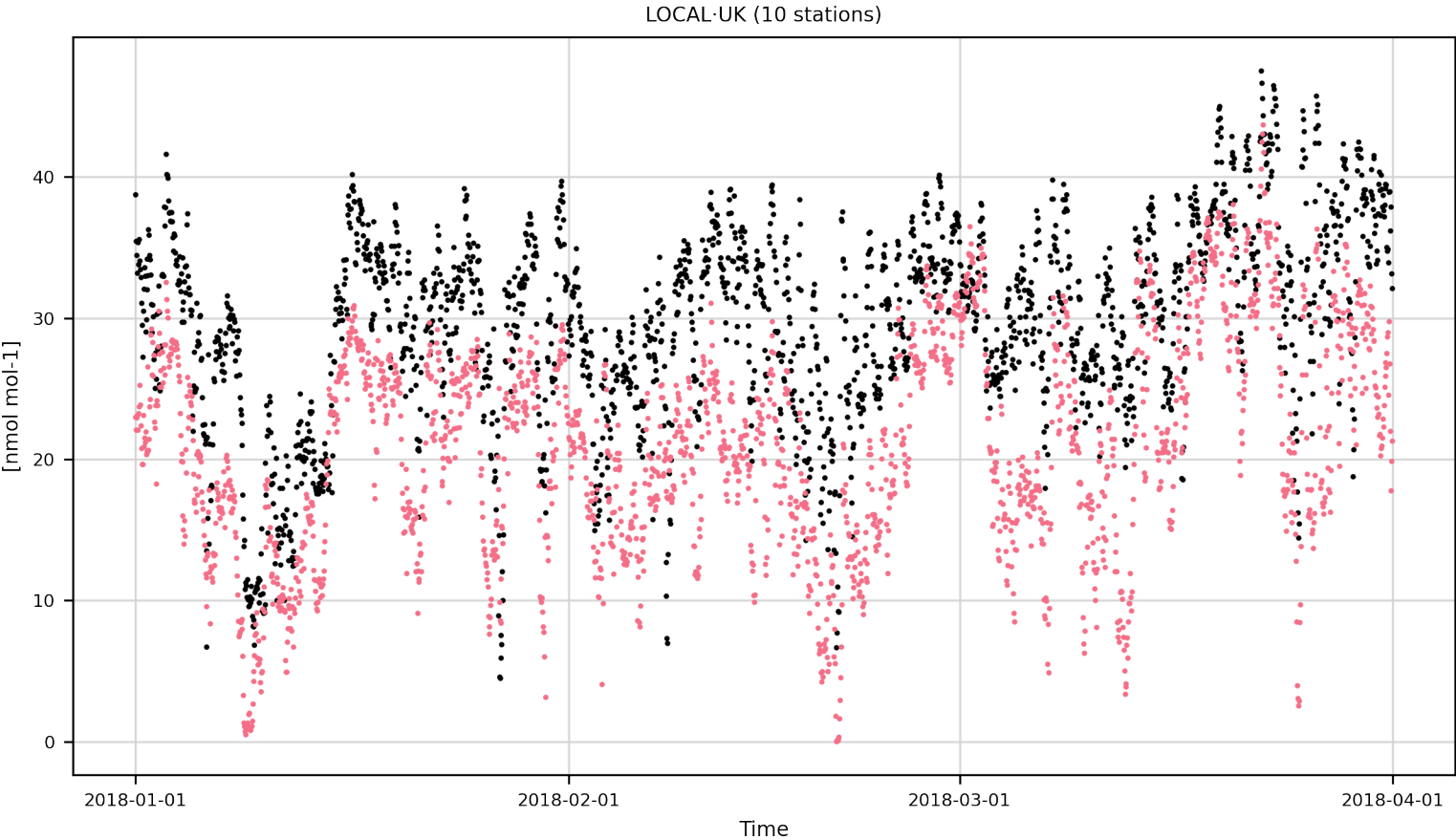


MONARCH
LOCAL·UK (10 stations)





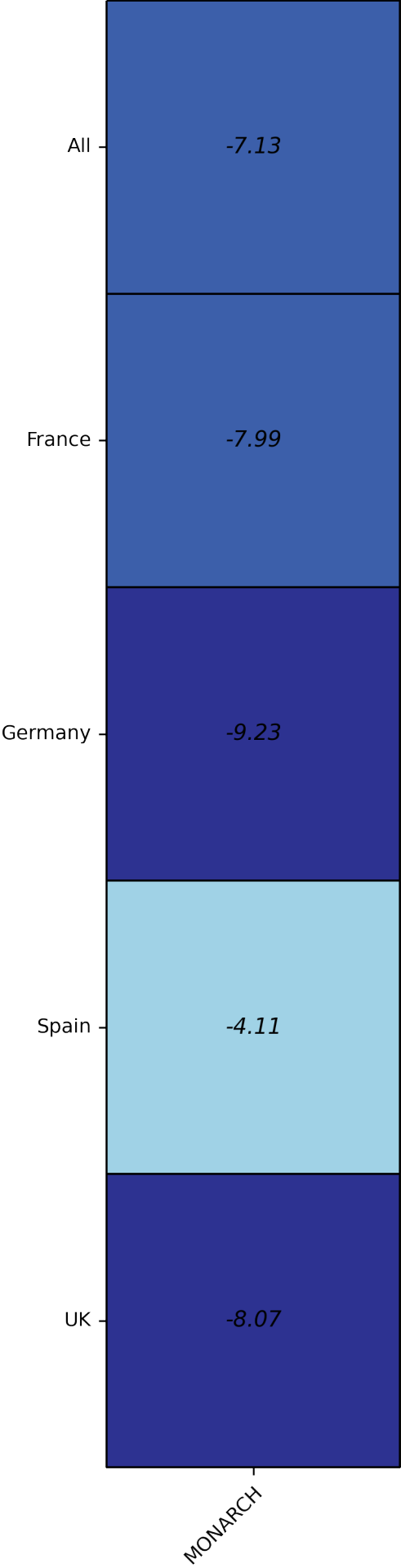
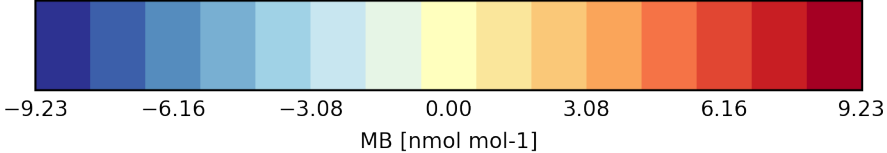




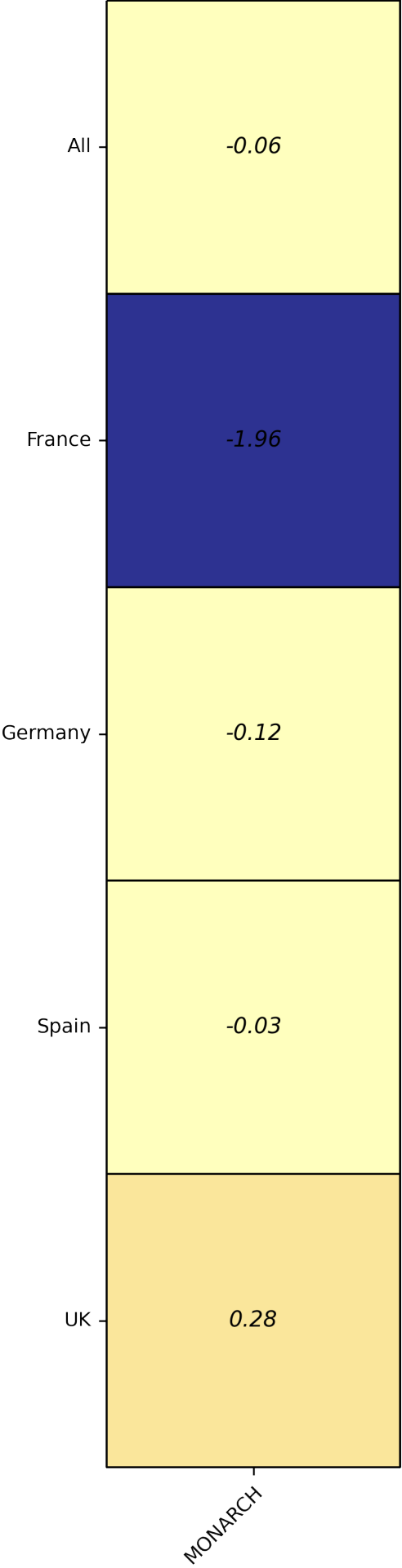
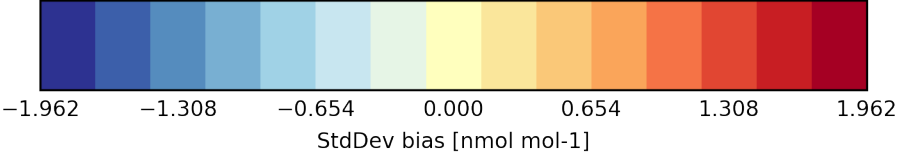
		Mean	StdDev	p5	Median	p95
All	observations	30.15	8.78	13.67	31.07	42.90
	MONARCH	25.25	8.84	9.70	25.48	38.21
France	observations	26.37	11.18	4.67	27.52	41.94
	MONARCH	18.37	9.22	1.79	19.54	31.94
Germany	observations	29.61	9.02	12.20	30.65	42.80
	MONARCH	19.55	8.86	4.86	19.65	35.43
Spain	observations	34.07	7.06	21.57	34.40	46.15
	MONARCH	31.00	7.66	19.42	30.79	43.19
UK	observations	29.19	8.52	13.80	30.34	41.01
	MONARCH	19.72	8.93	5.41	19.85	34.37

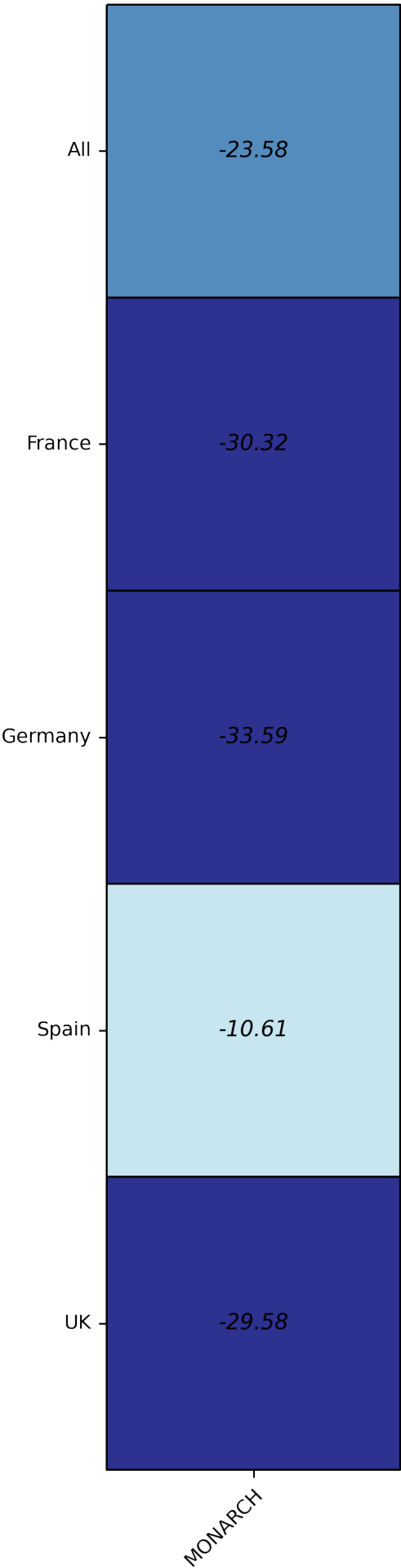
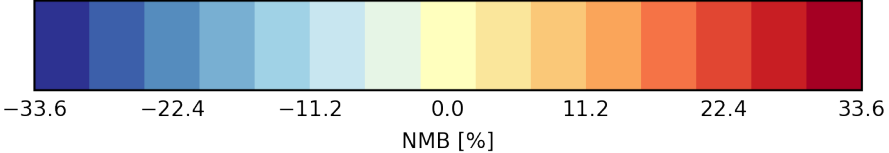
		Mean_bias	StdDev_bias	NMB	NME	RMSE	r
All	MONARCH	-7.13	-0.06	-23.58	31.50	10.54	0.66
France	MONARCH	-7.99	-1.96	-30.32	34.19	10.73	0.77
Germany	MONARCH	-9.23	-0.12	-33.59	36.01	11.80	0.72
Spain	MONARCH	-4.11	-0.03	-10.61	20.41	9.14	0.65
UK	MONARCH	-8.07	0.28	-29.58	32.45	10.63	0.70

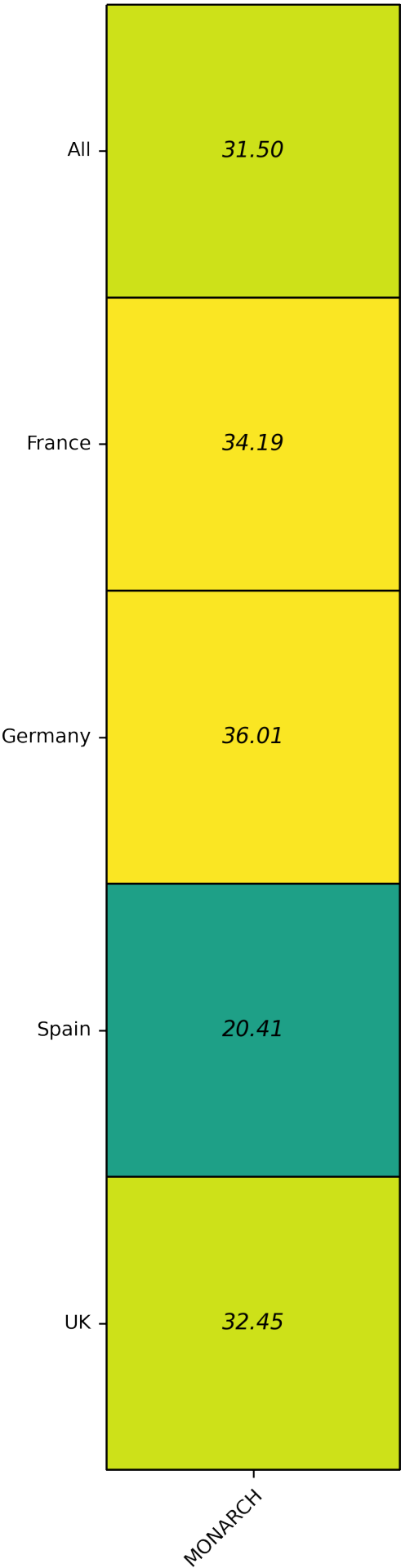
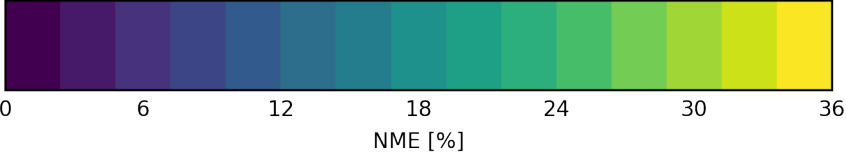
Heatmap MB (Summary)
EBAS|sconco3



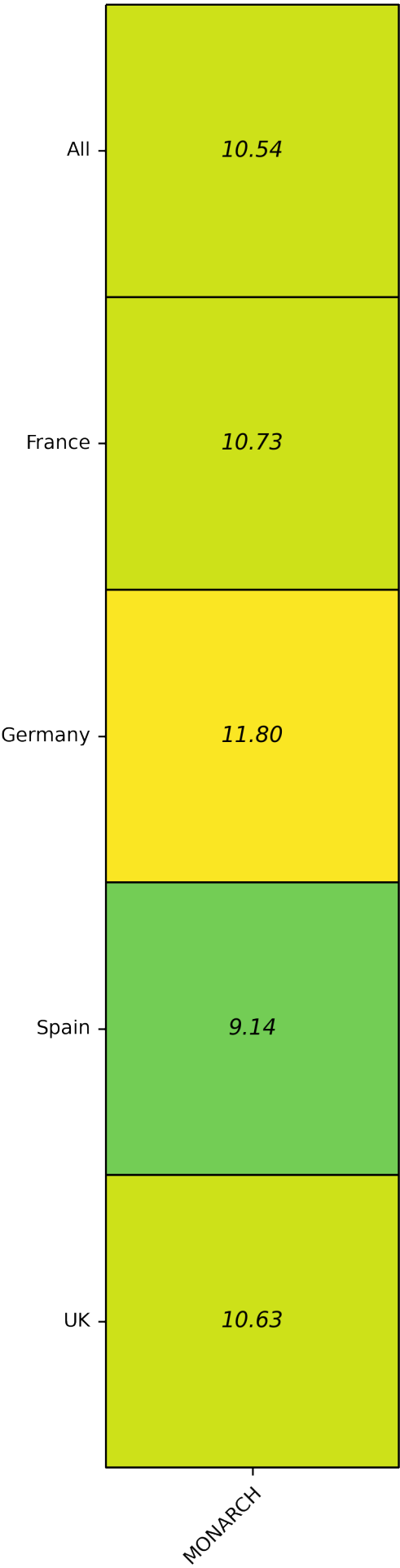
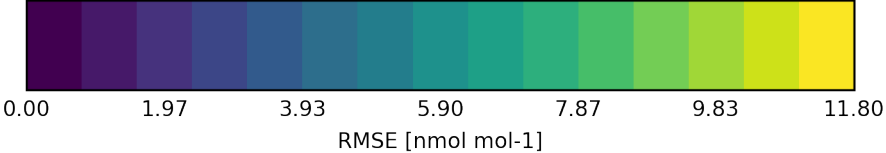
Heatmap StdDev bias (Summary)
EBAS|sconco3



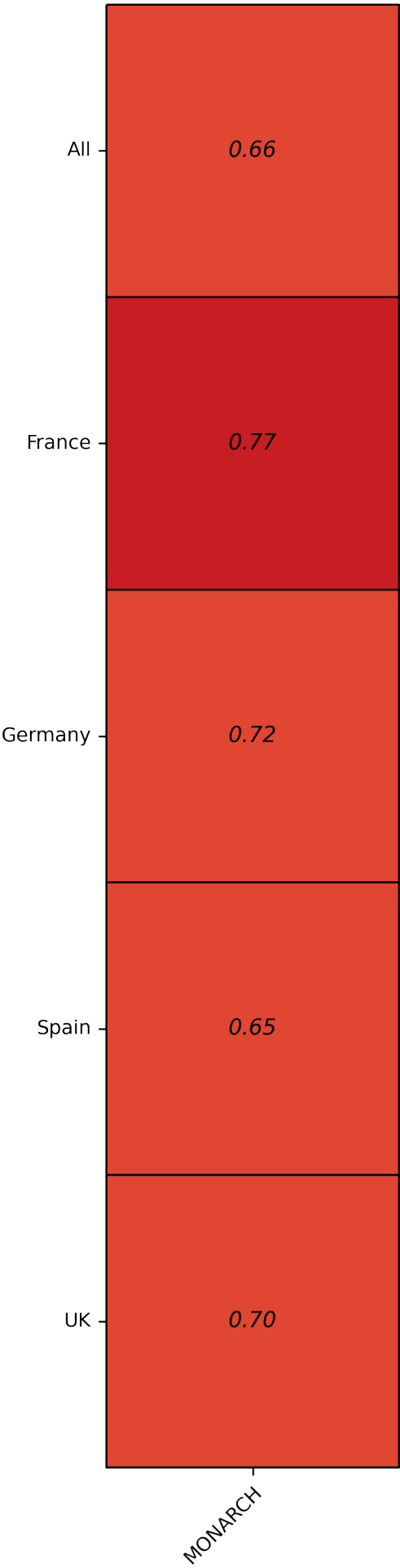
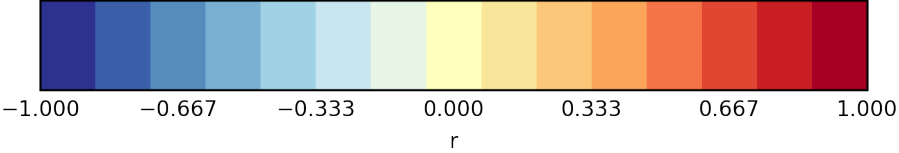


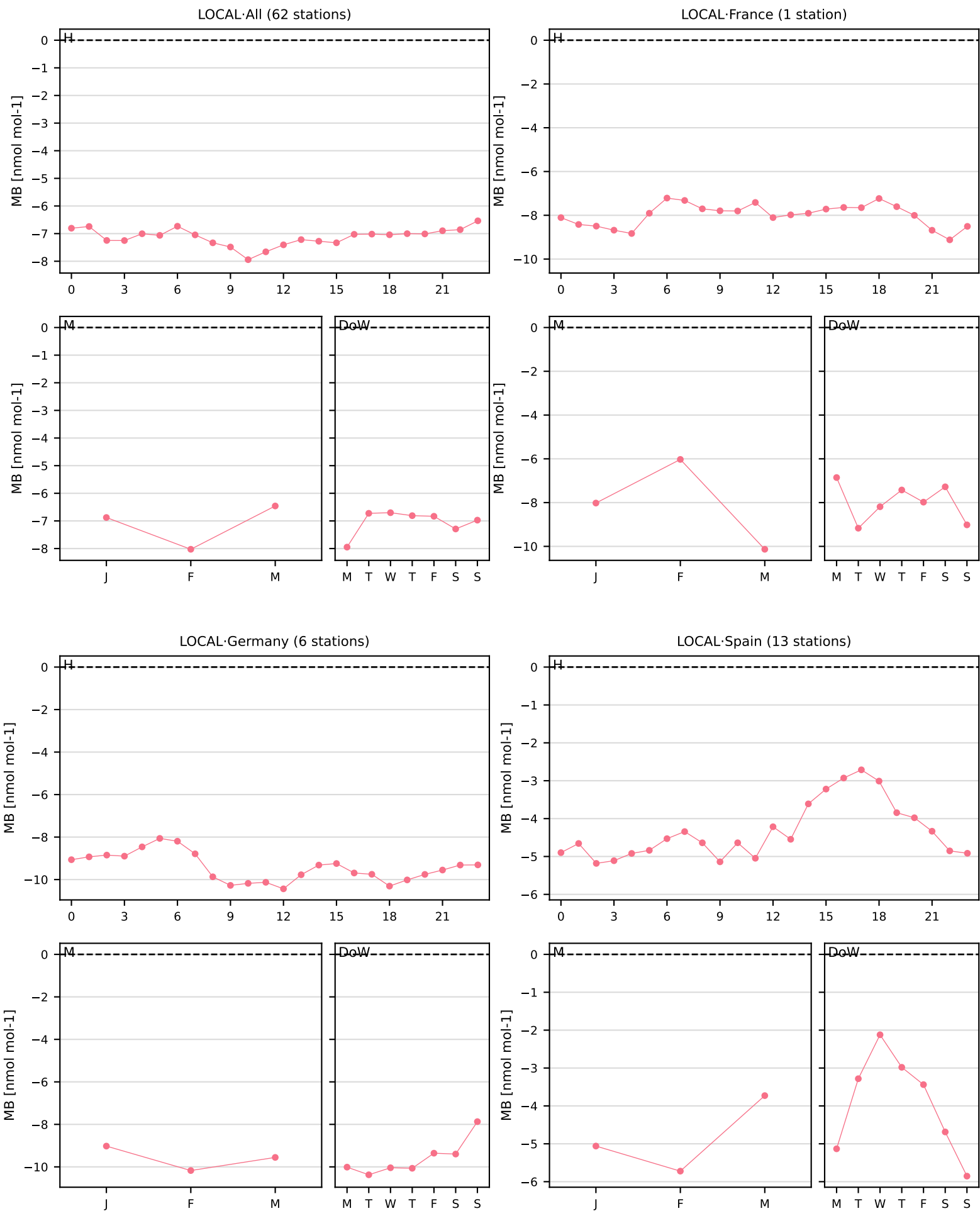


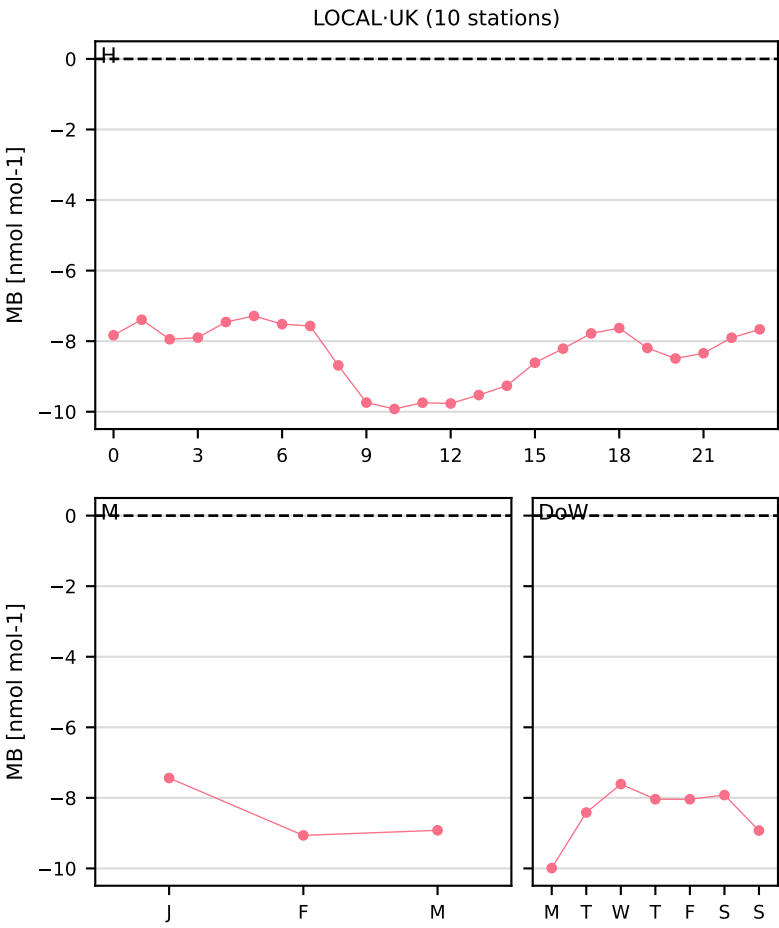
Heatmap RMSE (Summary)
EBAS|sconco3



Heatmap r (Summary)
EBAS|sconco3

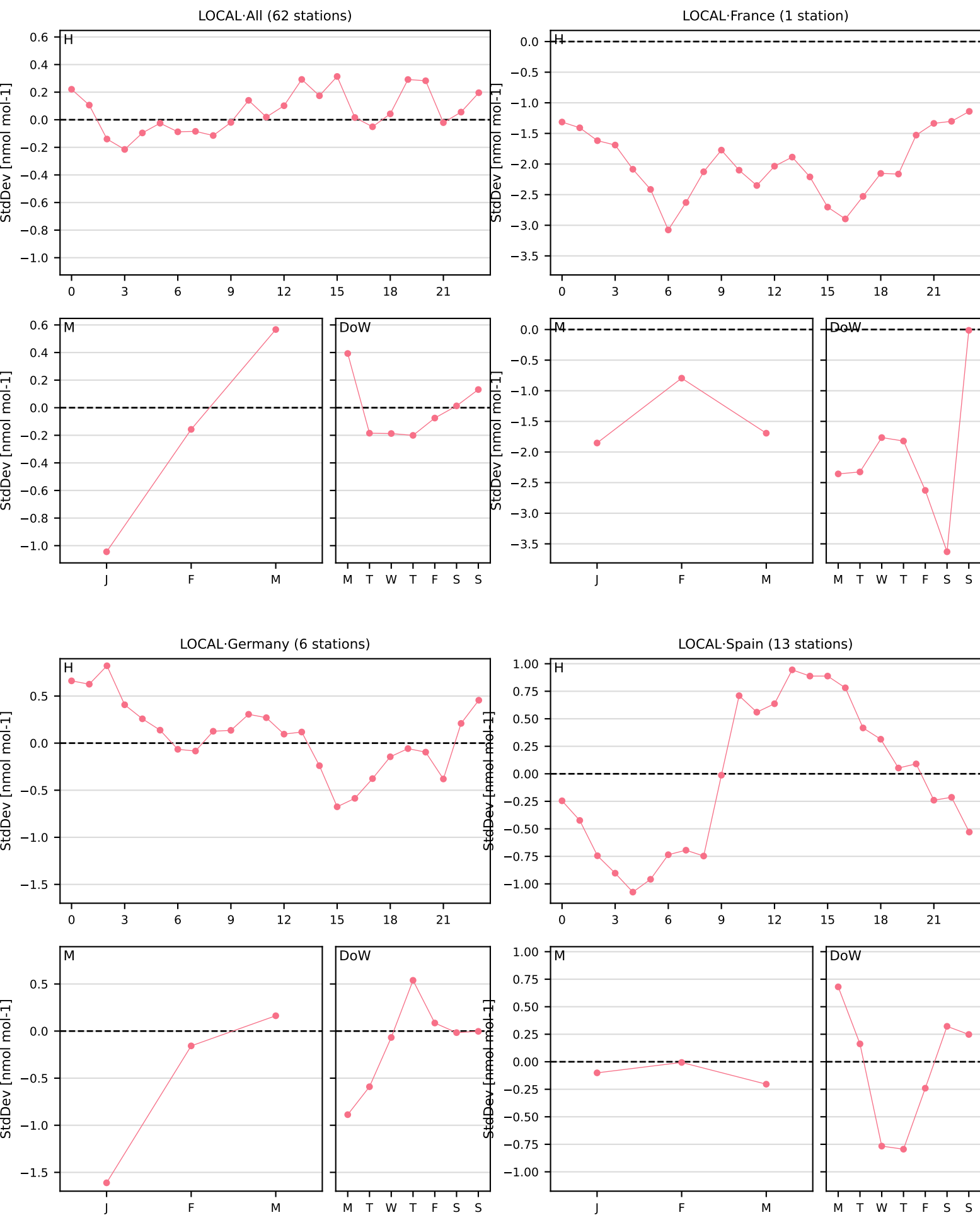


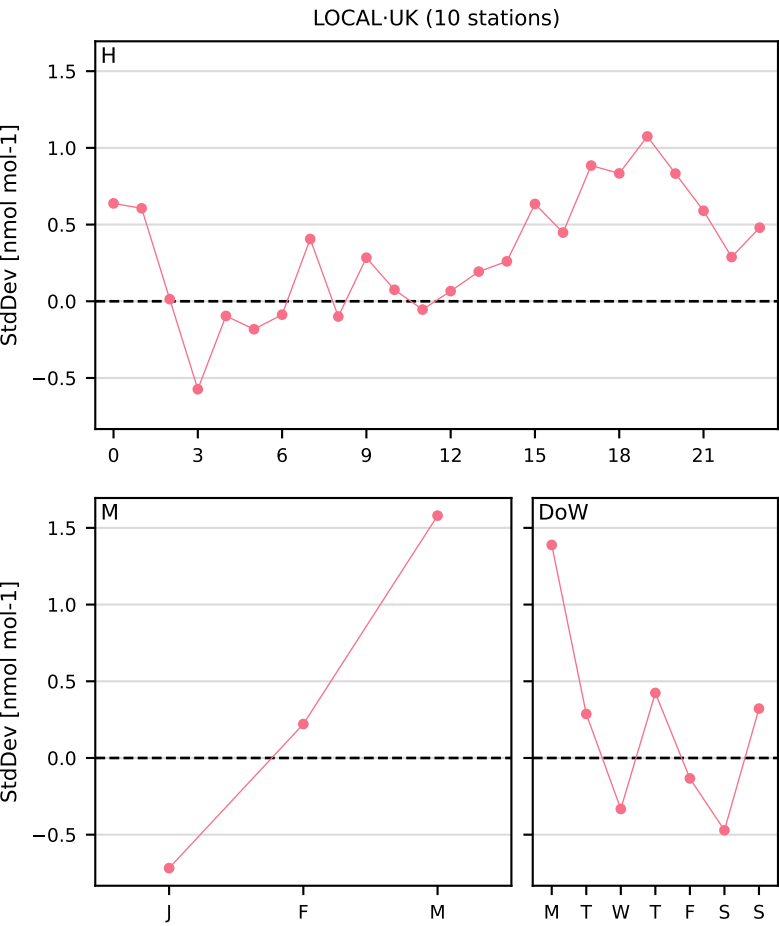




Periodic StdDev bias (Summary)
EBAS|sconco3

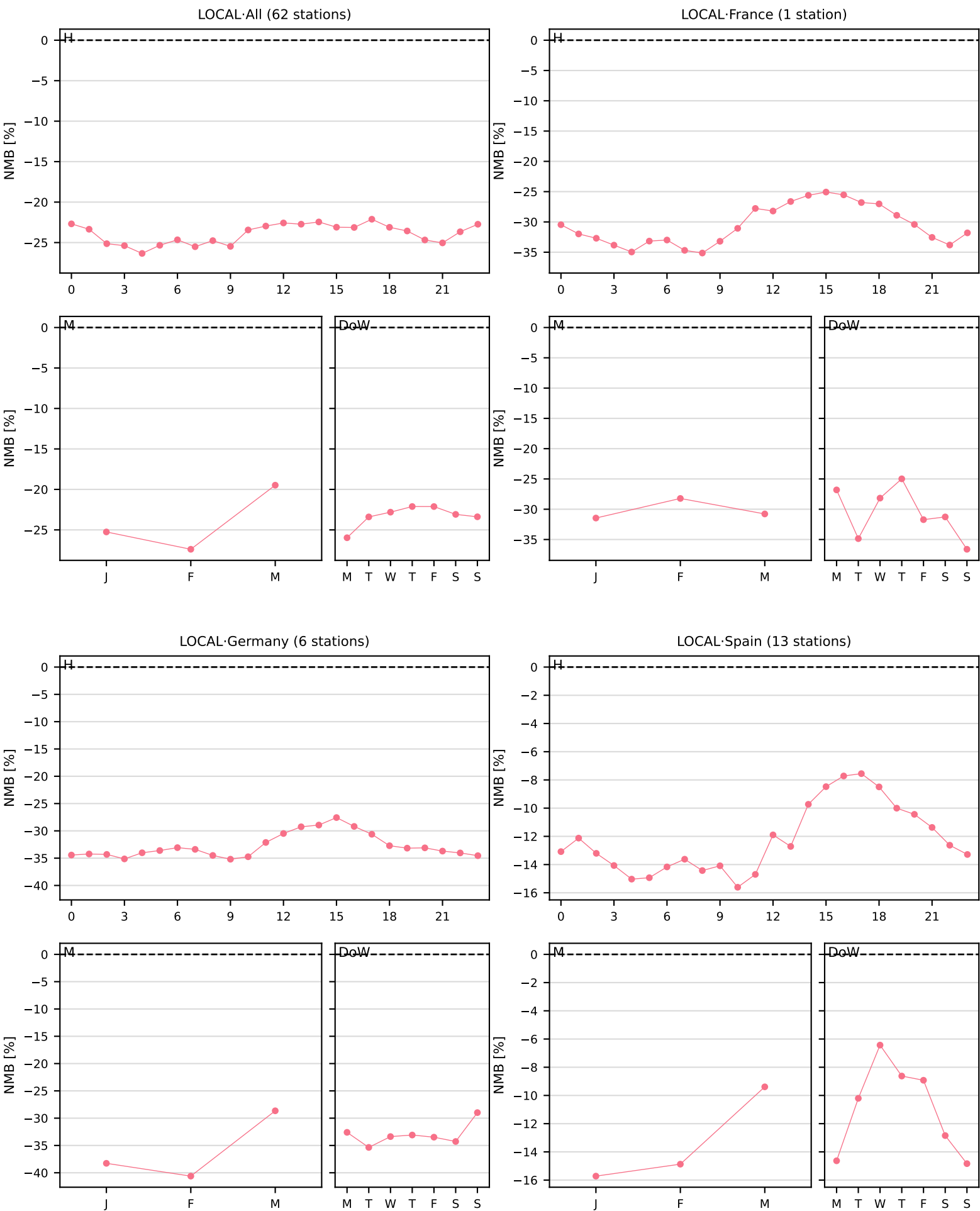
MONARCH

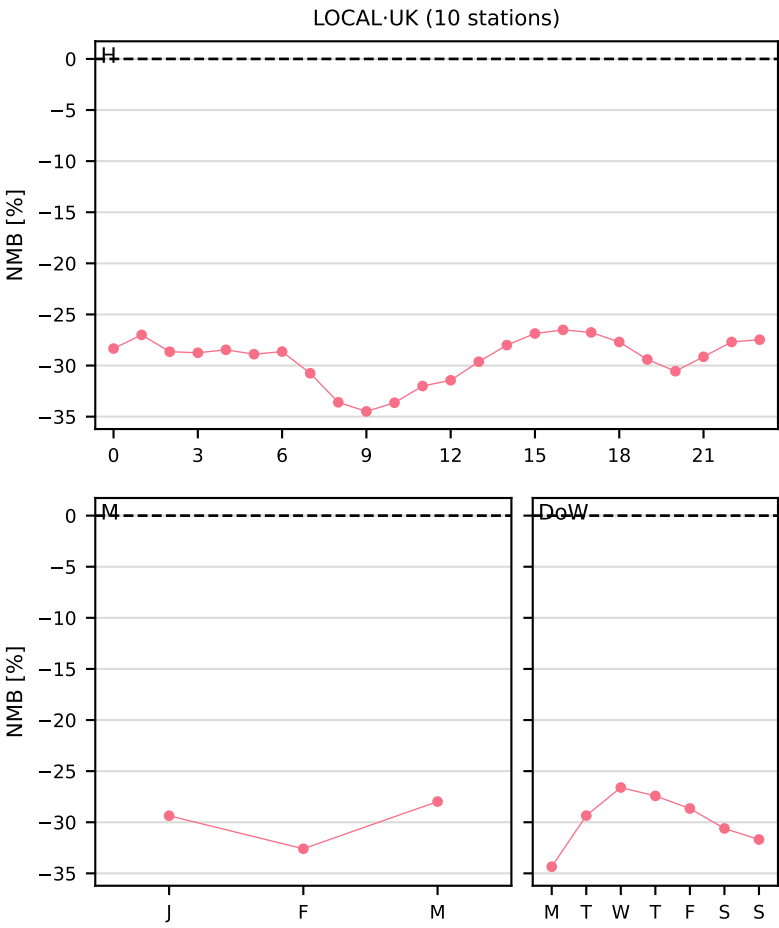




Periodic NMB (Summary)
EBAS|sconco3

MONARCH

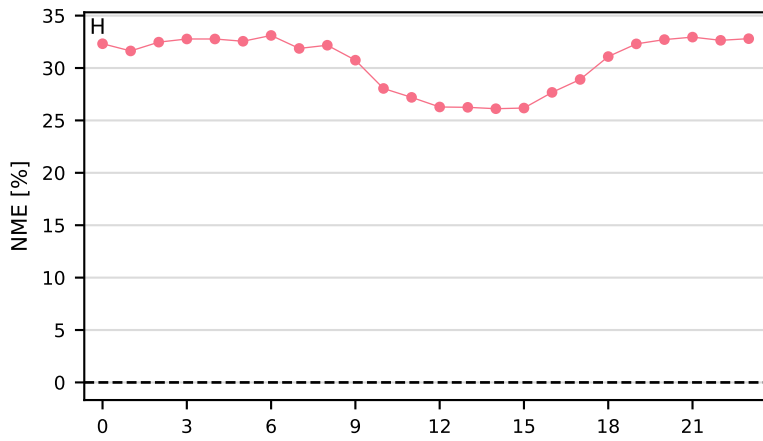




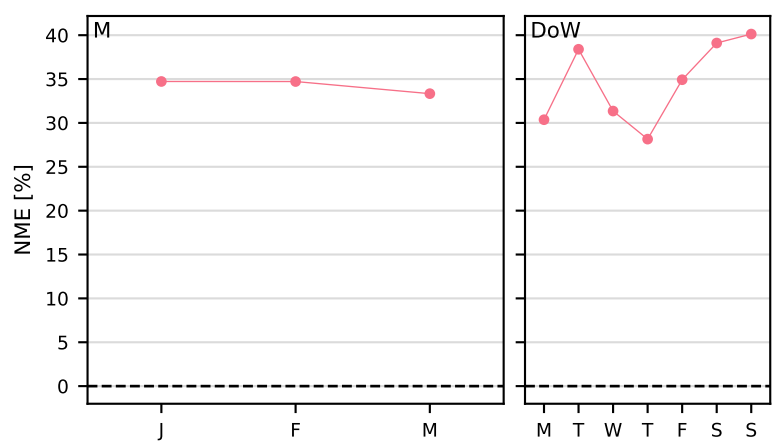
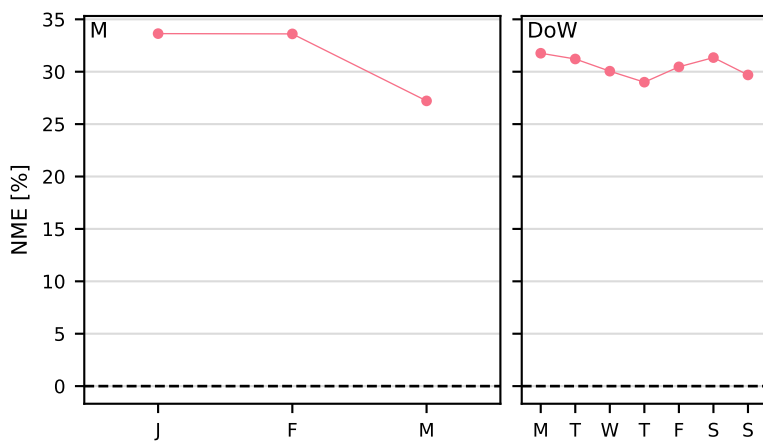
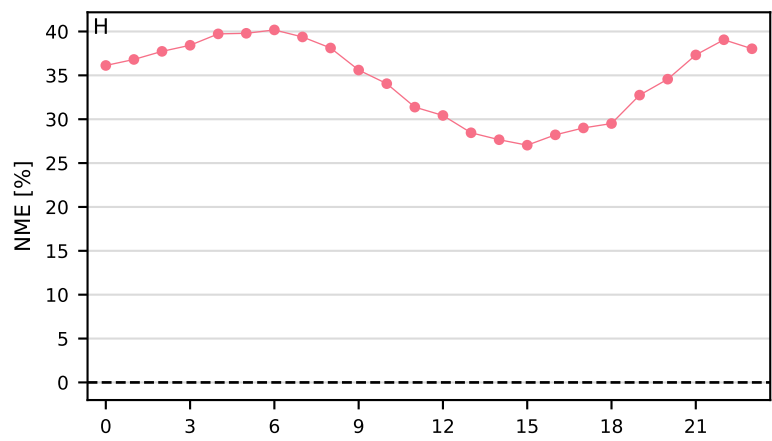
Periodic NME (Summary)
EBAS|sconco3

MONARCH

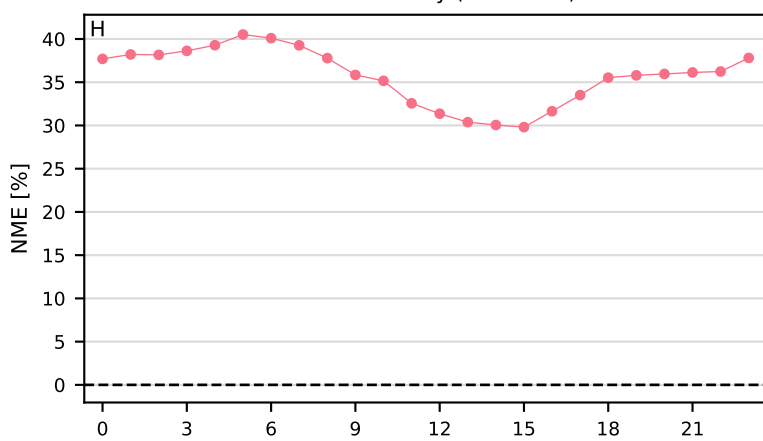
LOCAL·All (62 stations)



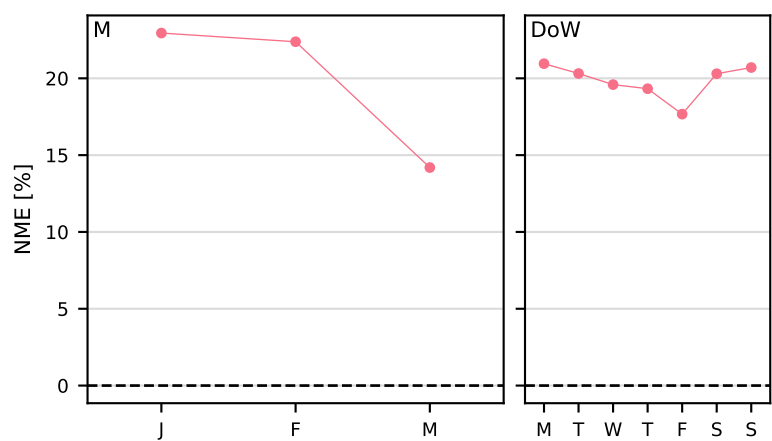
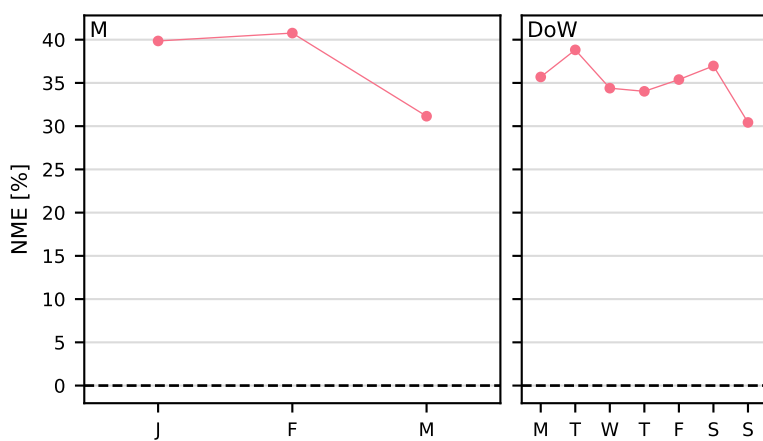
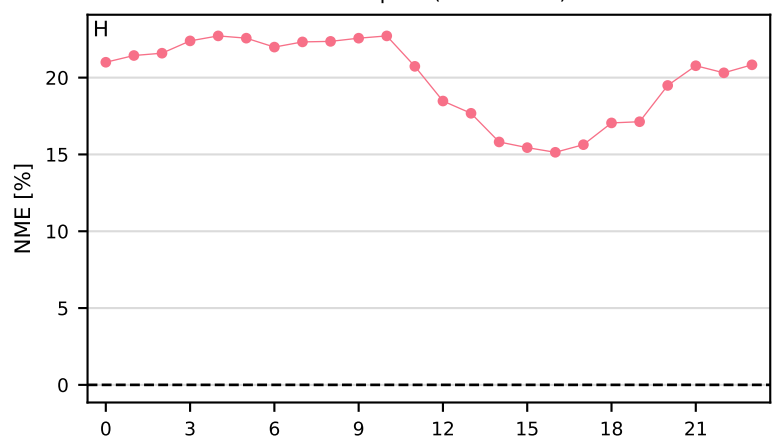
LOCAL·France (1 station)



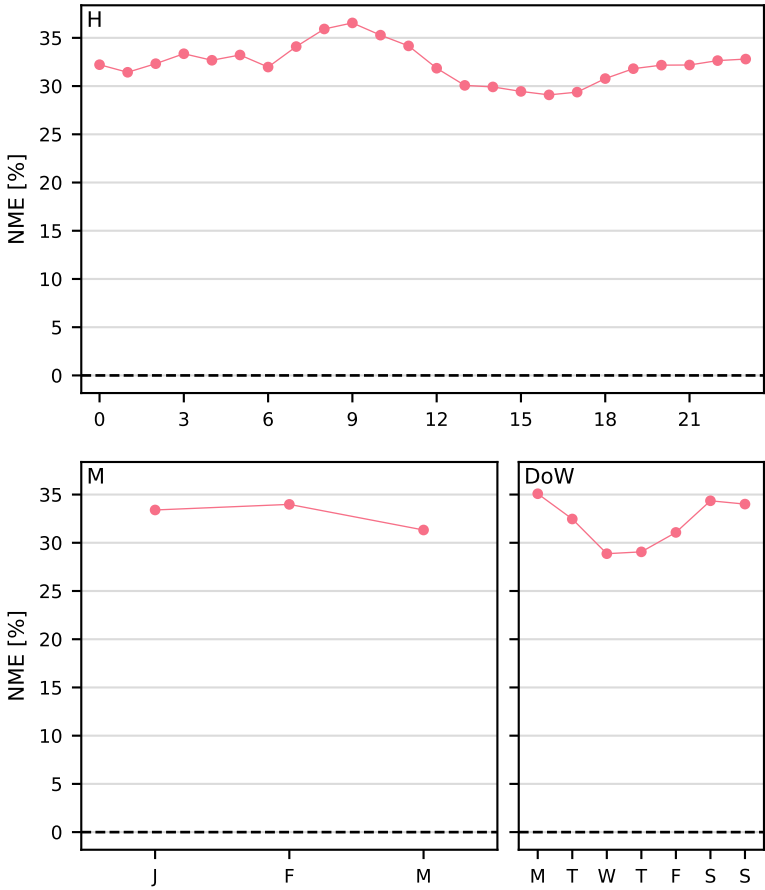
LOCAL·Germany (6 stations)



LOCAL·Spain (13 stations)



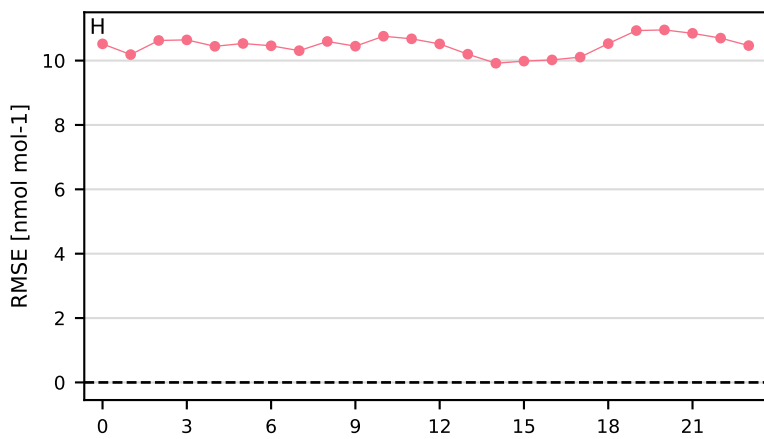
LOCAL·UK (10 stations)



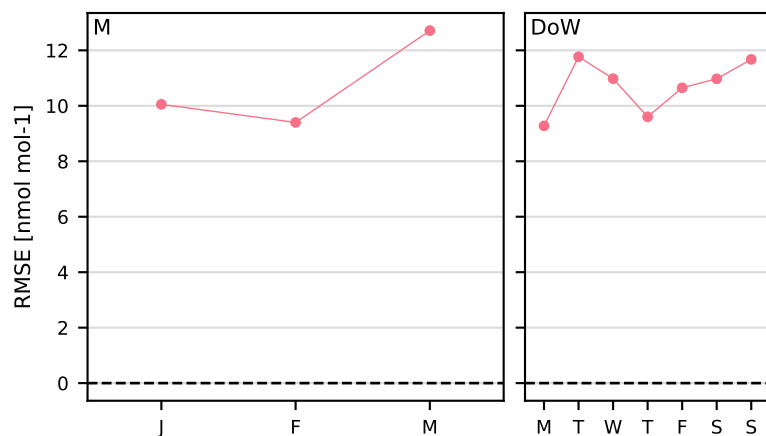
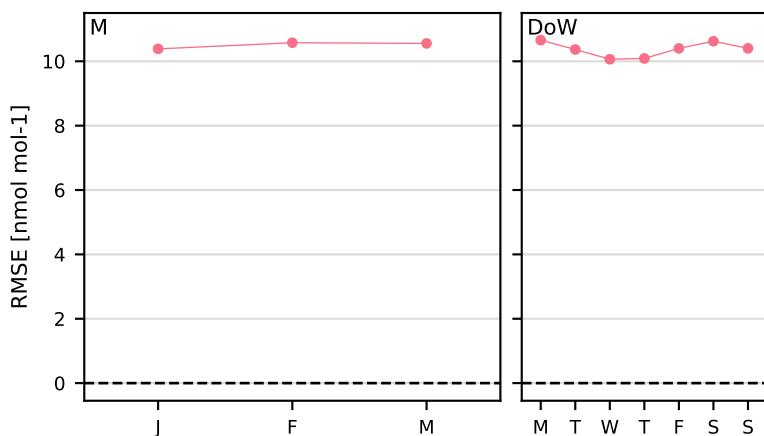
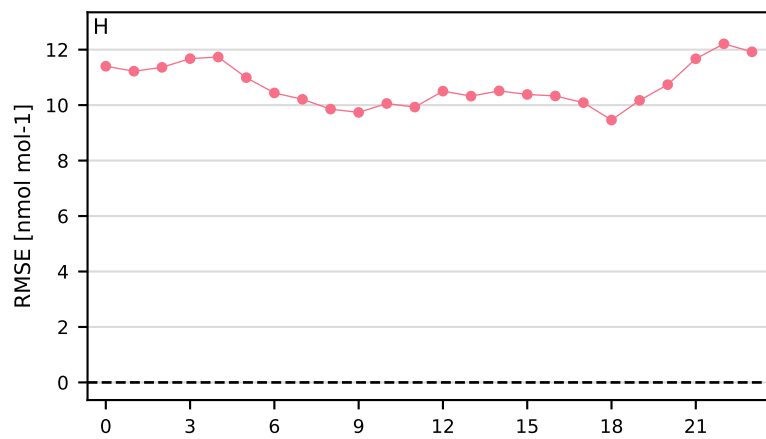
Periodic RMSE (Summary)
EBAS|sconco3

● MONARCH

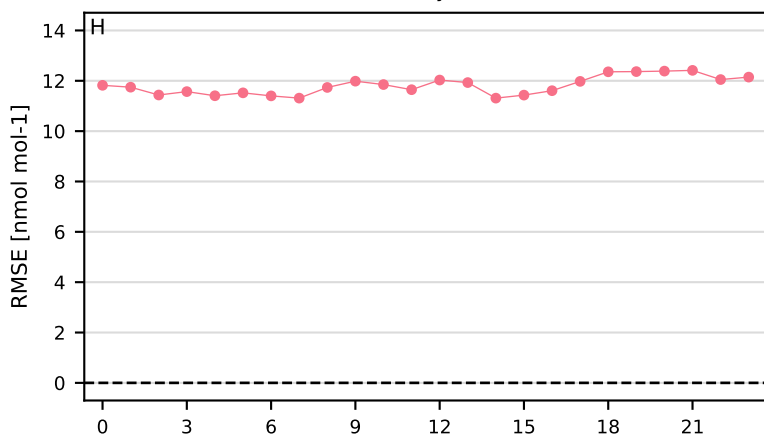
LOCAL·All (62 stations)



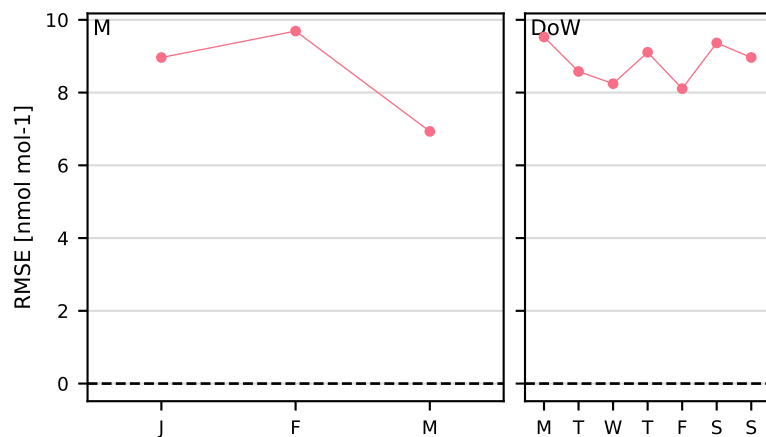
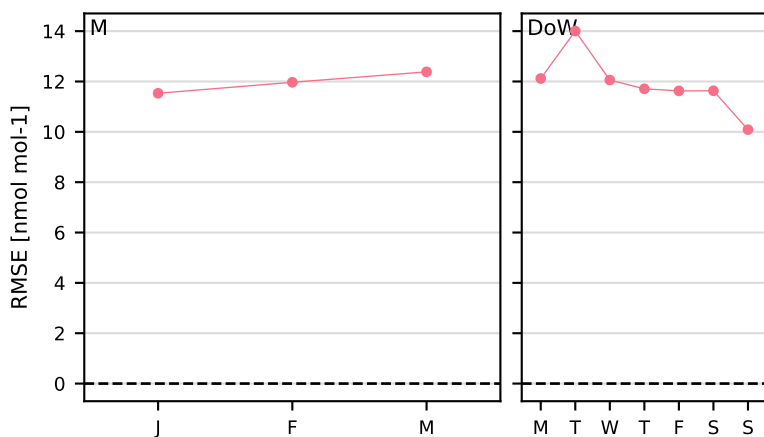
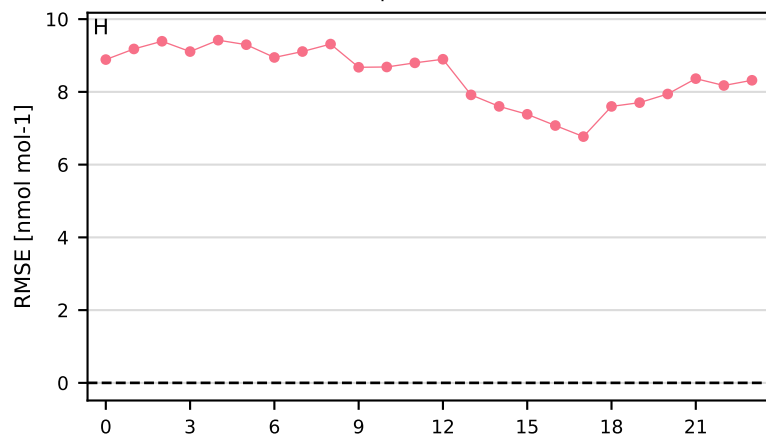
LOCAL·France (1 station)



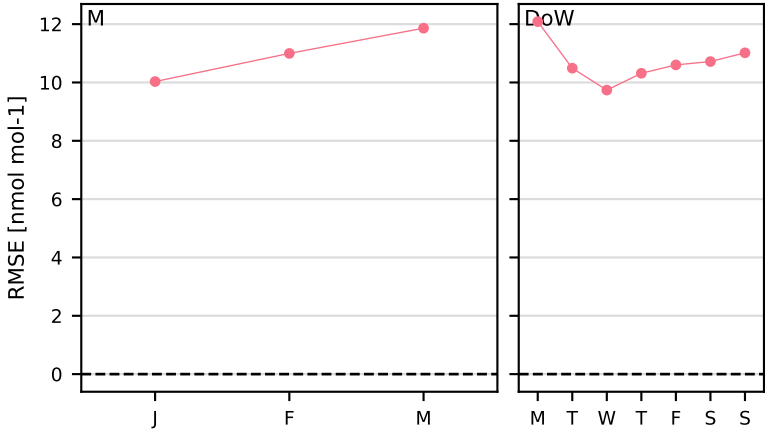
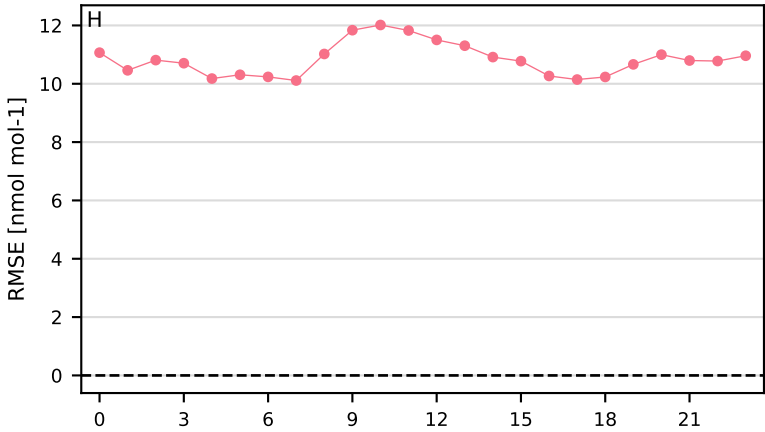
LOCAL·Germany (6 stations)



LOCAL·Spain (13 stations)



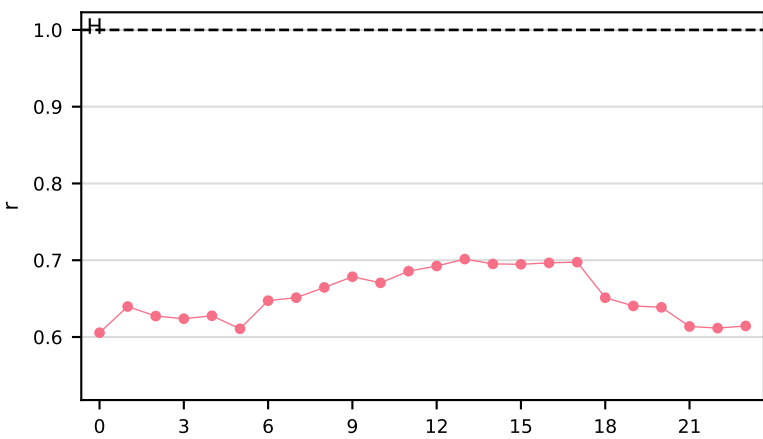
LOCAL·UK (10 stations)



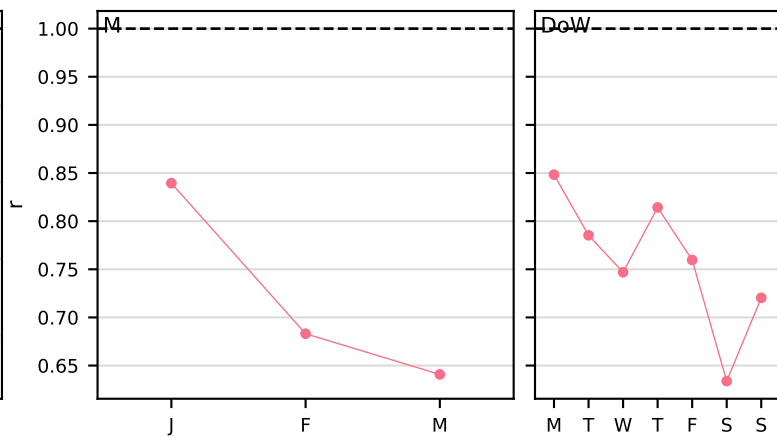
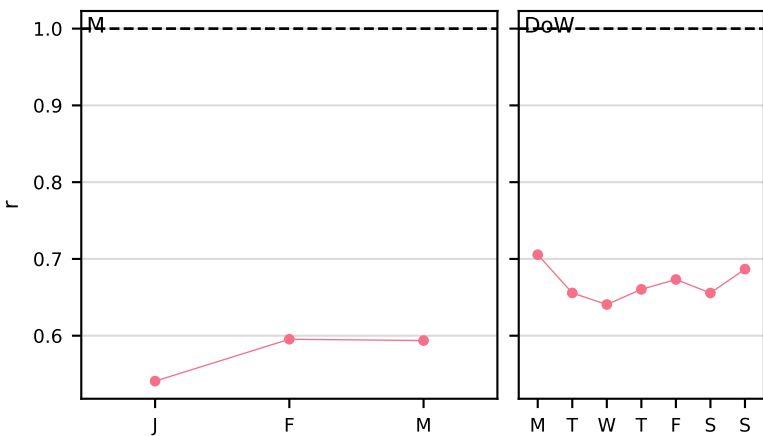
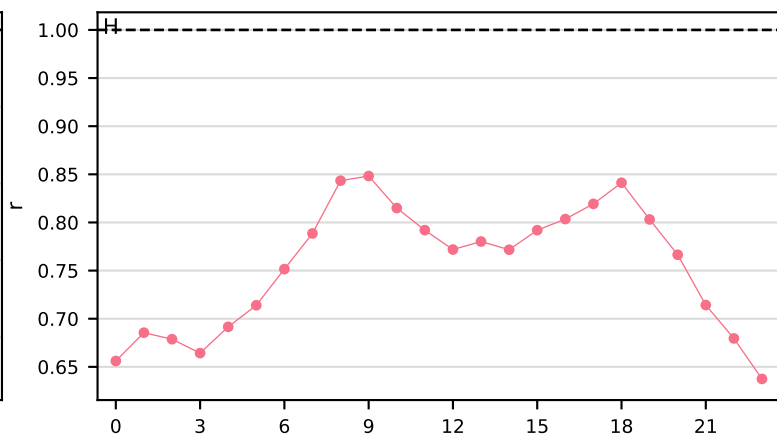
Periodic r (Summary)
EBAS|sconco3

MONARCH

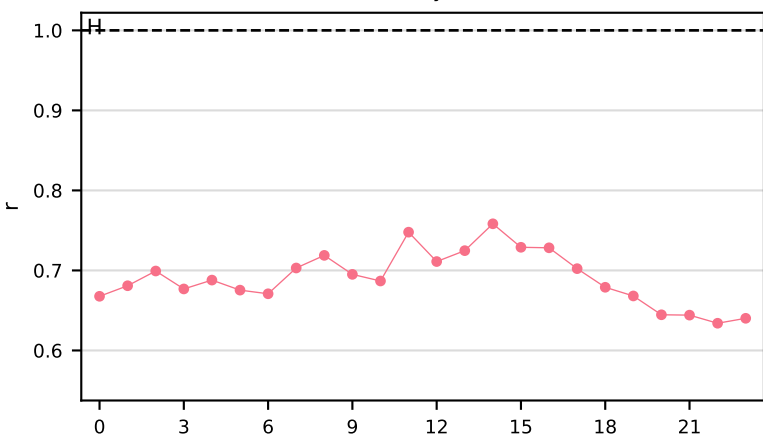
LOCAL·All (62 stations)



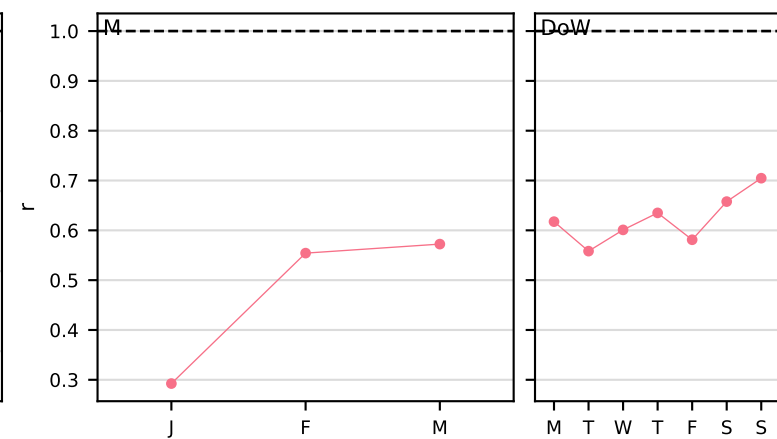
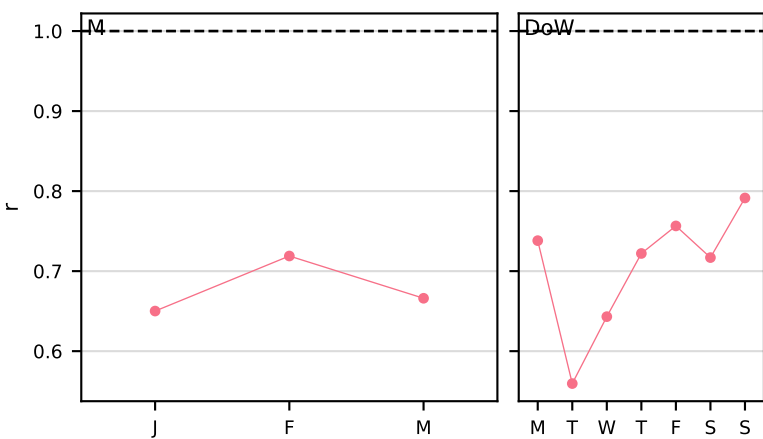
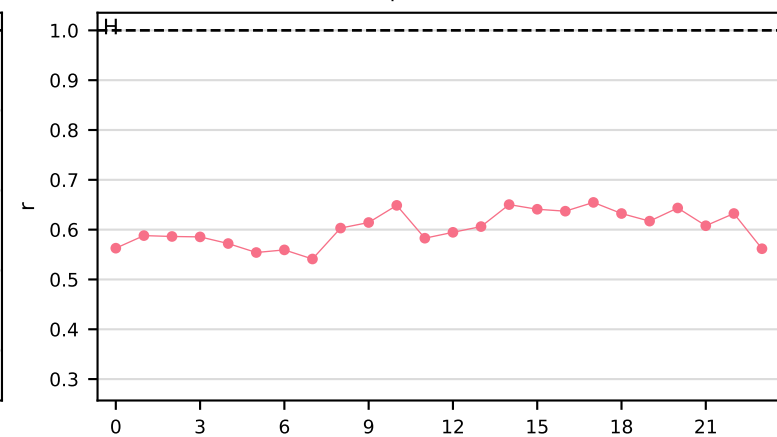
LOCAL·France (1 station)

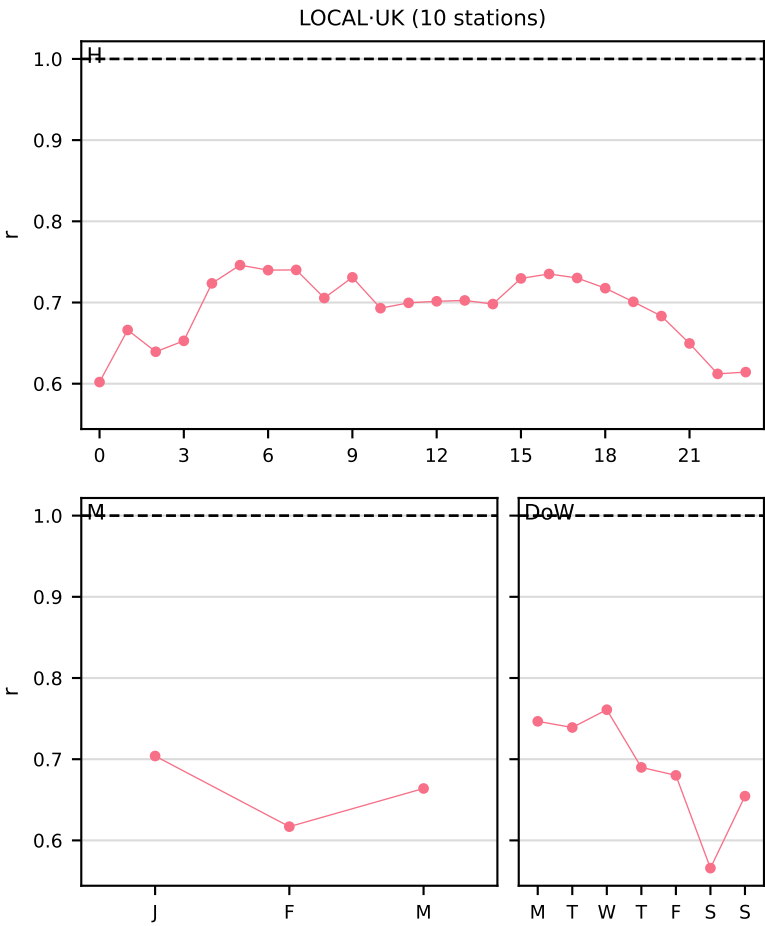


LOCAL·Germany (6 stations)



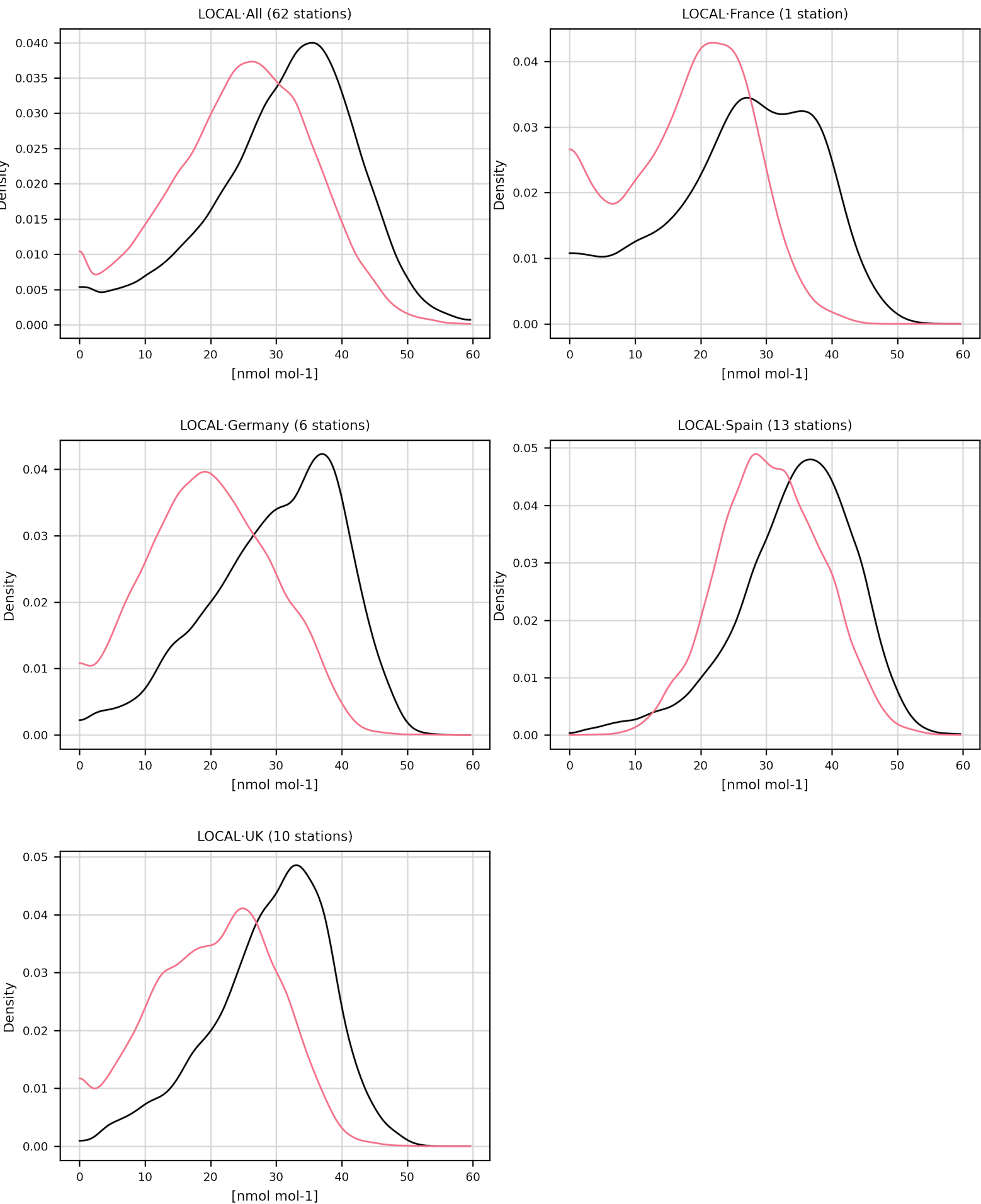
LOCAL·Spain (13 stations)

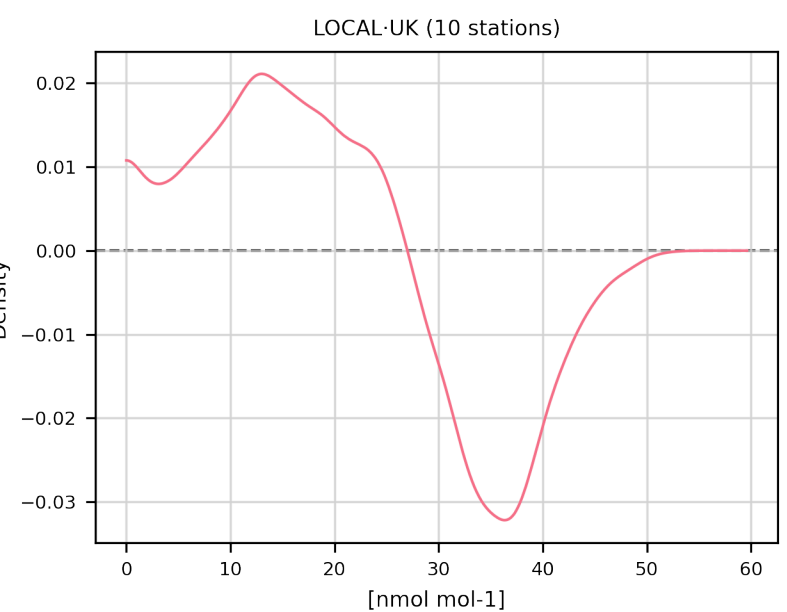
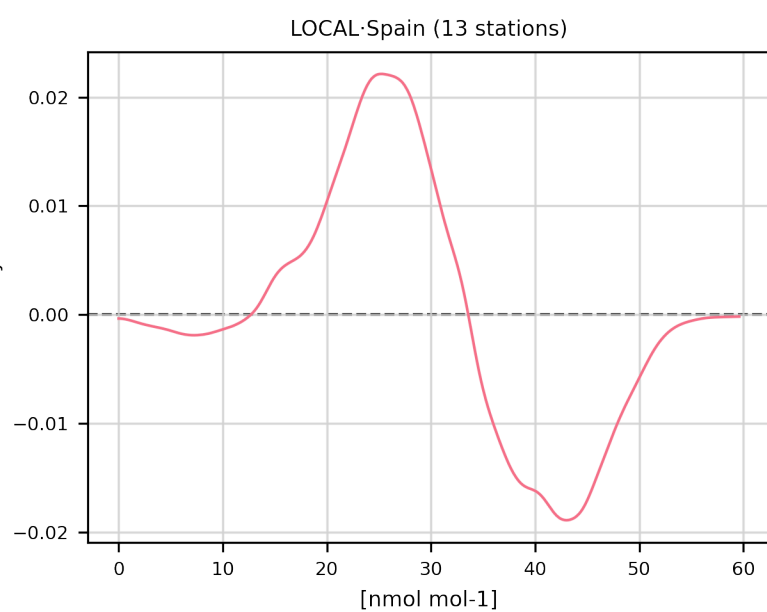
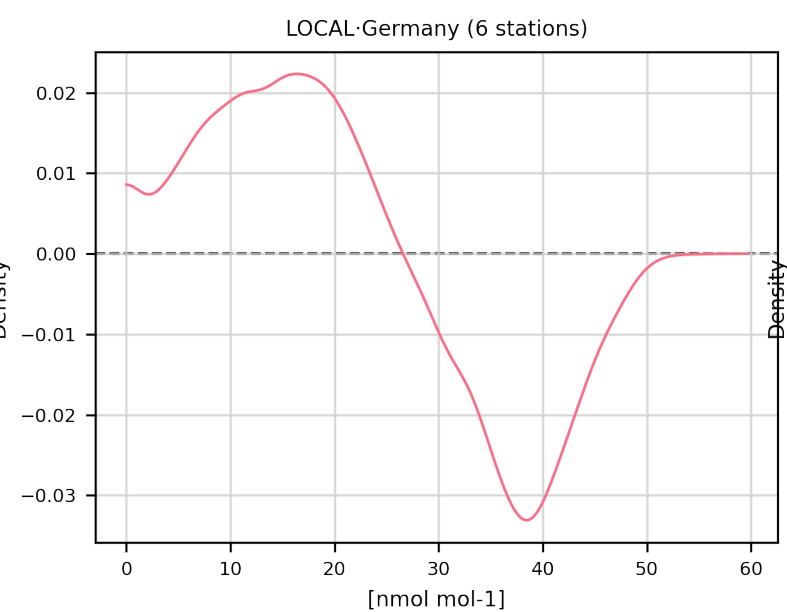
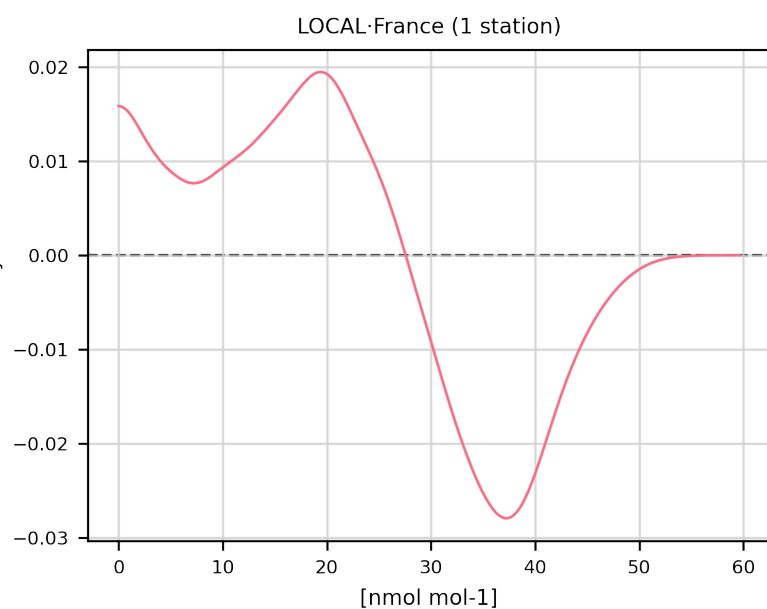
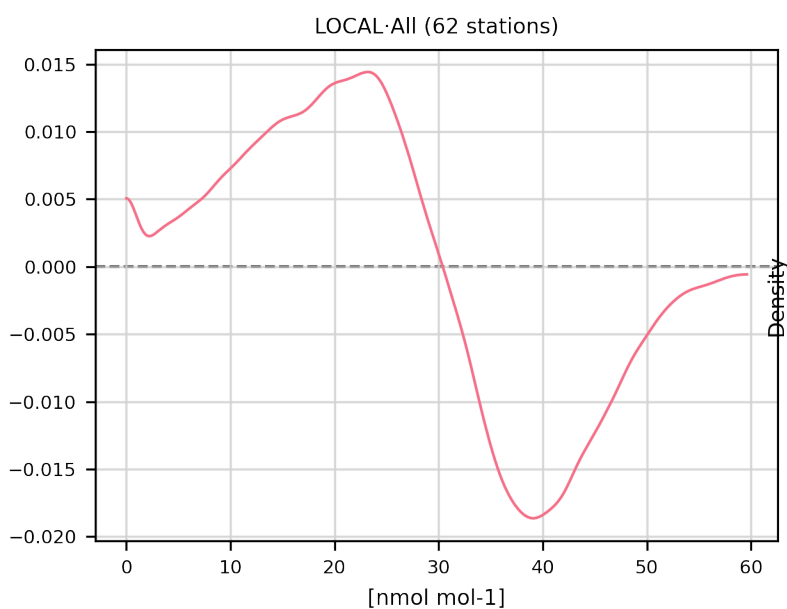


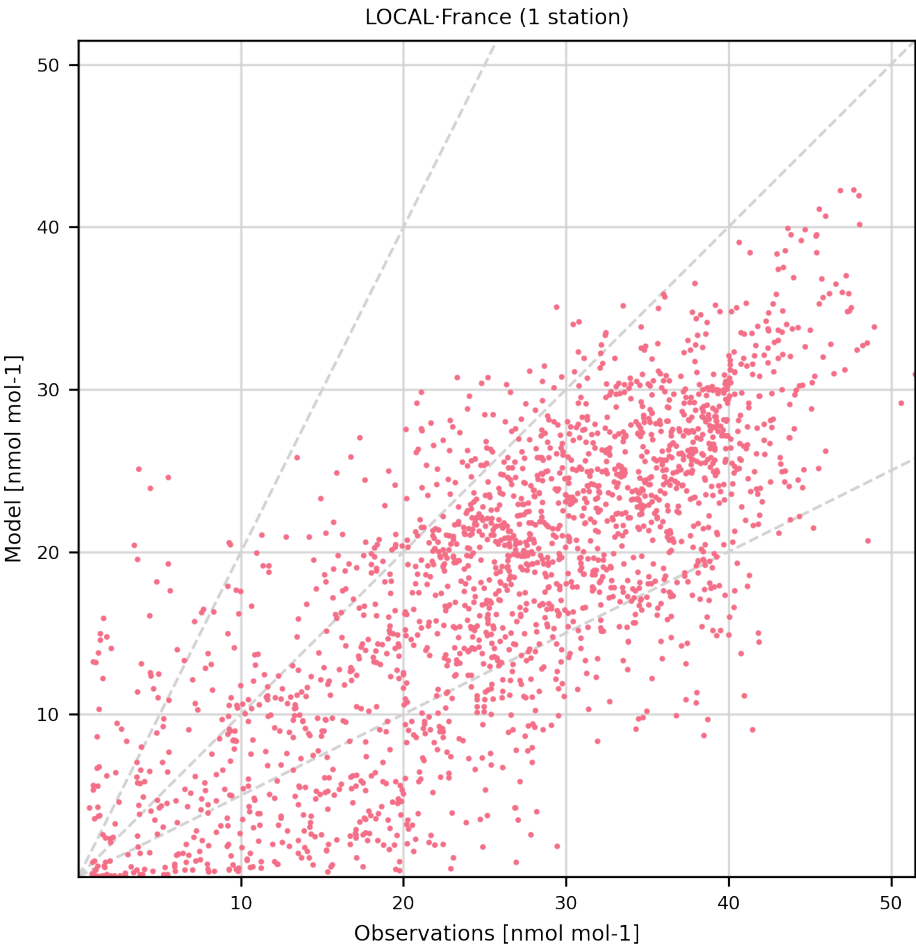
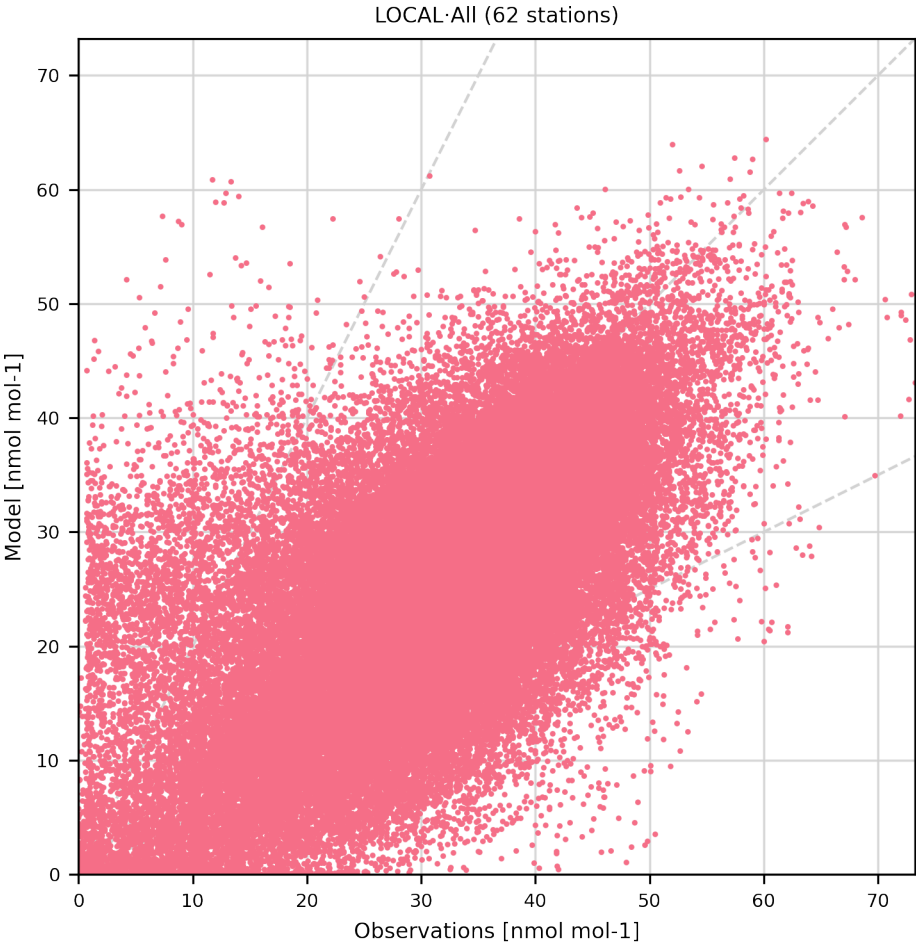


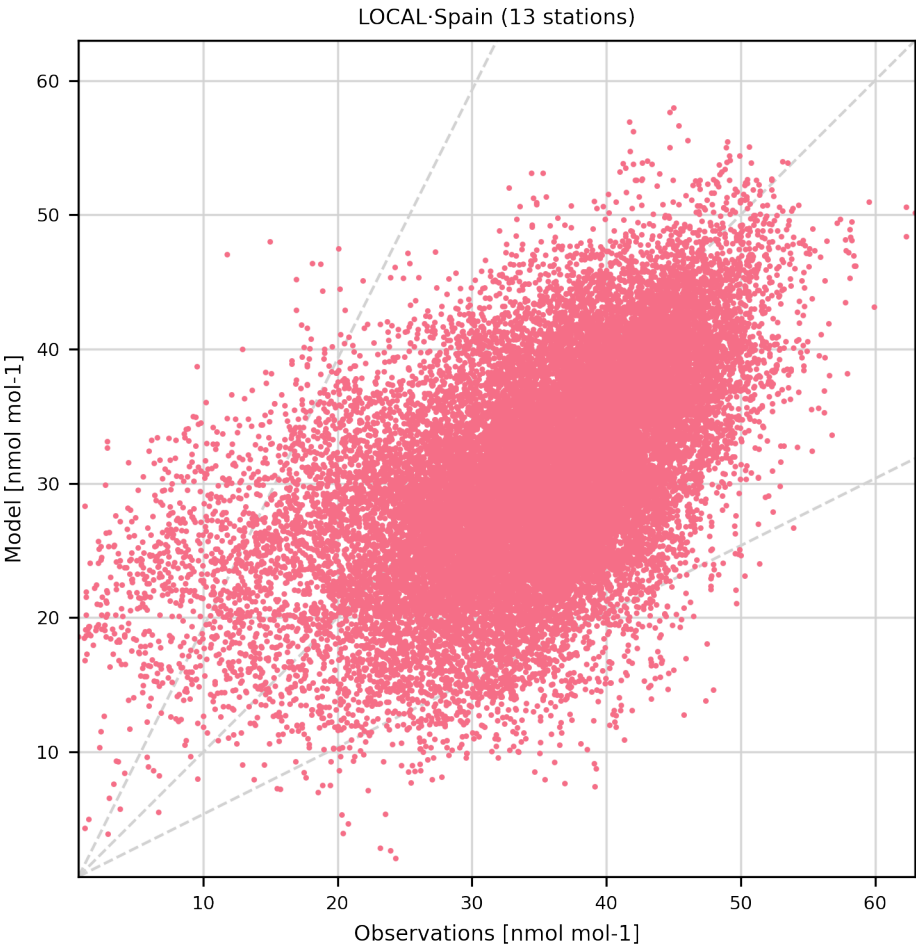
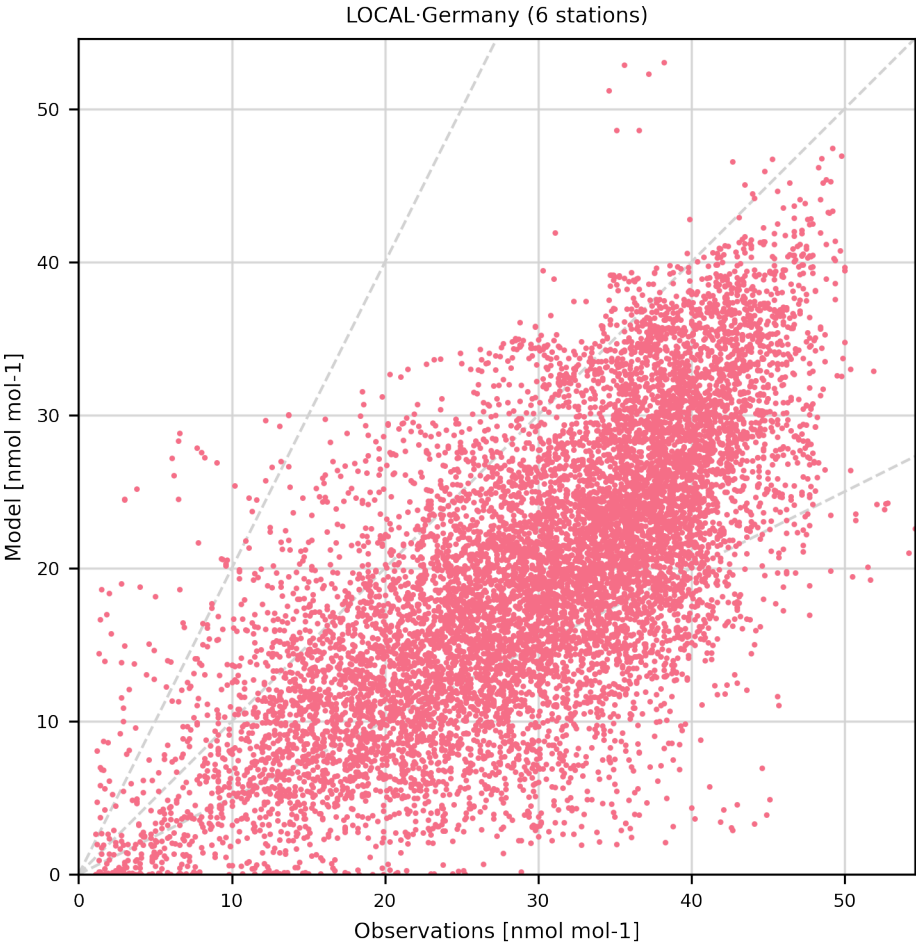
Distribution (Summary)
EBAS|sconco3

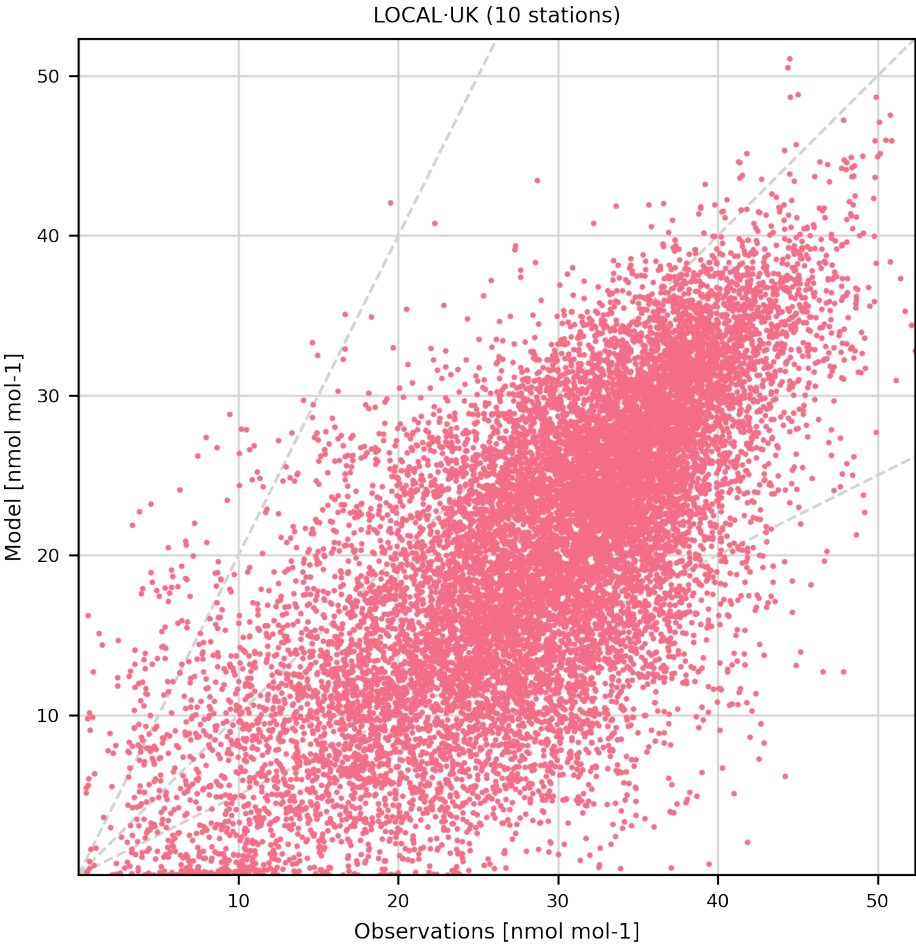
● observations ● MONARCH



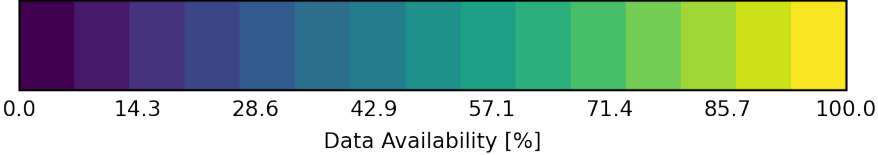




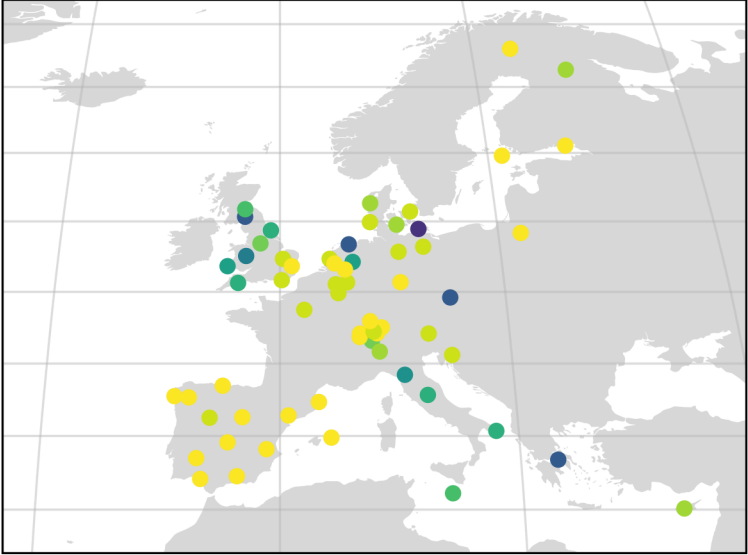




Map Data Availability (Summary)
EBAS|sconcn02



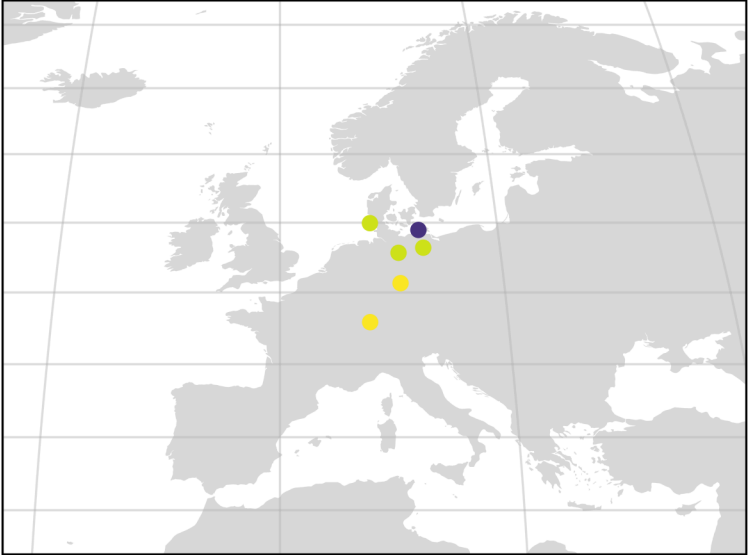
observations
LOCAL·All (62 stations)



observations
LOCAL·France (1 station)



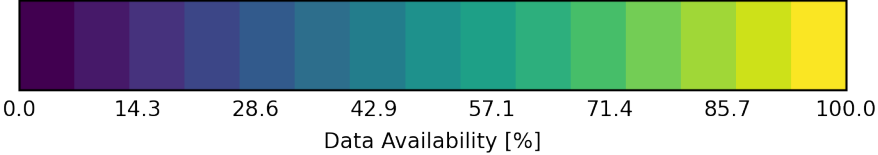
observations
LOCAL·Germany (6 stations)



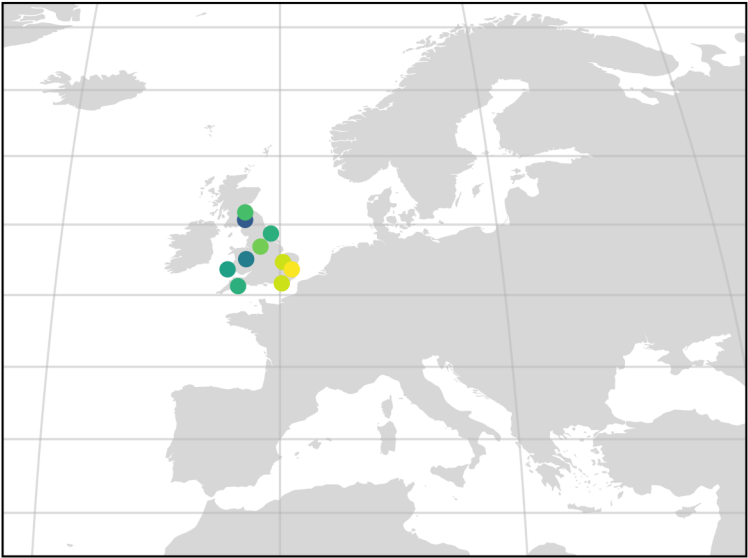
observations
LOCAL·Spain (13 stations)



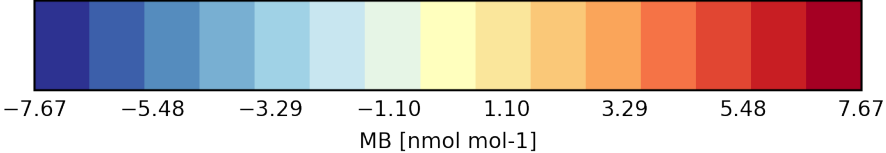
Map Data Availability (Summary)
EBAS|sconcn02



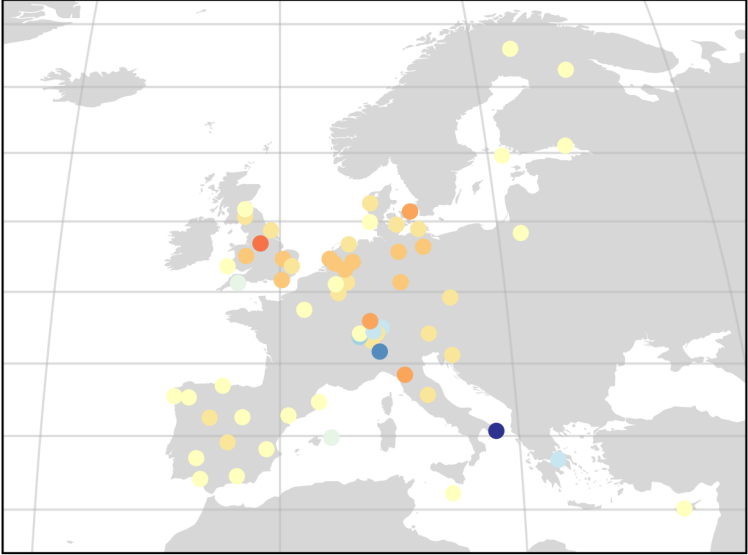
observations
LOCAL·UK (10 stations)



Map MB (Summary)
EBAS|sconcn02



MONARCH
LOCAL·All (62 stations)



MONARCH
LOCAL·France (1 station)



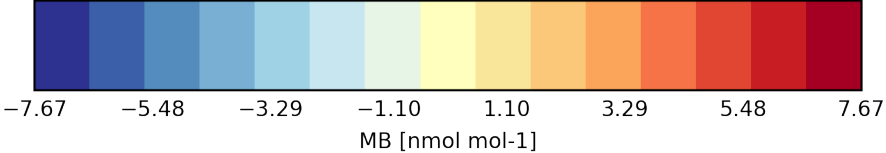
MONARCH
LOCAL·Germany (6 stations)



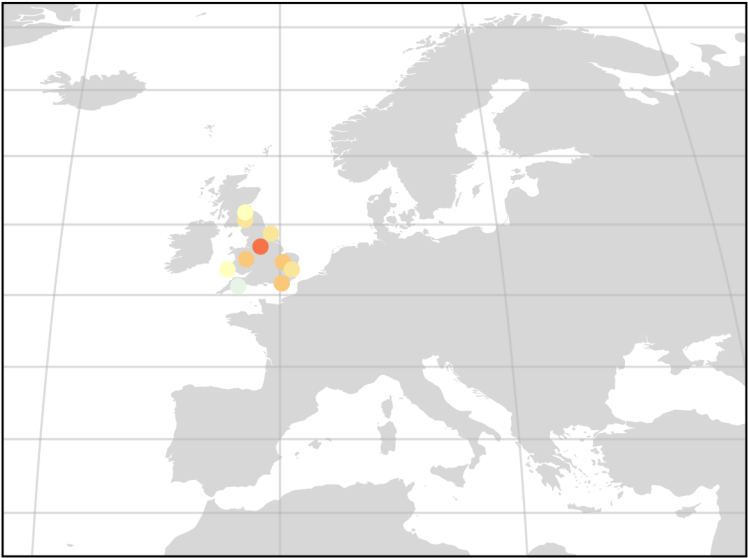
MONARCH
LOCAL·Spain (13 stations)



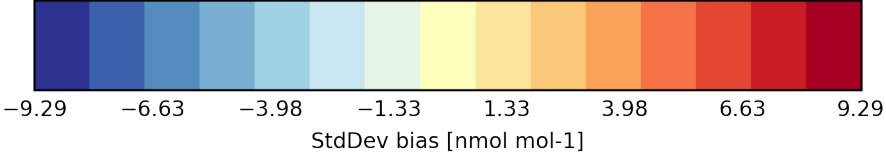
Map MB (Summary)
EBAS|sconcn02



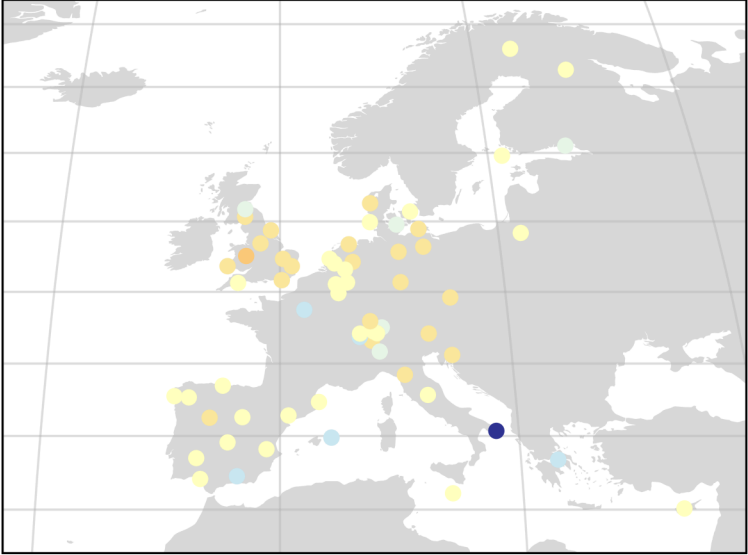
MONARCH
LOCAL·UK (10 stations)



Map StdDev bias (Summary)
EBAS|sconcnO2



MONARCH
LOCAL·All (62 stations)



MONARCH
LOCAL·France (1 station)



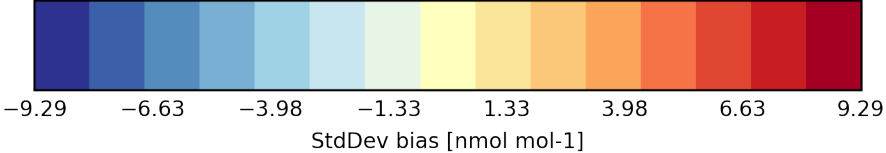
MONARCH
LOCAL·Germany (6 stations)



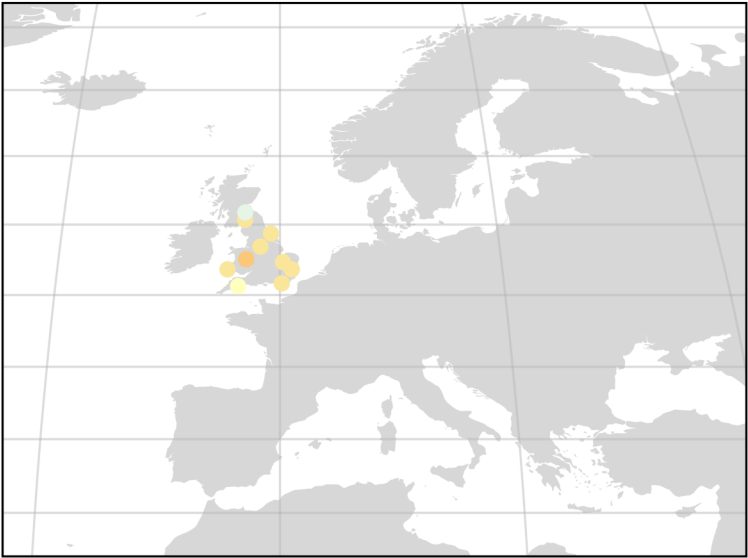
MONARCH
LOCAL·Spain (13 stations)



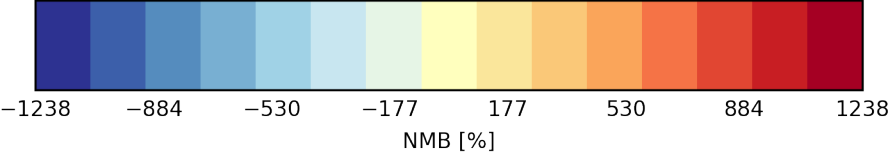
Map StdDev bias (Summary)
EBAS|sconcnO2



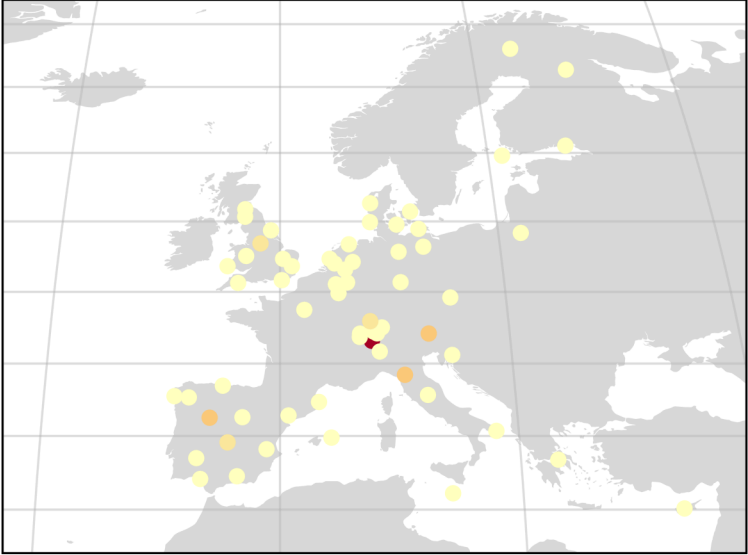
MONARCH
LOCAL·UK (10 stations)



Map NMB (Summary)
EBAS|sconcn02



MONARCH
LOCAL·All (62 stations)



MONARCH
LOCAL·France (1 station)



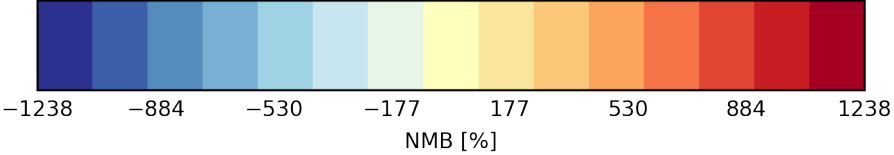
MONARCH
LOCAL·Germany (6 stations)



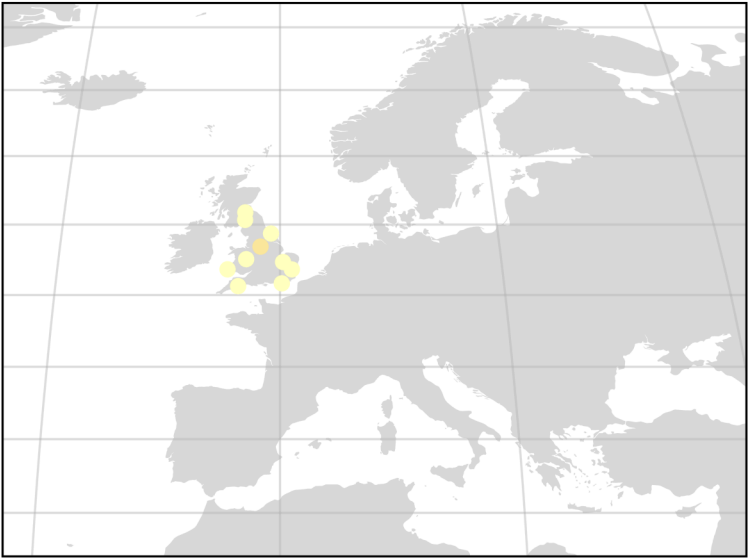
MONARCH
LOCAL·Spain (13 stations)



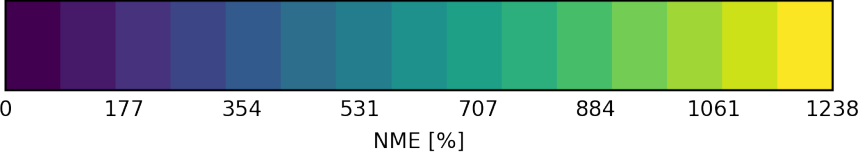
Map NMB (Summary)
EBAS|sconcn02



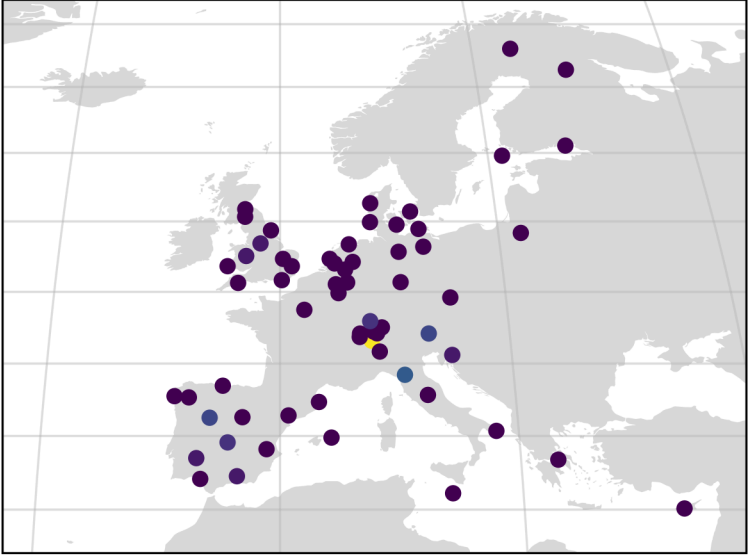
MONARCH
LOCAL·UK (10 stations)



Map NME (Summary)
EBAS|sconcn02



MONARCH
LOCAL·All (62 stations)



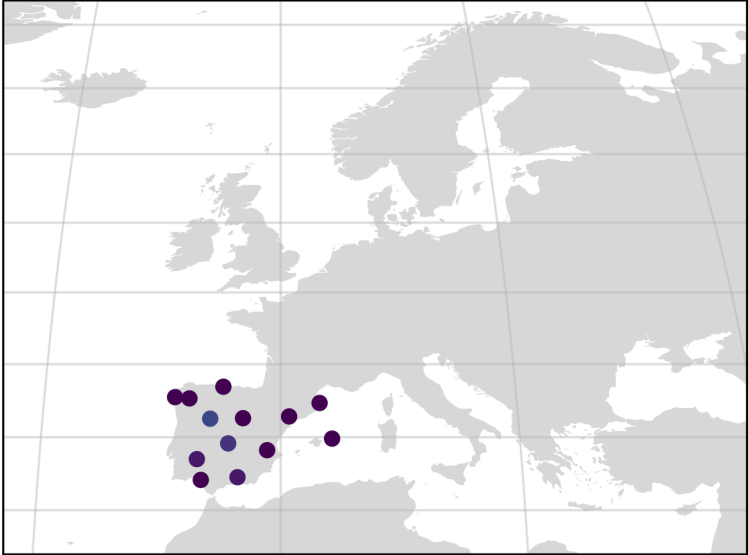
MONARCH
LOCAL·France (1 station)



MONARCH
LOCAL·Germany (6 stations)



MONARCH
LOCAL·Spain (13 stations)

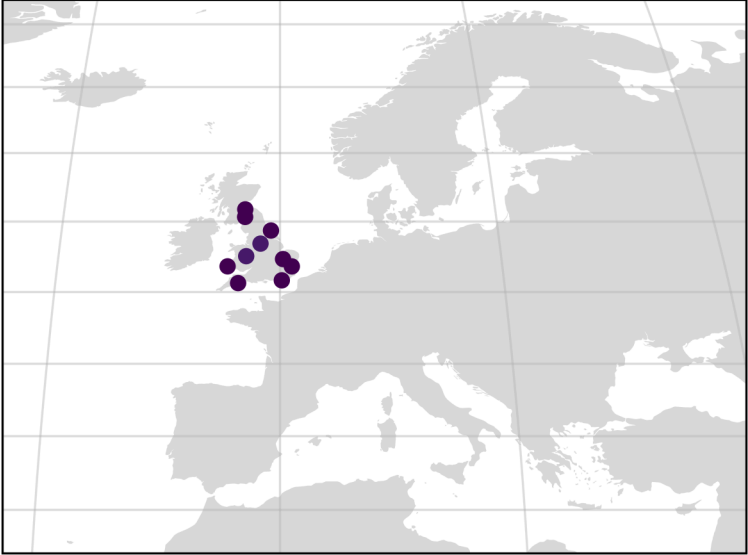


Map NME (Summary)
EBAS|sconcn02

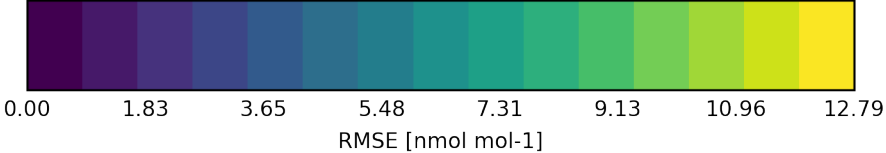


NME [%]

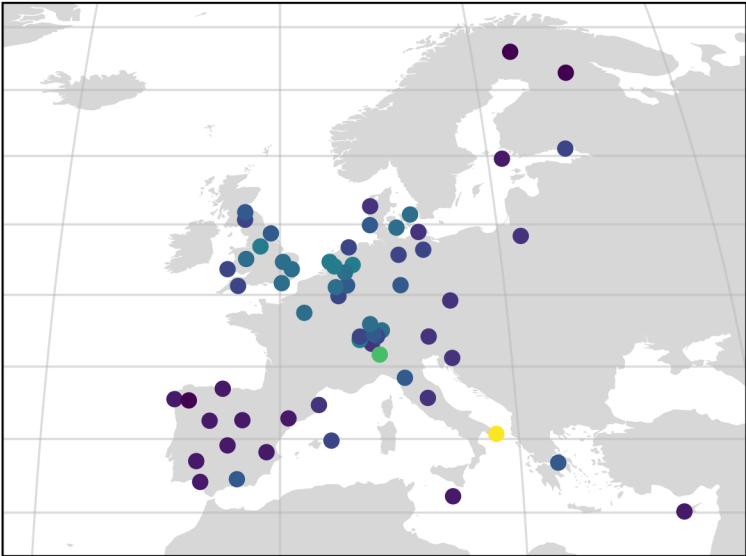
MONARCH
LOCAL·UK (10 stations)



Map RMSE (Summary)
EBAS|sconcnO2



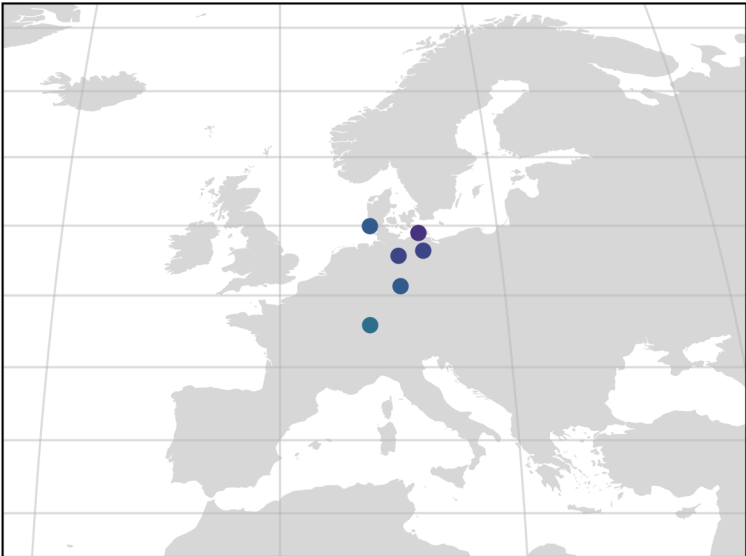
MONARCH
LOCAL·All (62 stations)



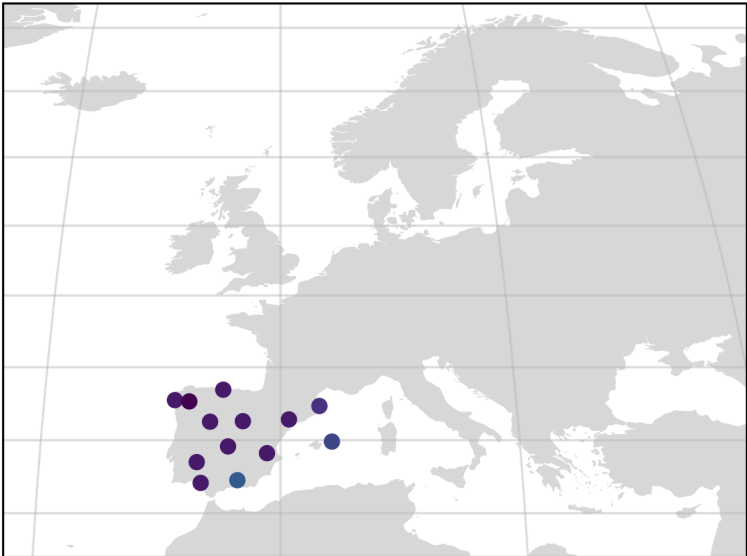
MONARCH
LOCAL·France (1 station)



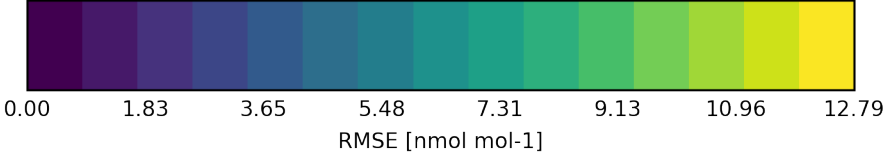
MONARCH
LOCAL·Germany (6 stations)



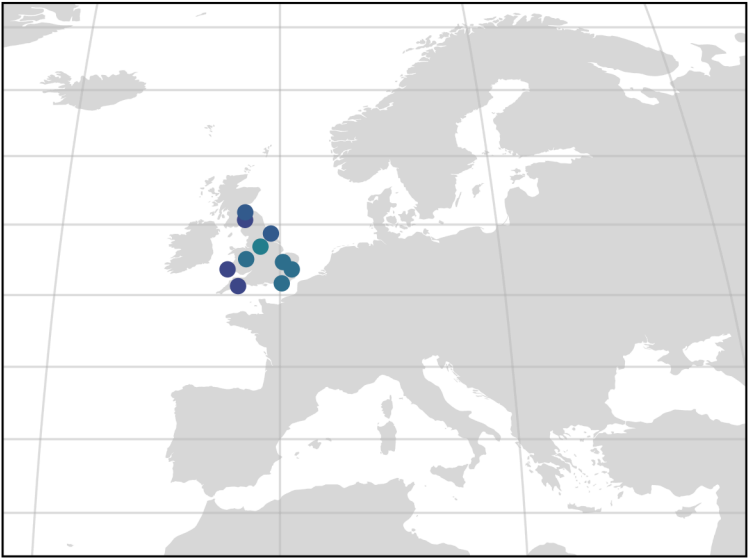
MONARCH
LOCAL·Spain (13 stations)



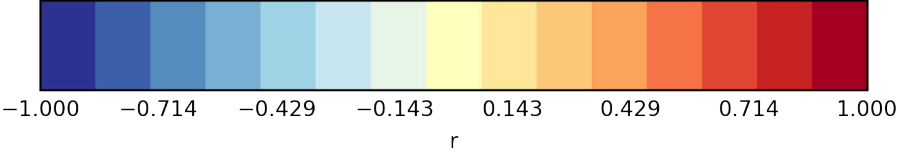
Map RMSE (Summary)
EBAS|sconcnO2



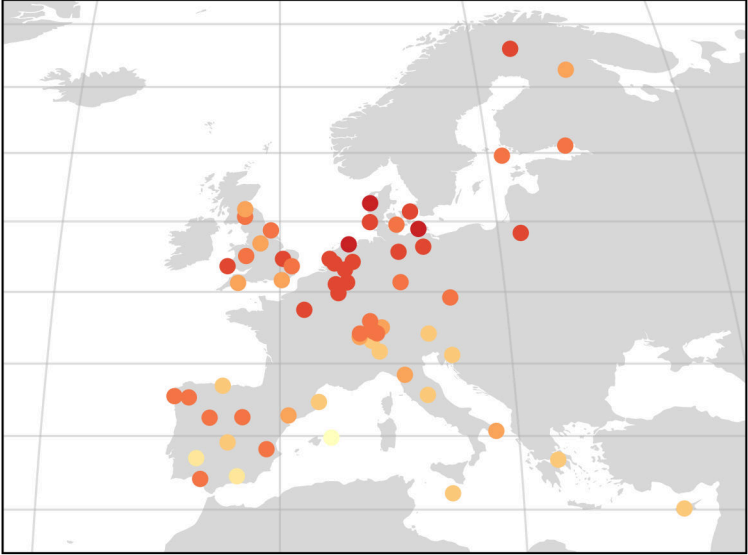
MONARCH
LOCAL-UK (10 stations)



Map r (Summary)
EBAS|sconcn02



MONARCH
LOCAL·All (62 stations)



MONARCH
LOCAL·France (1 station)



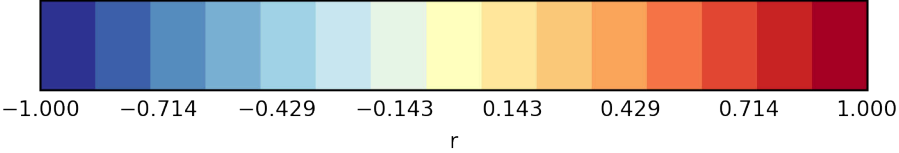
MONARCH
LOCAL·Germany (6 stations)



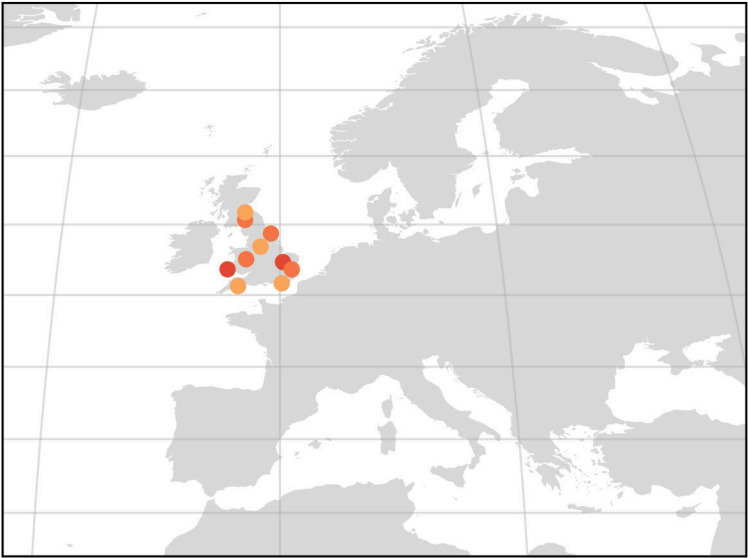
MONARCH
LOCAL·Spain (13 stations)

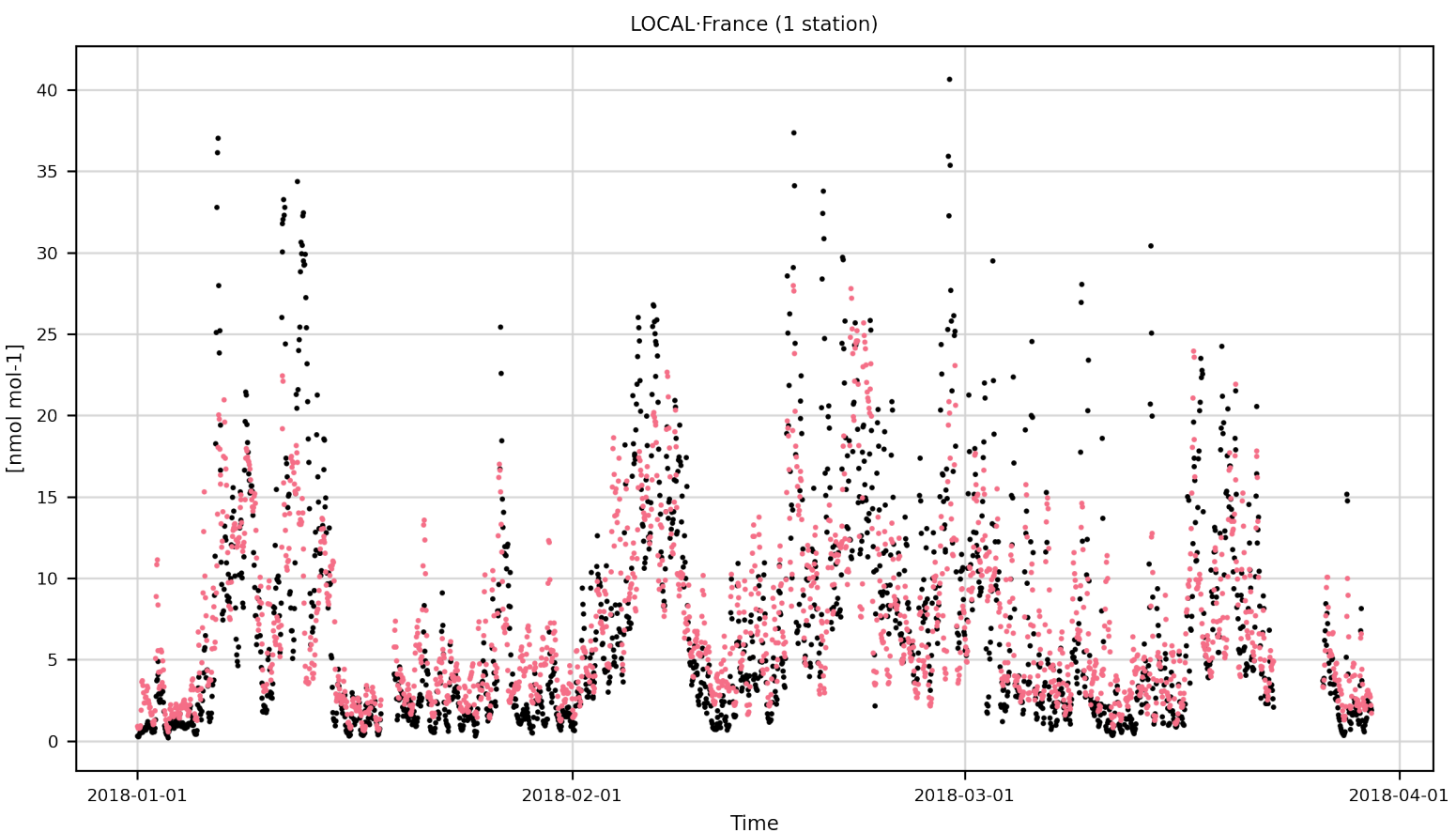
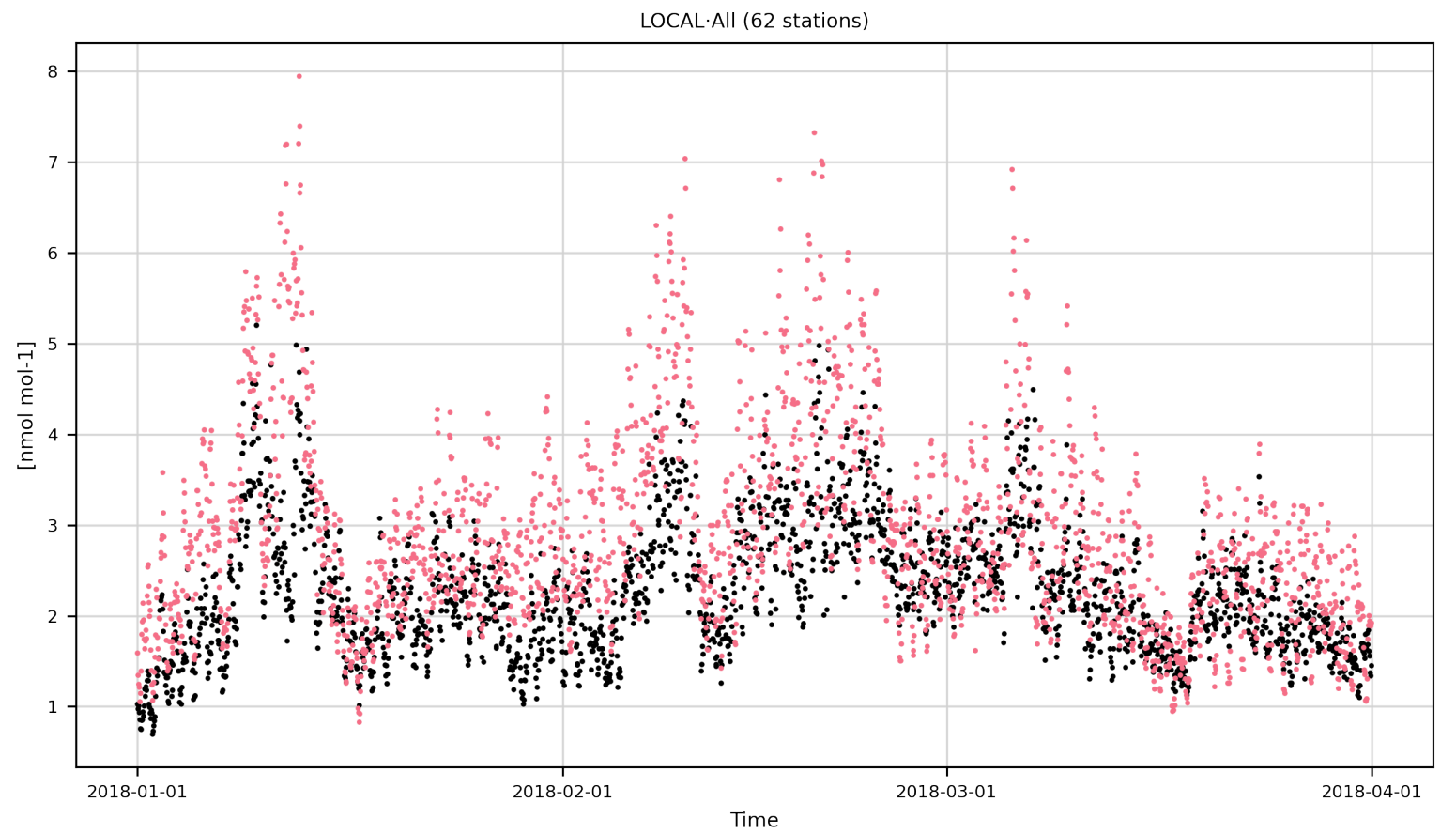


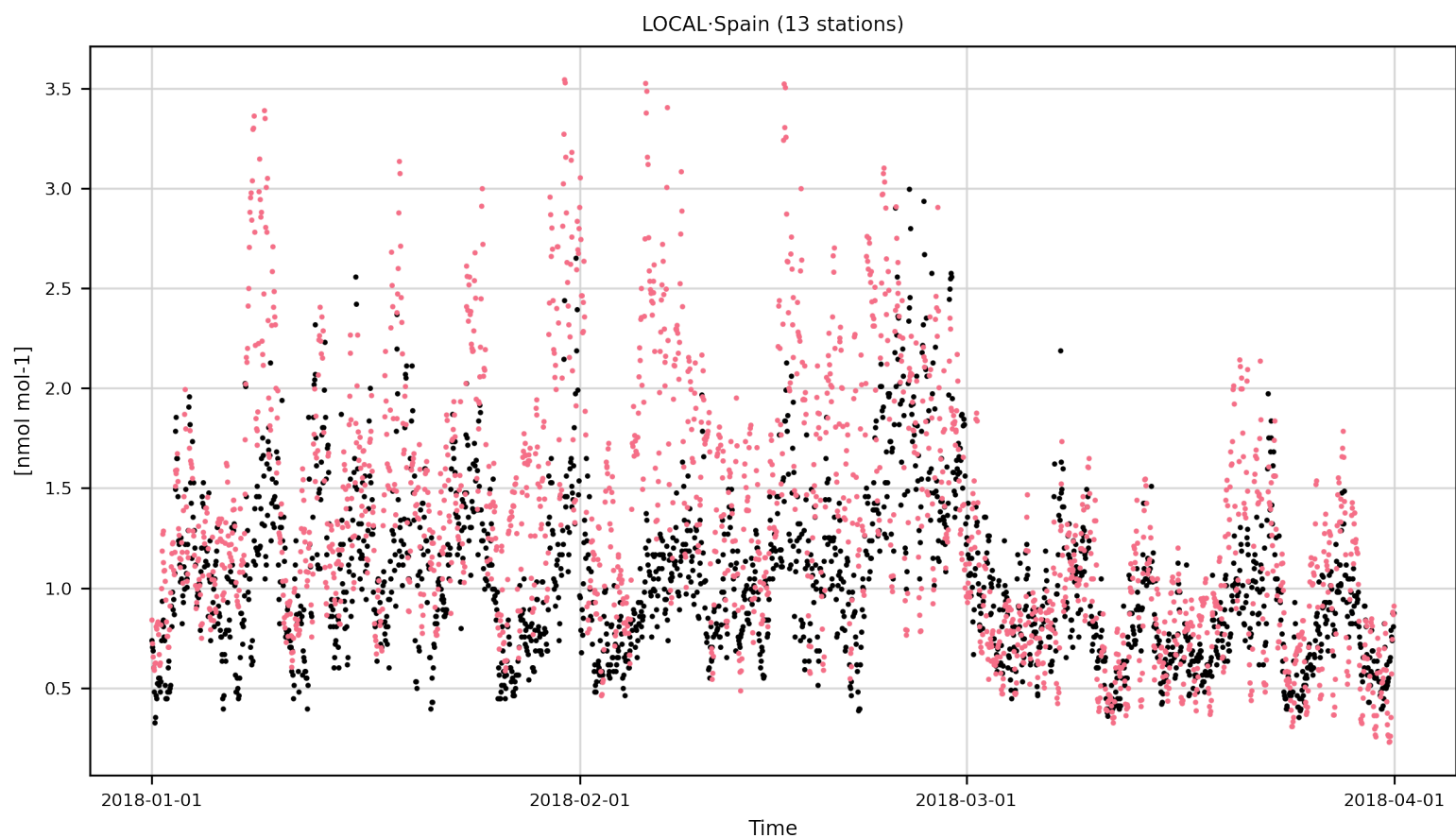
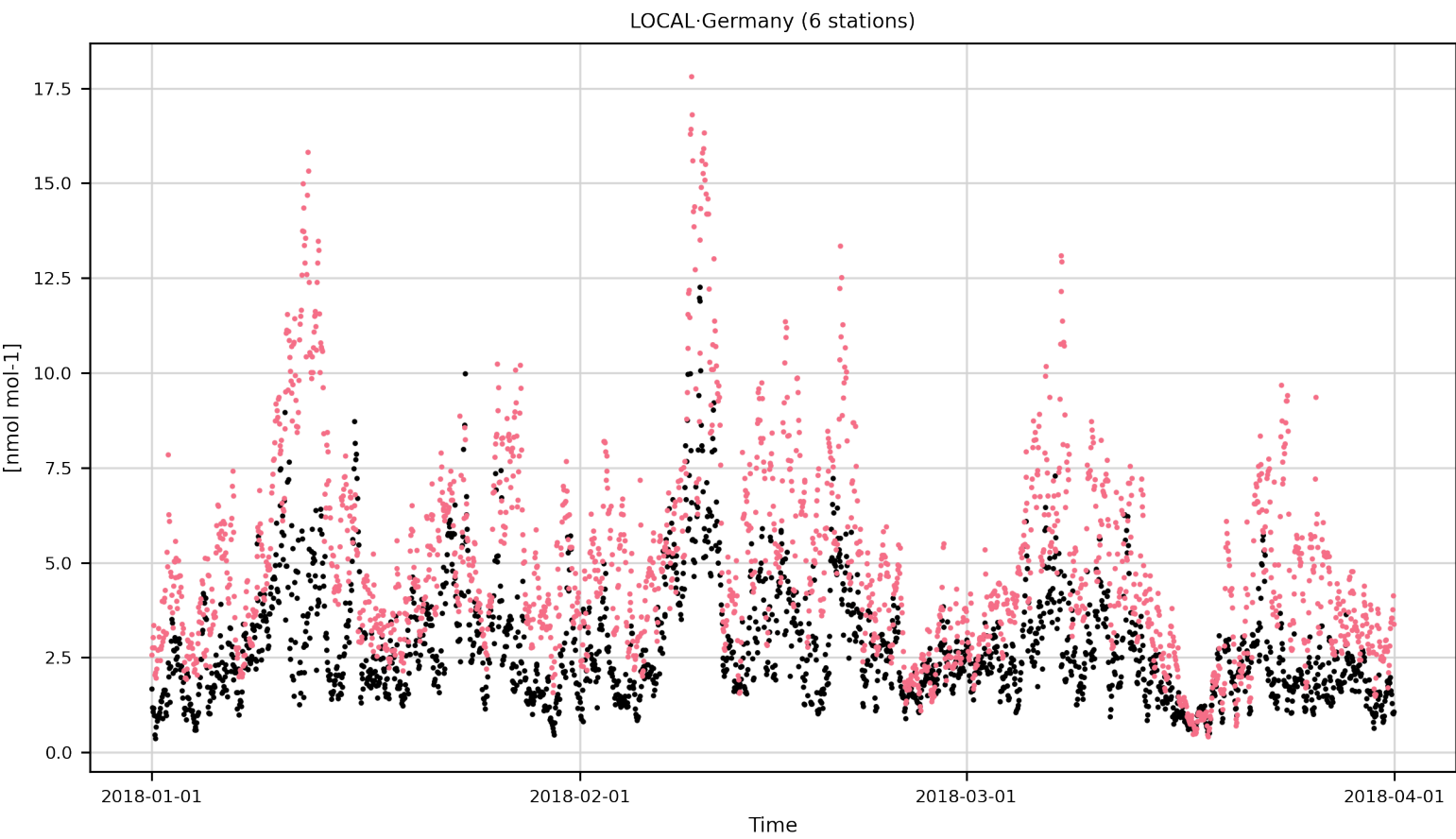
Map r (Summary)
EBAS|sconcn02

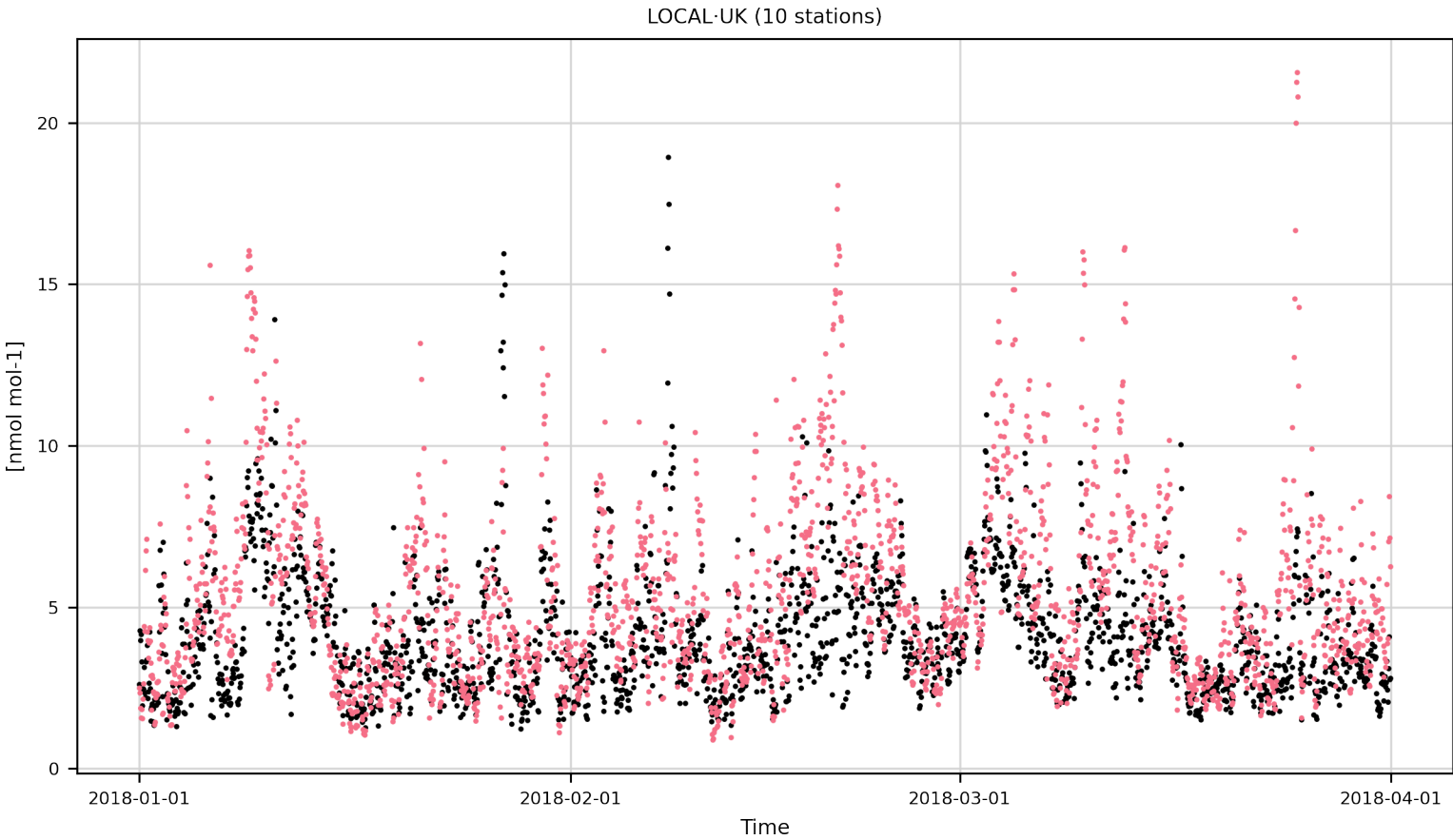


MONARCH
LOCAL·UK (10 stations)





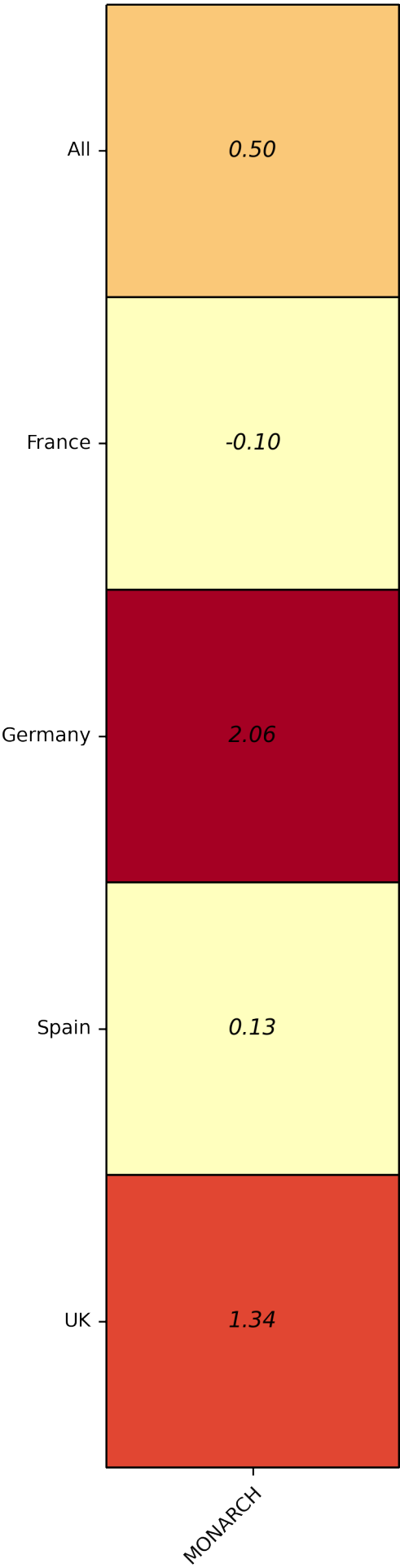
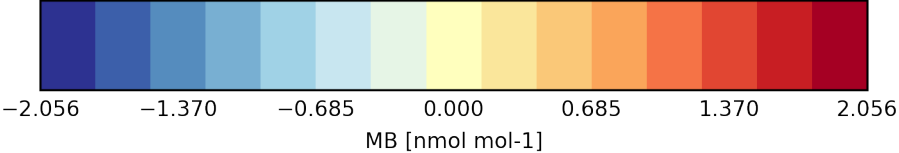




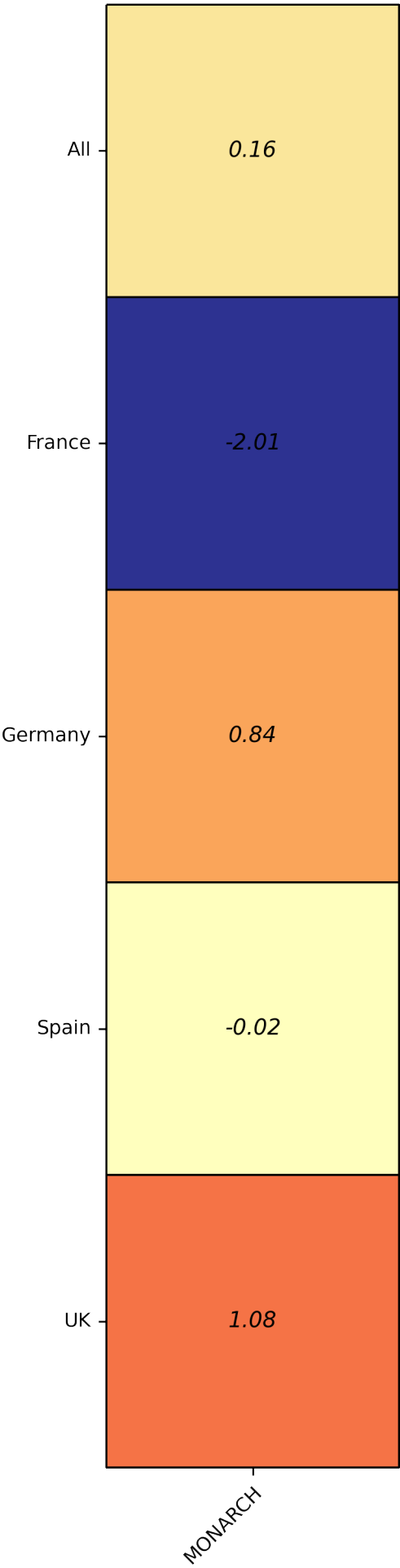
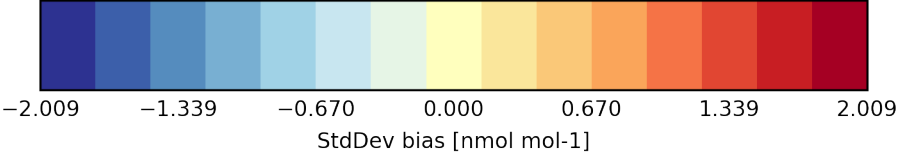
		Mean	StdDev	p5	Median	p95
All	observations	3.16	2.78	0.72	2.39	8.35
	MONARCH	3.71	2.86	0.87	3.00	9.38
France	observations	7.31	7.06	0.76	4.60	22.01
	MONARCH	7.21	5.06	1.67	5.63	17.38
Germany	observations	3.05	2.28	0.55	2.48	7.63
	MONARCH	5.13	3.25	1.33	4.20	11.79
Spain	observations	1.28	1.04	0.33	1.03	3.07
	MONARCH	1.37	1.01	0.38	1.11	3.33
UK	observations	4.39	3.18	1.34	3.26	10.54
	MONARCH	5.92	4.20	1.22	4.71	14.74

		Mean_bias	StdDev_bias	NMB	NME	RMSE	r
All	MONARCH	0.50	0.16	15.13	57.89	3.03	0.48
France	MONARCH	-0.10	-2.01	-1.33	43.36	4.84	0.73
Germany	MONARCH	2.06	0.84	62.05	71.81	3.35	0.61
Spain	MONARCH	0.13	-0.02	9.48	59.86	1.36	0.42
UK	MONARCH	1.34	1.08	30.46	58.78	4.13	0.48

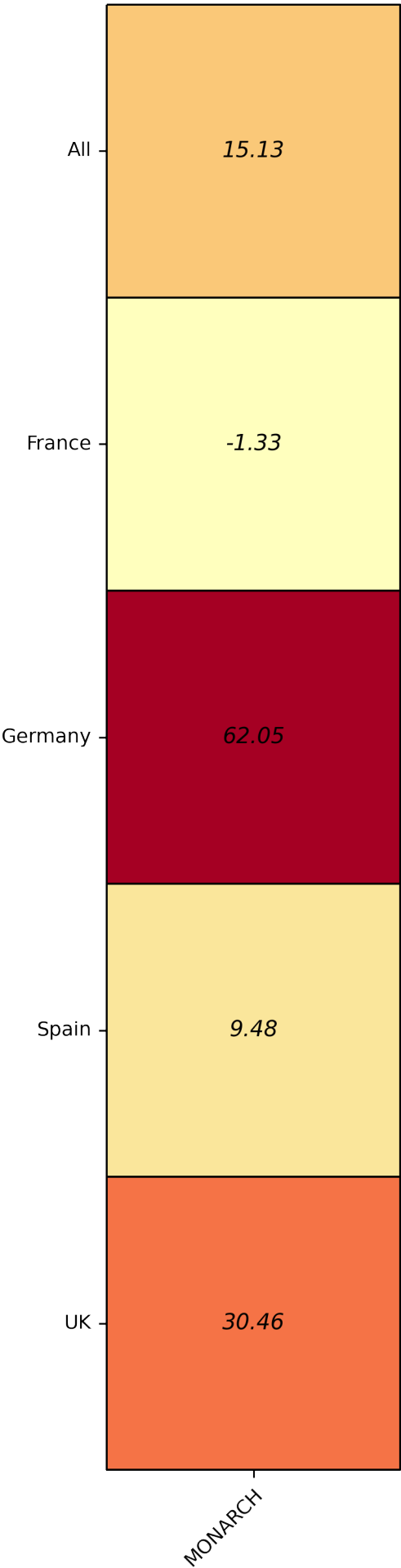
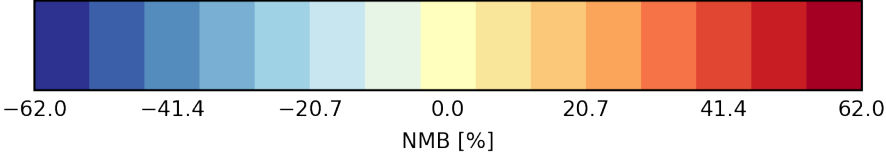
Heatmap MB (Summary)
EBAS|sconcnO2



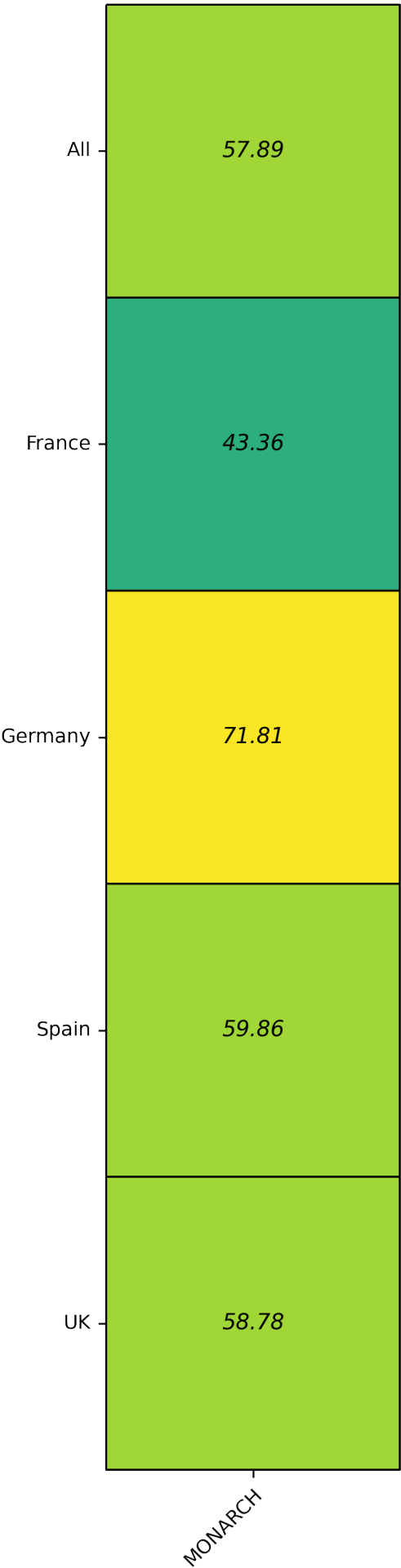
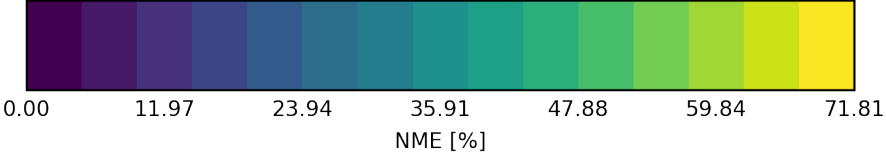
Heatmap StdDev bias (Summary)
EBAS|sconcnO2



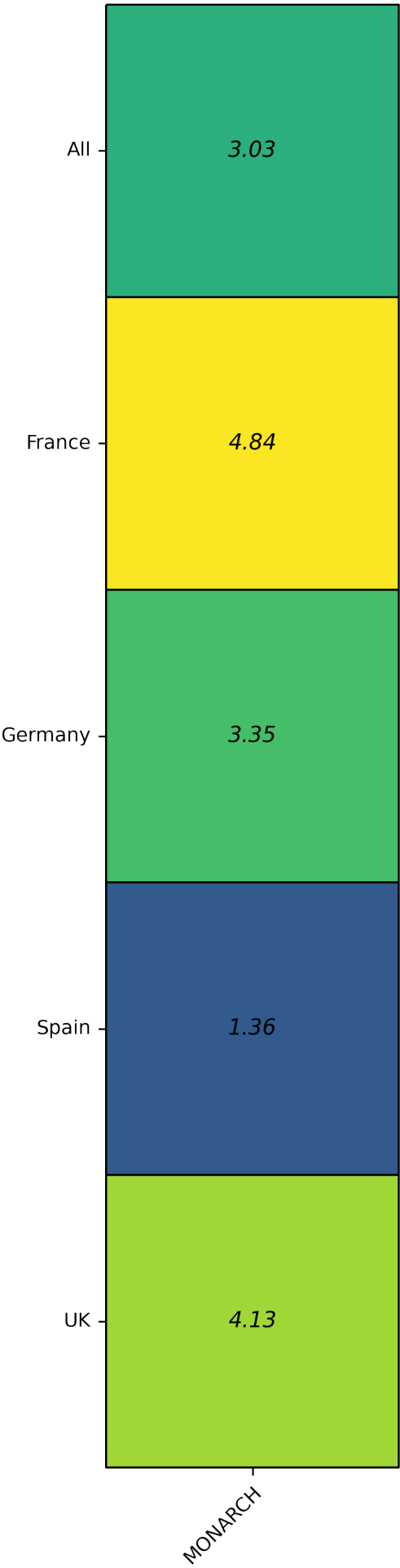
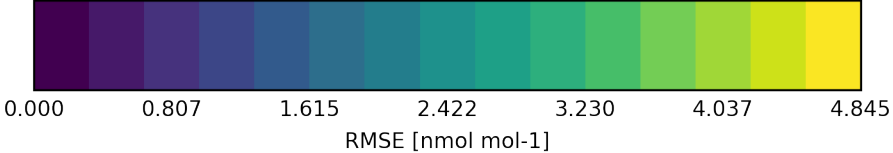
Heatmap NMB (Summary)
EBAS|sconcn02



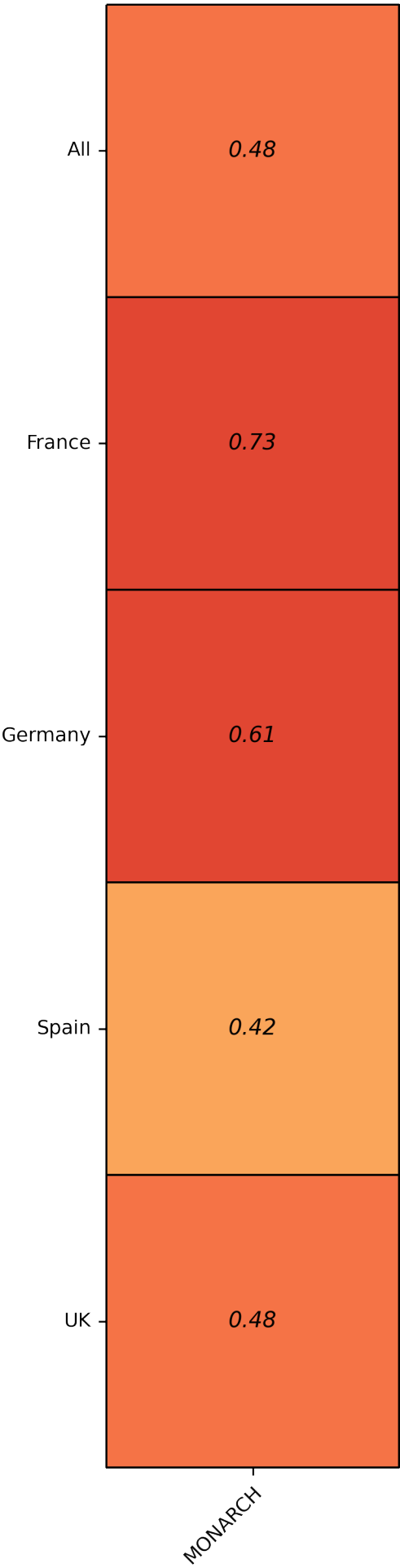
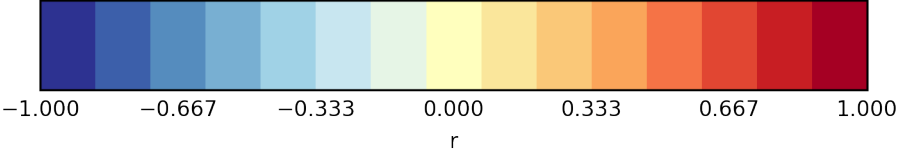
Heatmap NME (Summary)
EBAS|sconcn02



Heatmap RMSE (Summary)
EBAS|sconcnO2



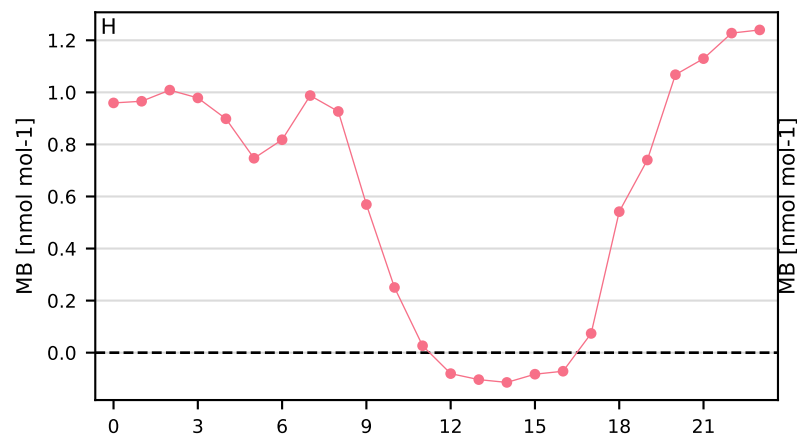
Heatmap r (Summary)
EBAS|sconcn2



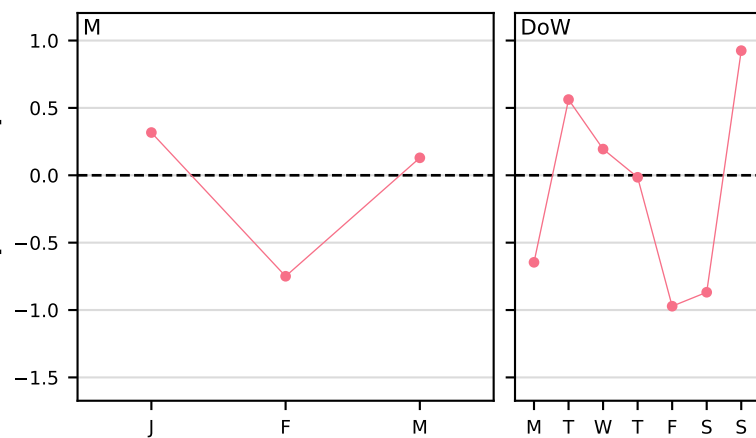
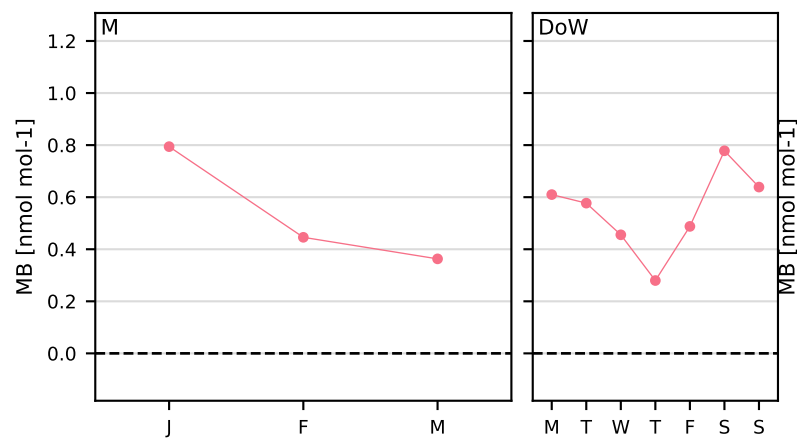
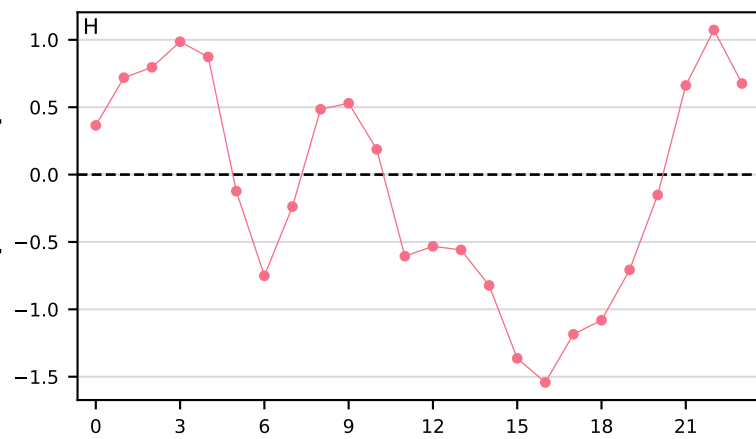
Periodic MB (Summary)
EBAS|sconcn02

● MONARCH

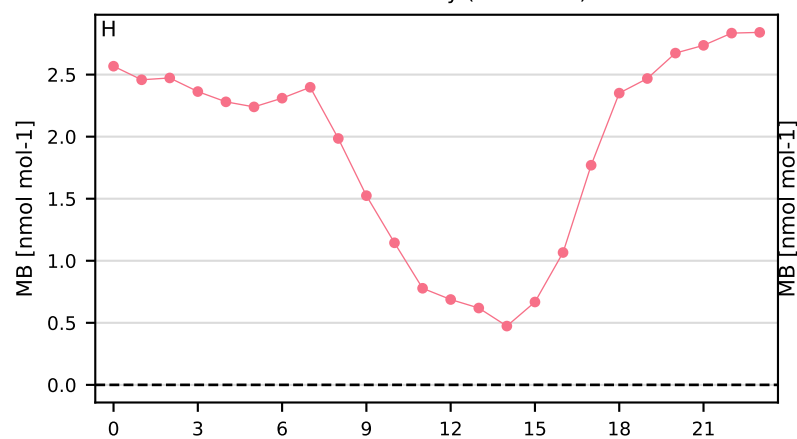
LOCAL·All (62 stations)



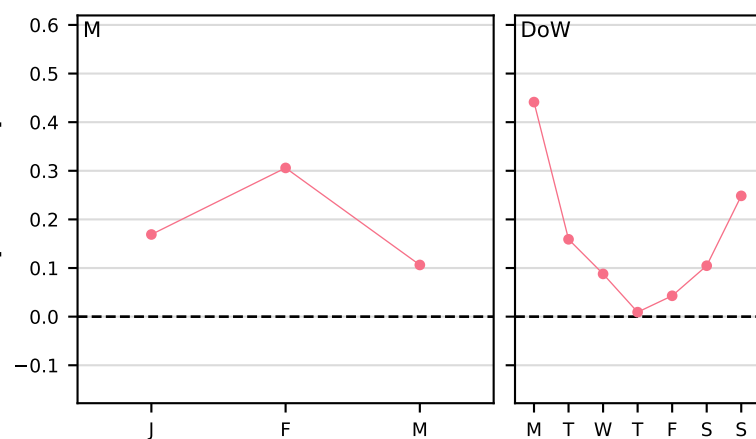
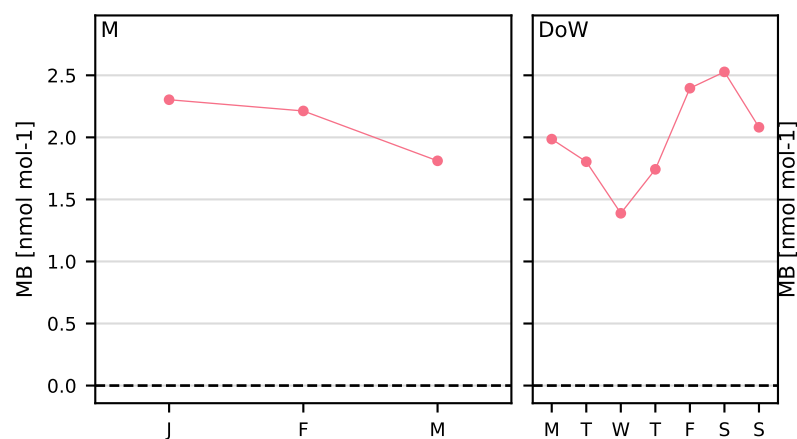
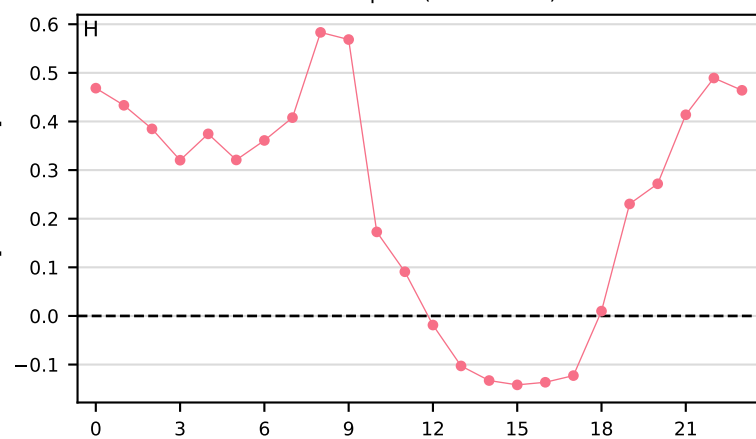
LOCAL·France (1 station)



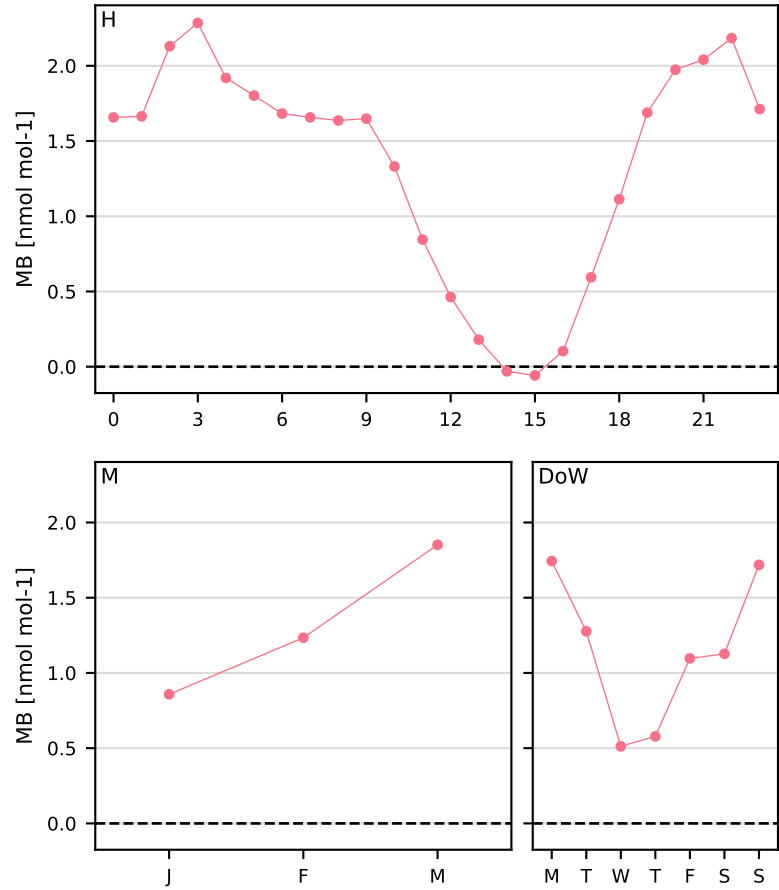
LOCAL·Germany (6 stations)



LOCAL·Spain (13 stations)

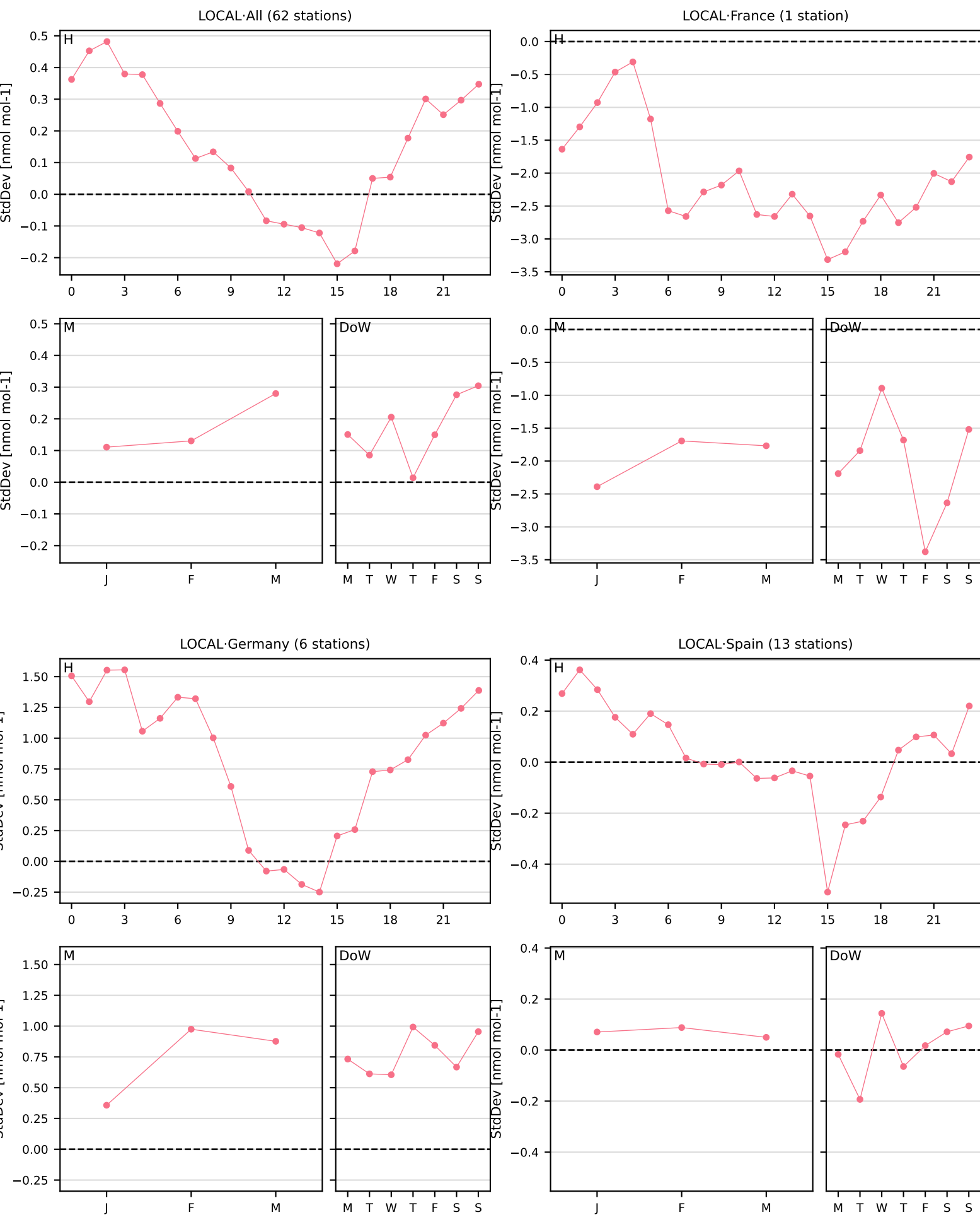


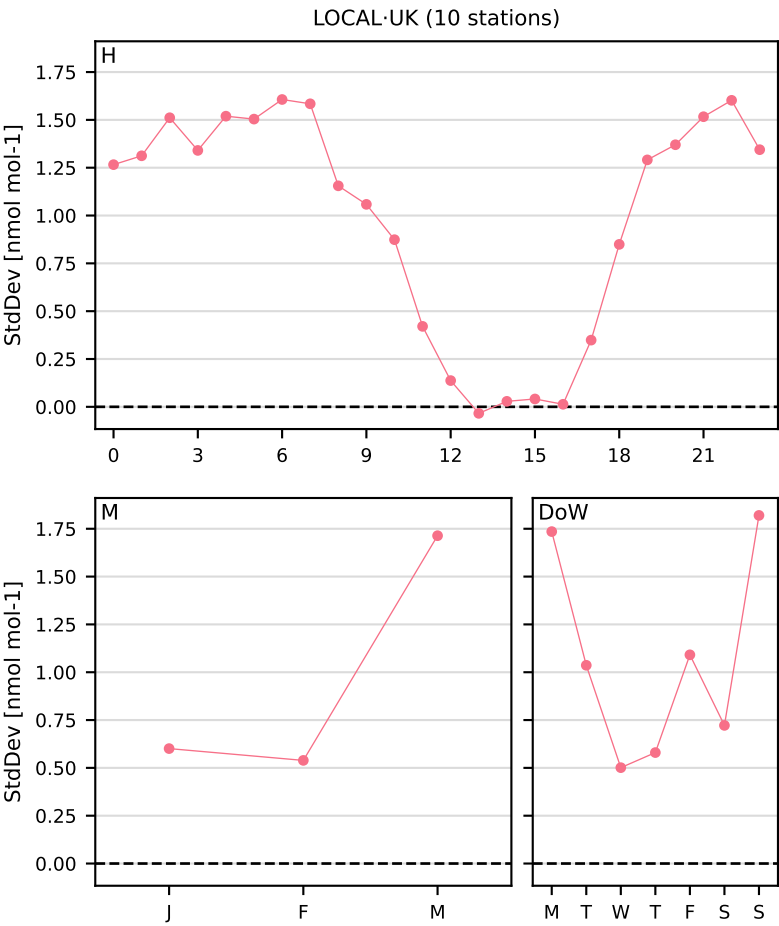
LOCAL·UK (10 stations)



Periodic StdDev bias (Summary)
EBAS|sconcn02

MONARCH

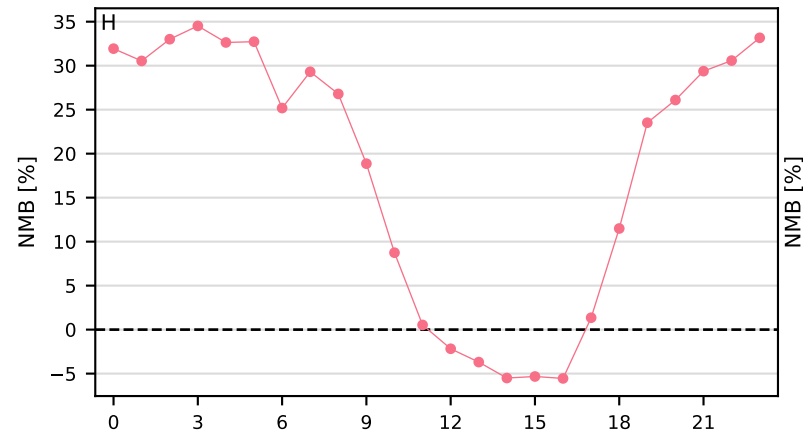




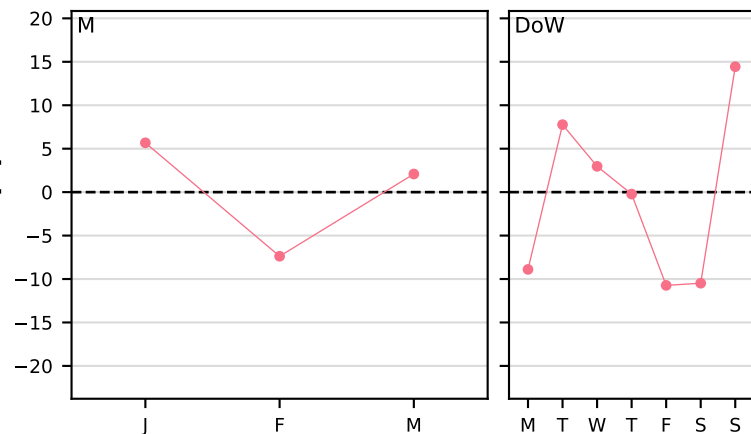
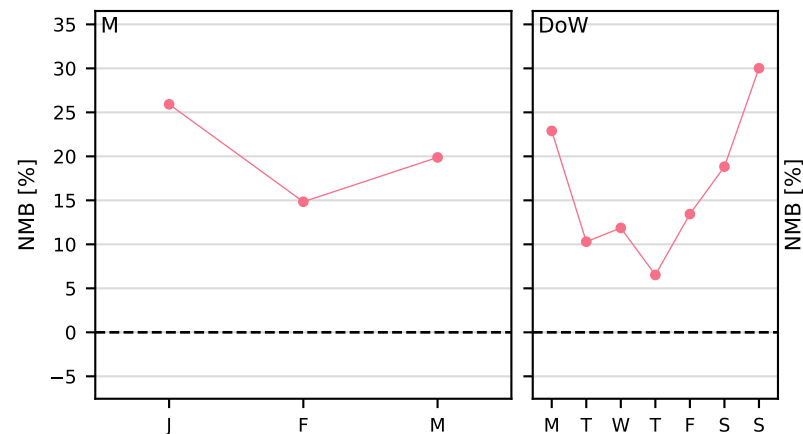
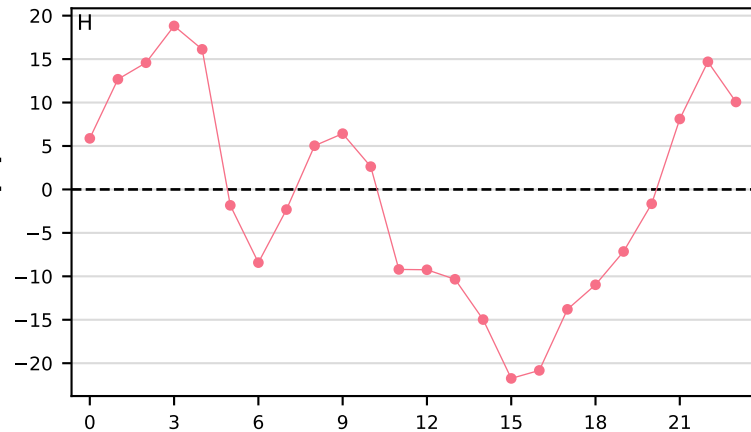
Periodic NMB (Summary)
EBAS|sconcn02

● MONARCH

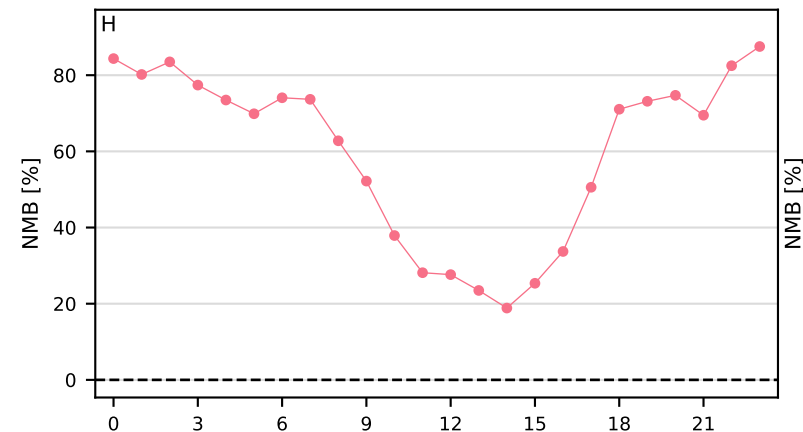
LOCAL·All (62 stations)



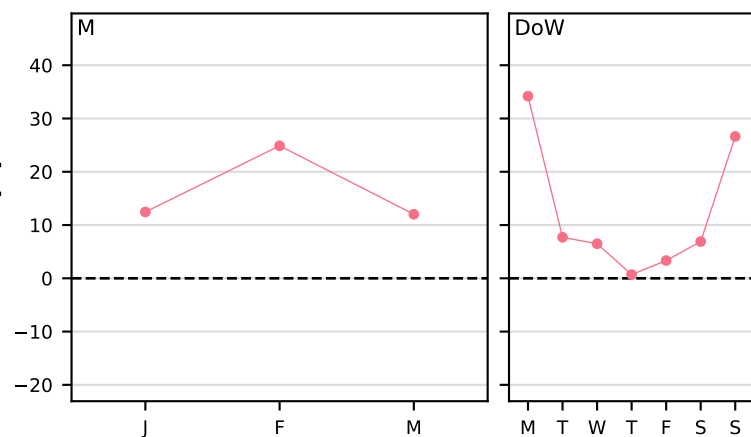
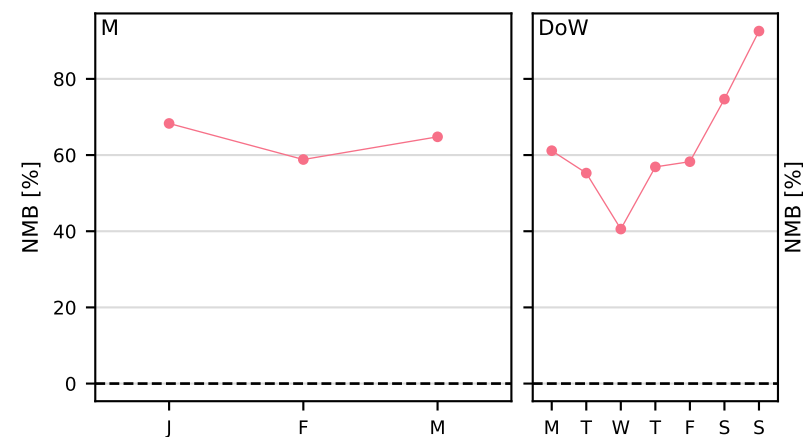
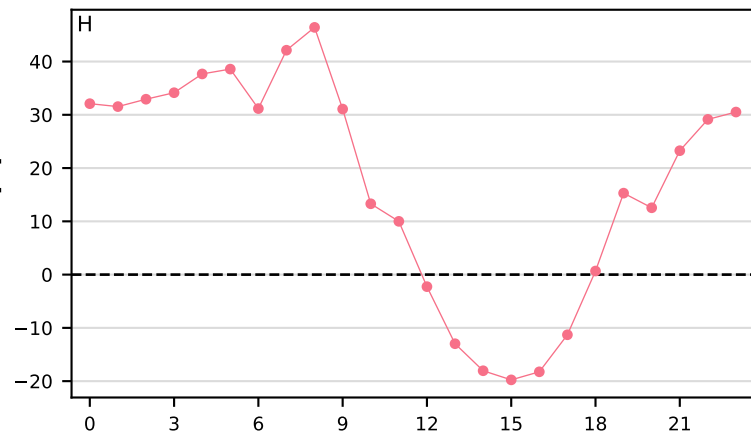
LOCAL·France (1 station)



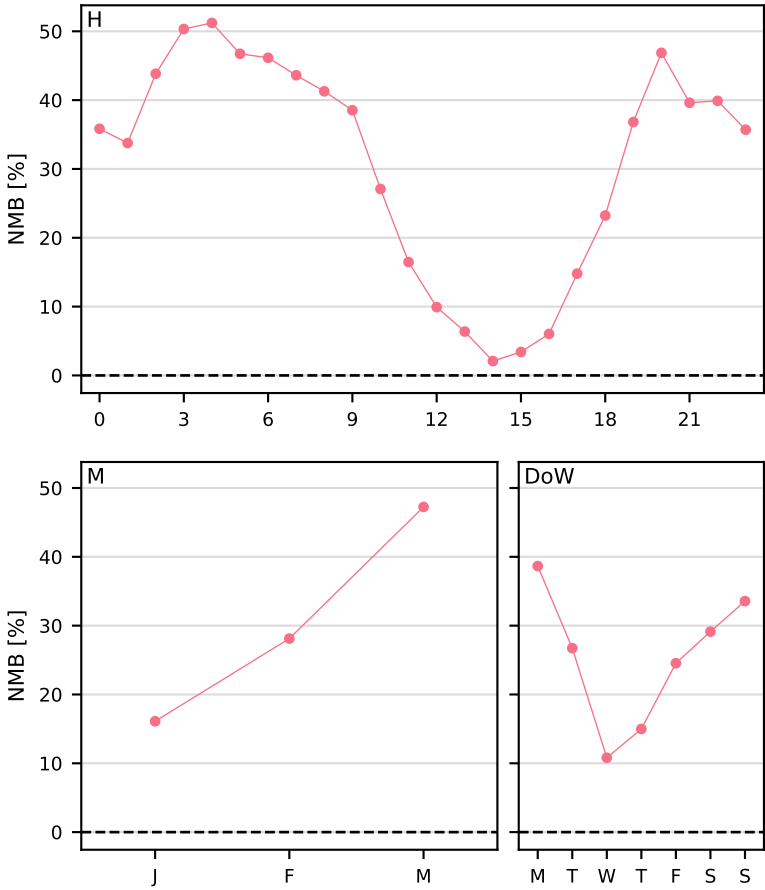
LOCAL·Germany (6 stations)



LOCAL·Spain (13 stations)



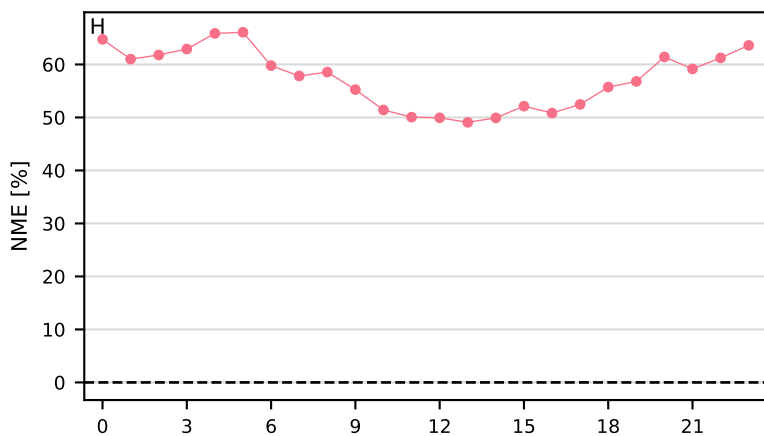
LOCAL·UK (10 stations)



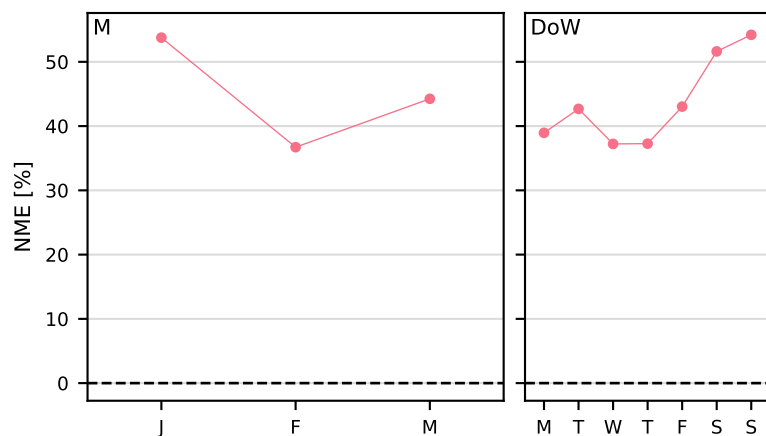
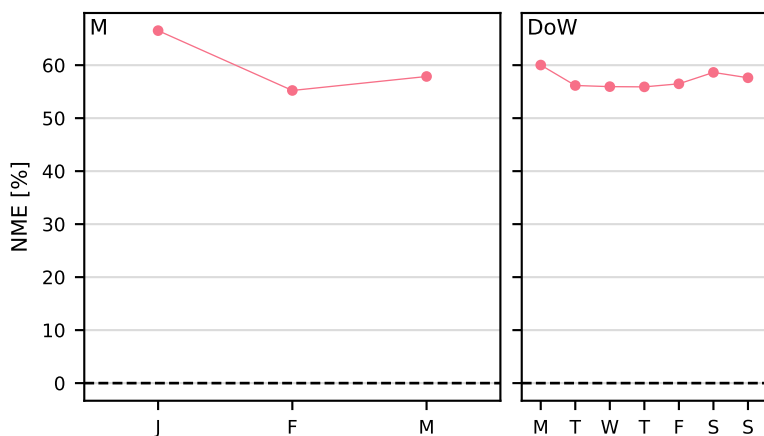
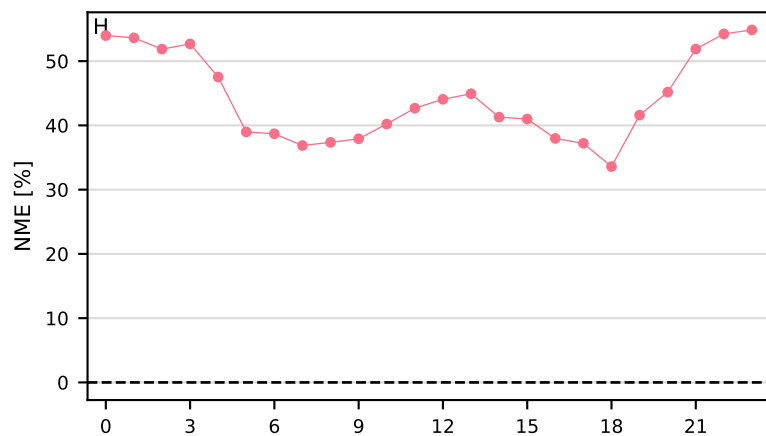
Periodic NME (Summary)
EBAS|sconcn02

● MONARCH

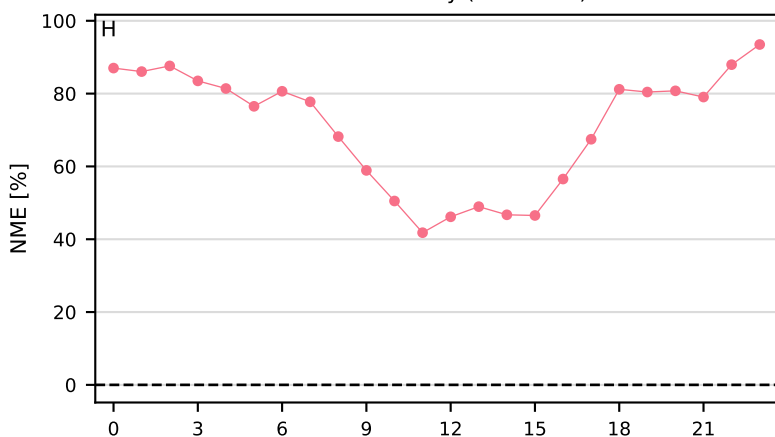
LOCAL·All (62 stations)



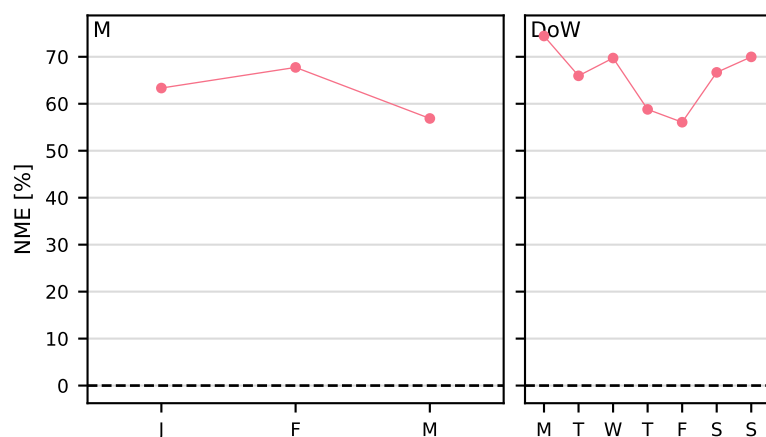
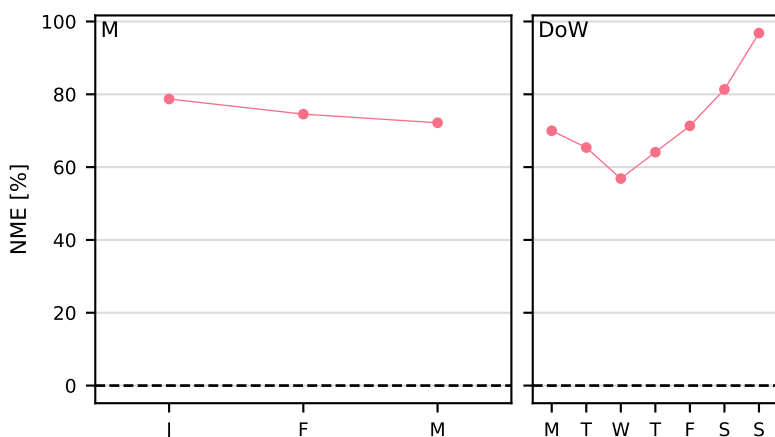
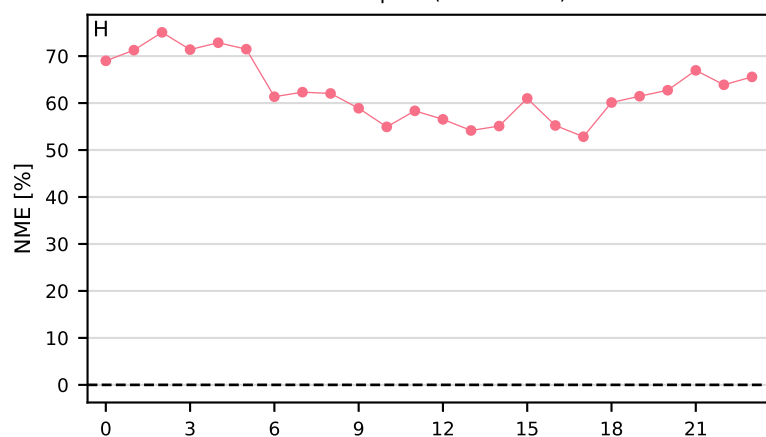
LOCAL·France (1 station)

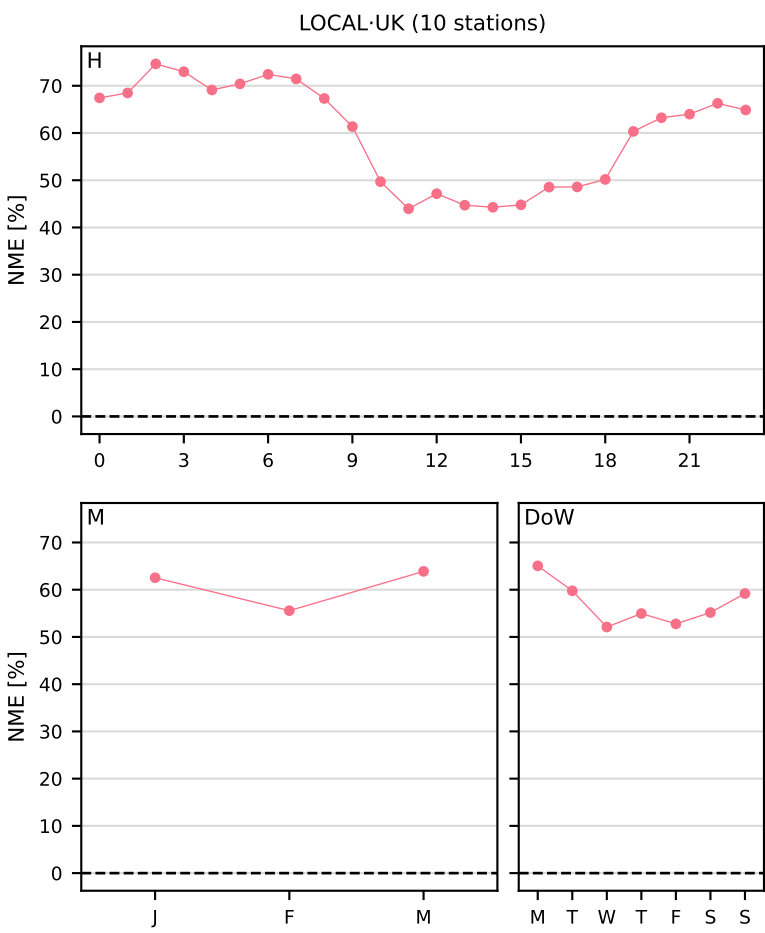


LOCAL·Germany (6 stations)



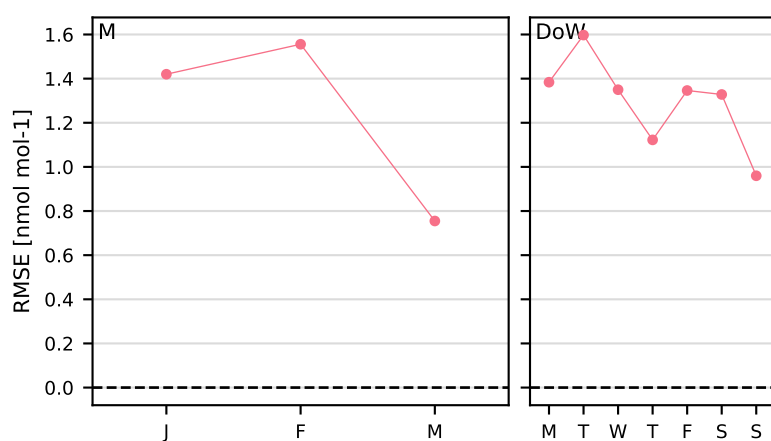
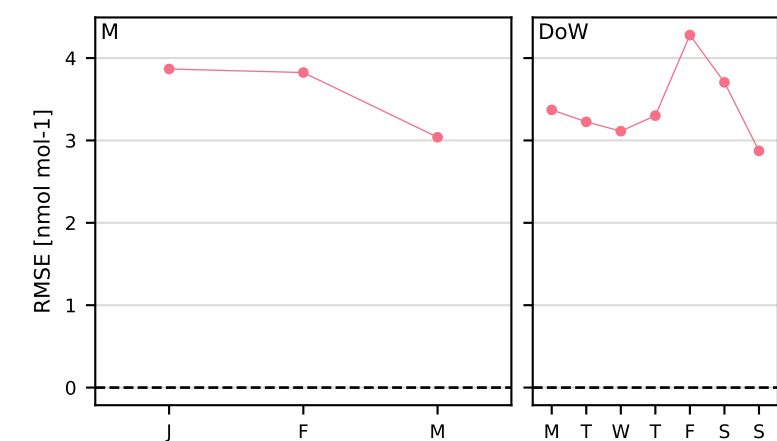
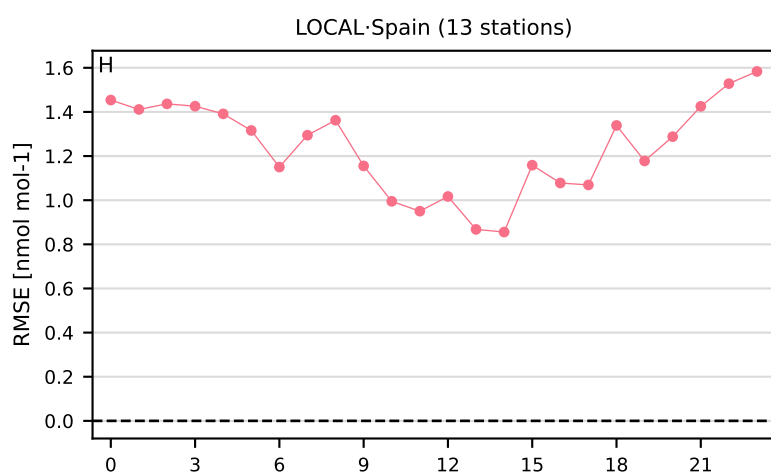
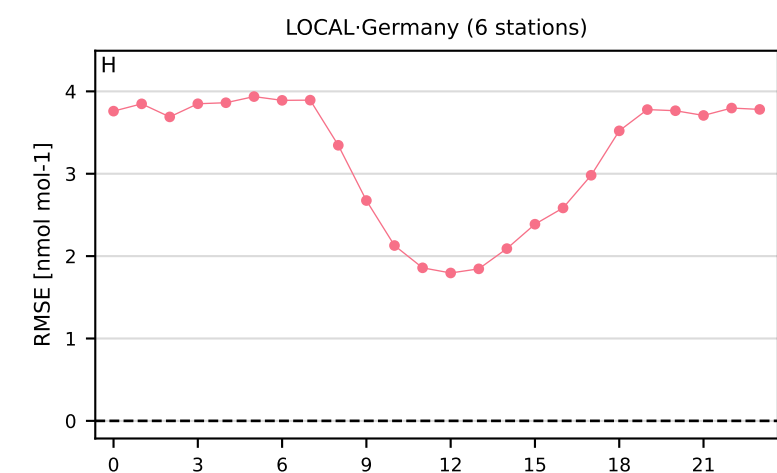
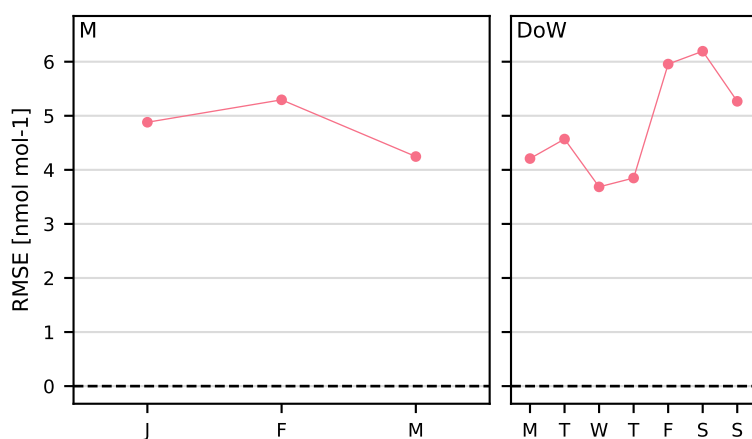
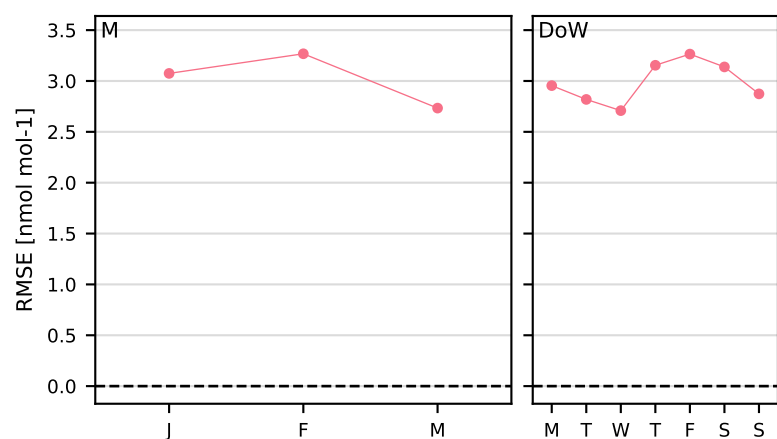
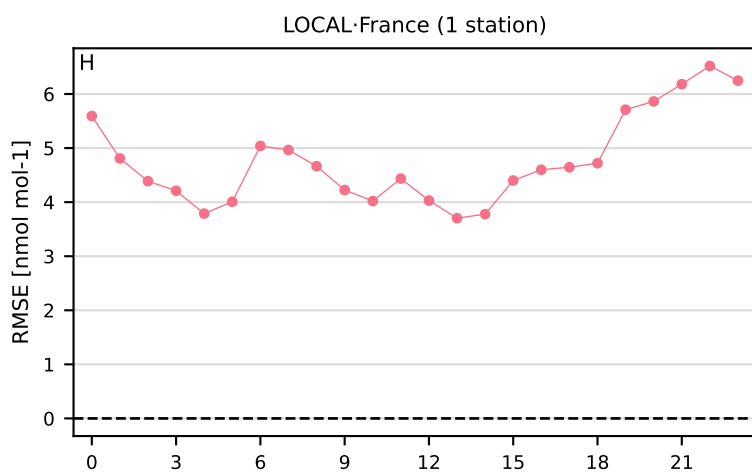
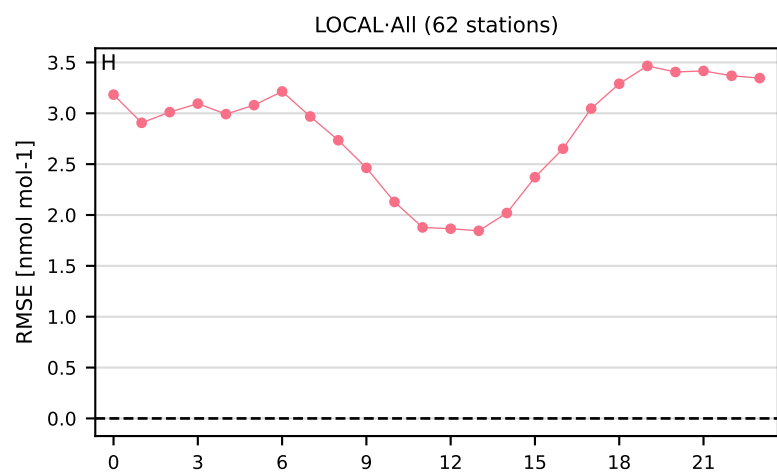
LOCAL·Spain (13 stations)



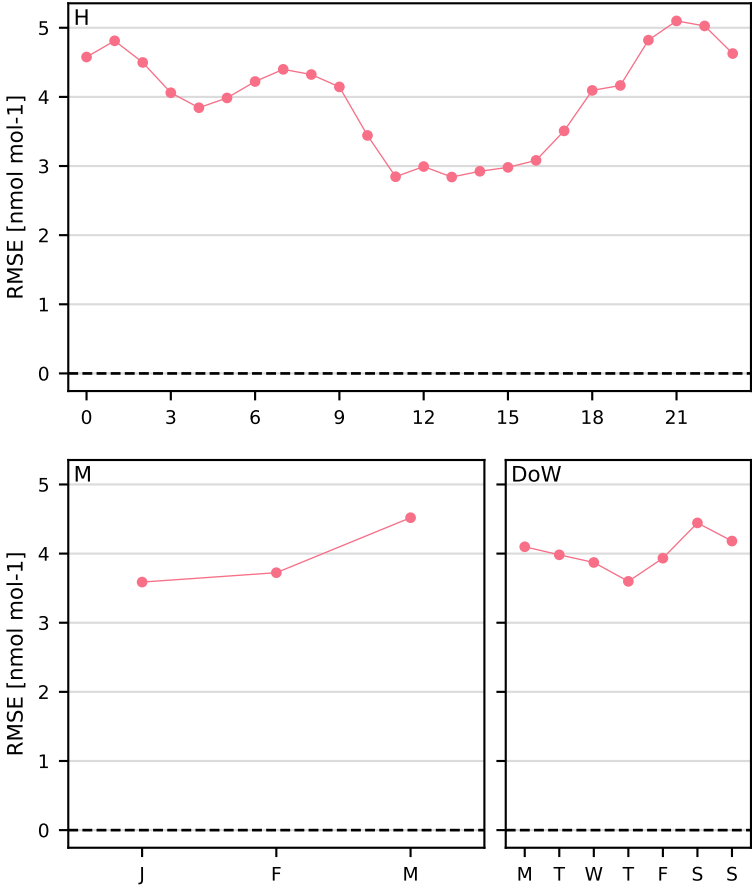


Periodic RMSE (Summary)
EBAS|sconcn02

MONARCH



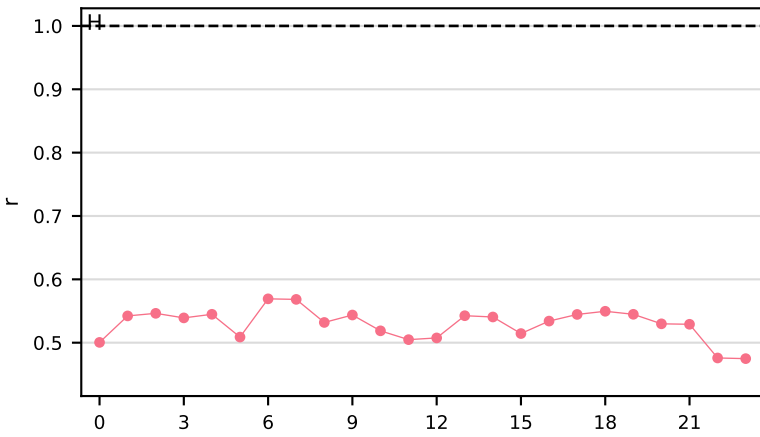
LOCAL·UK (10 stations)



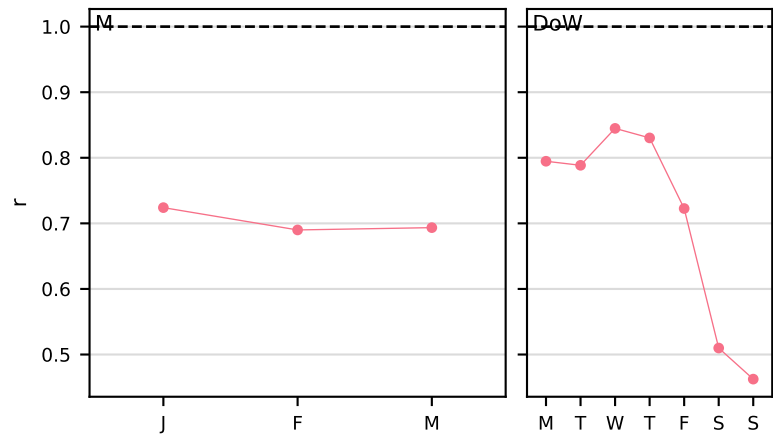
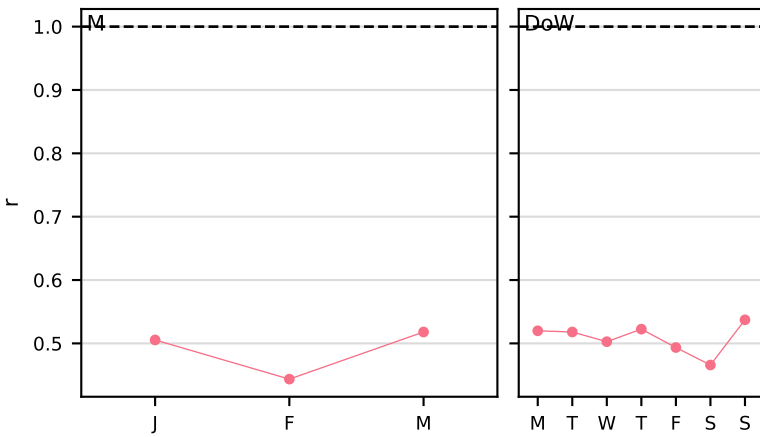
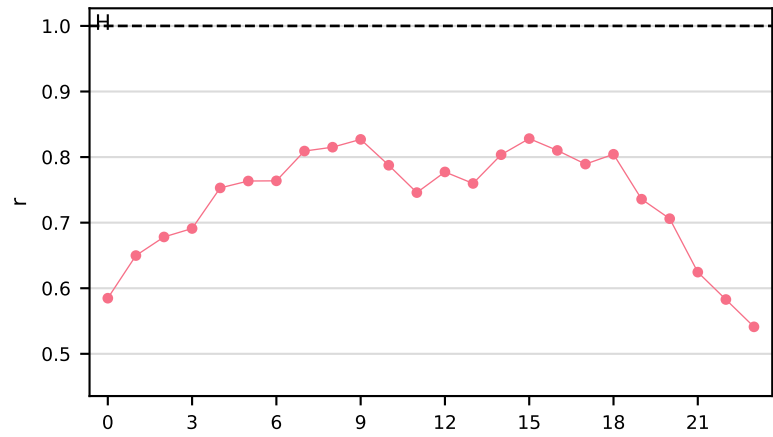
Periodic r (Summary)
EBAS|sconcn02

MONARCH

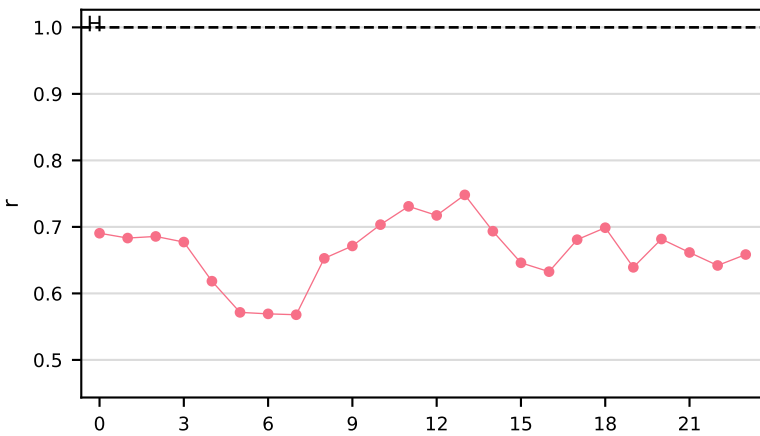
LOCAL·All (62 stations)



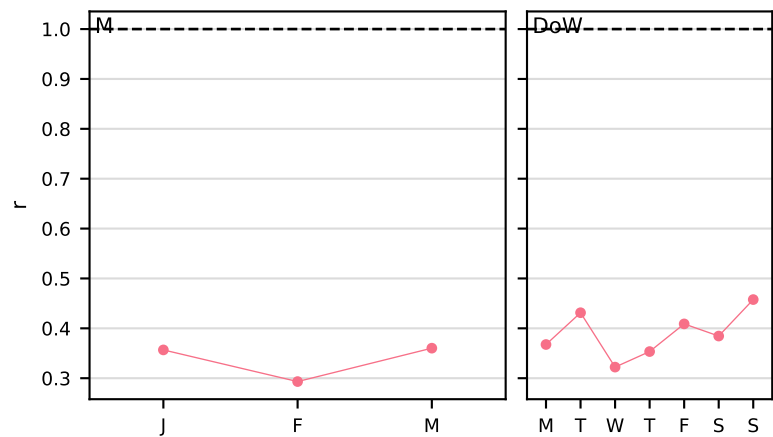
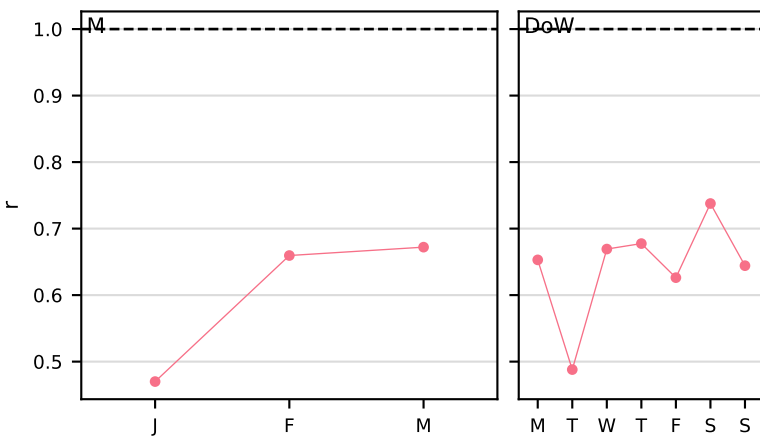
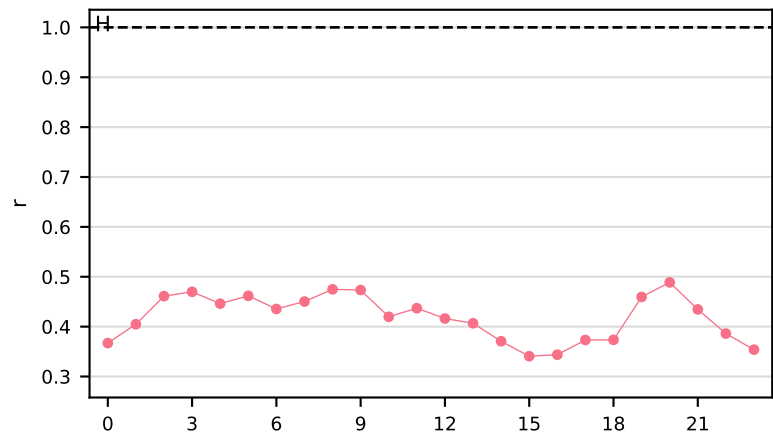
LOCAL·France (1 station)



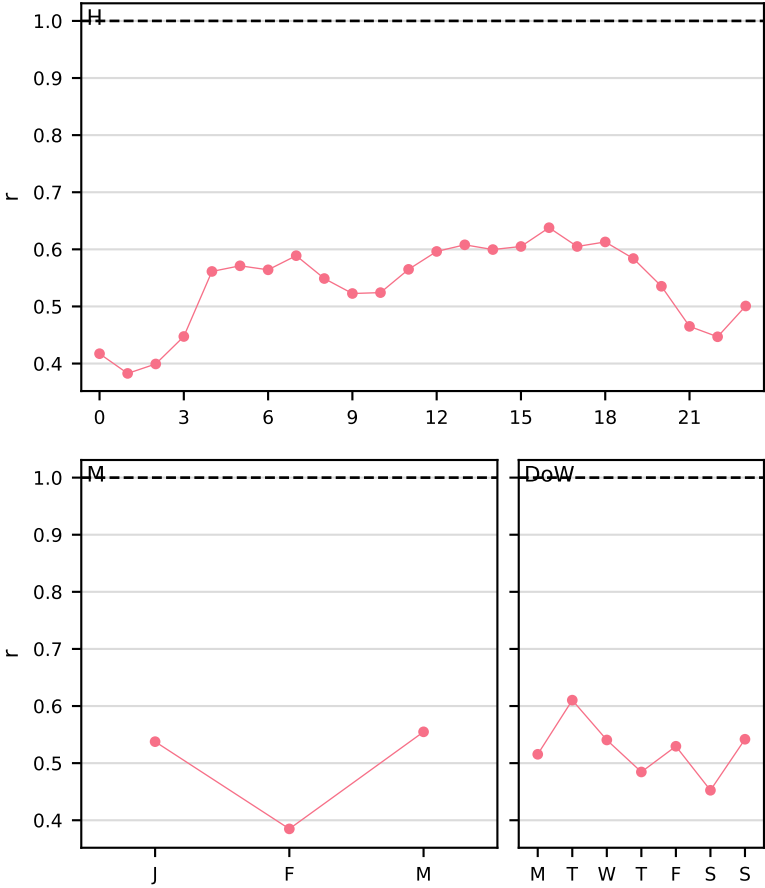
LOCAL·Germany (6 stations)



LOCAL·Spain (13 stations)



LOCAL·UK (10 stations)



Distribution (Summary)
EBAS|sconcn02

● observations ● MONARCH

